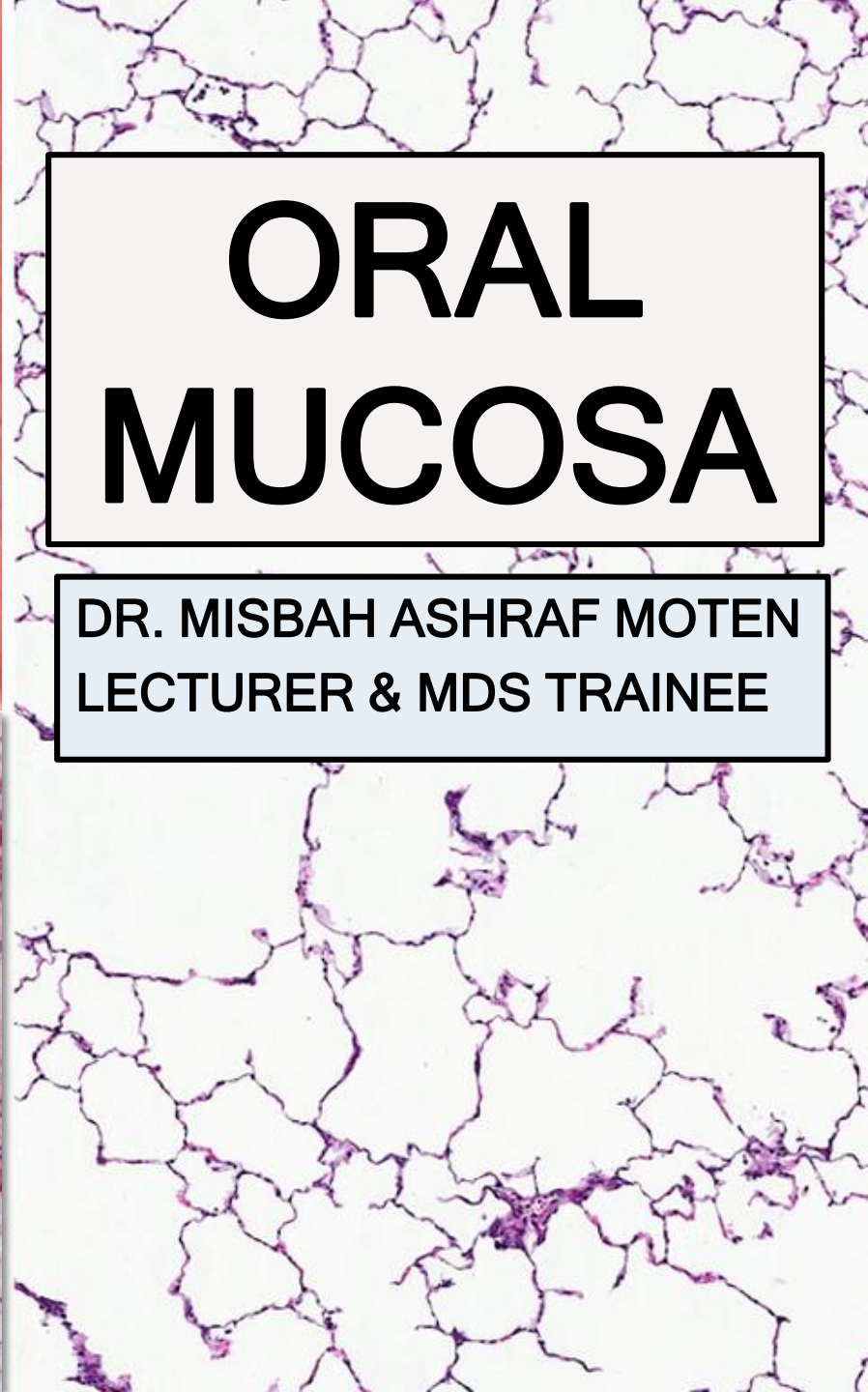
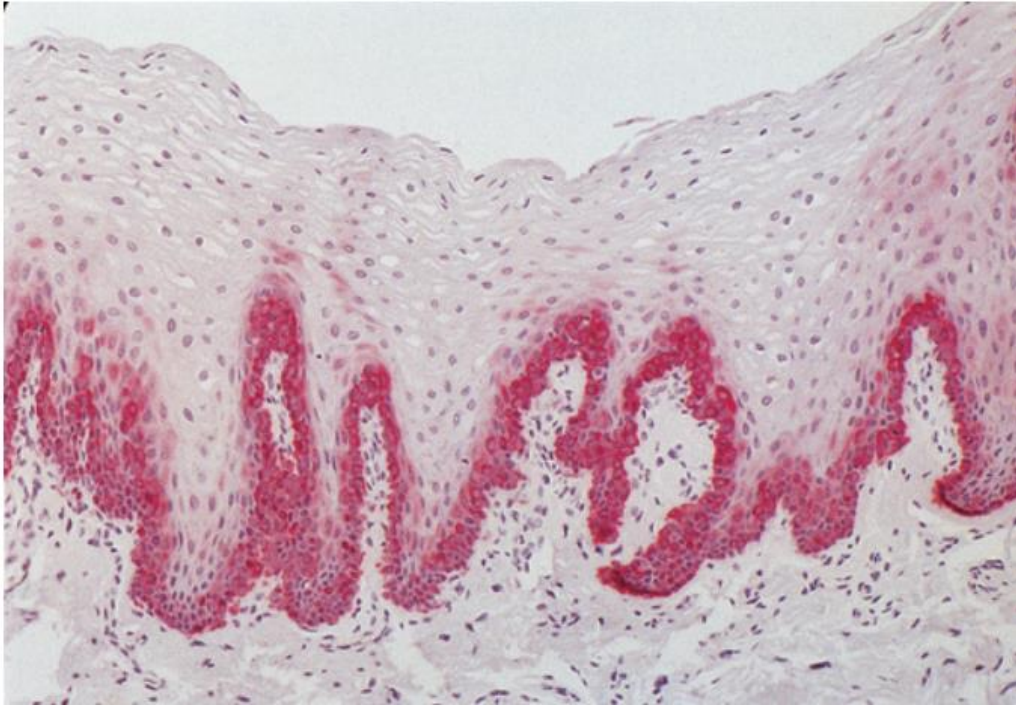




ORAL MUCOSA

DR. MISBAH ASHRAF MOTEN
LECTURER & MDS TRAINEE





LEARNING OUTCOMES



◆ By the end of this session, you will be able to: ◆



- 1** Identify the epithelial layers across oral regions and relate their structure to function.



- 2** Classify the types of keratinocytes and nonkeratinocytes.



- 3** Compare cytokeratin distribution in keratinised vs. nonkeratinised epithelium.



- 4** Define gustatory epithelium.



- 5** Describe the basement membrane's structure and its role in tumour spread.



- 6** Contrast the lamina propria form in masticatory vs. lining mucosa.



- 7** List the constituents of the submucosa, its oral locations, and the salivary gland distribution within it.



- 8** Define free gingiva, attached gingiva, crevicular epithelium, and junctional epithelium.



- 9** Explain the uniqueness of the junctional epithelium and its underlying lamina propria in relation to function.



- 10** Define gingival crevicular fluid (GCF), its composition, and its primary role in health.



- 11** Explain how inflammation alters GCF production and composition, and note that it contains proteins which may act as biomarkers of disease progression.



- 12** Identify the types of oral mucosa lining the tongue, the different papillae, and the distribution of taste buds.



- 14** Recognise that a full appreciation of the normal appearance of the oral mucosa is essential for diagnosing pathological conditions in the oral cavity.

The background of the slide is composed of three distinct sections. The top section is a solid blue bar. Below it, the slide is split into two vertical panels. The left panel shows a pink-stained histological section of oral mucosa, likely showing the lamina propria and underlying connective tissue. The right panel shows a light-colored histological section, possibly of adipose tissue or a different layer of the mucosa, with a network of thin, purple-stained lines representing connective tissue or blood vessels. The title 'ORAL MUCOSA' is centered over the lower part of the blue bar and the top of the two panels.

ORAL MUCOSA

ORGANIZATION, CLASSIFICATION, FUNCTION

Mucous Membrane

Moist lining of the gastrointestinal tract, nasal passages & other body cavities that communicate with the exterior

In the oral cavity the lining is called as oral mucous membrane or “**oral mucosa**”





Why is the palate hard while the cheek is soft?

The Answer: Because Oral Mucosa is specialized according to function

Oral Mucosa - More Than Just a Mouth Lining

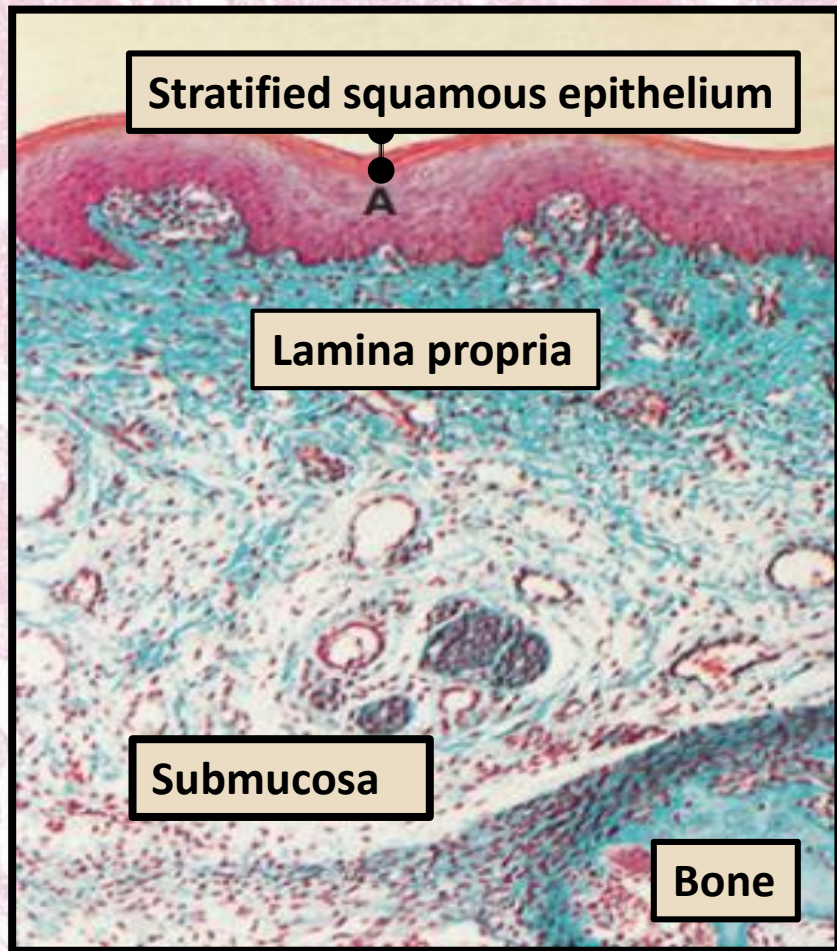


Fig. 14.1 Section showing regions of oral mucosa. A = stratified squamous epithelium; B = lamina propria; C = submucosa; D = bone (Masson's trichrome; $\times 35$).

- Basic pattern of an *epithelium* with an underlying connective tissue (*the lamina propria*)
- In many regions a third layer, the **submucosa**, lies between lamina propria and underlying bone (palate) or muscle (cheeks/lips).

THREE PERSONALITIES

• OF ORAL MUCOSA •



THE WARRIOR

MASTICATORY MUCOSA



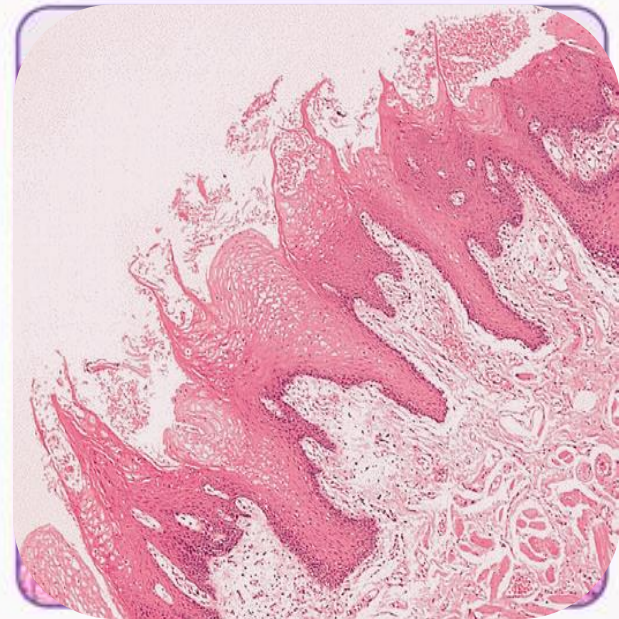
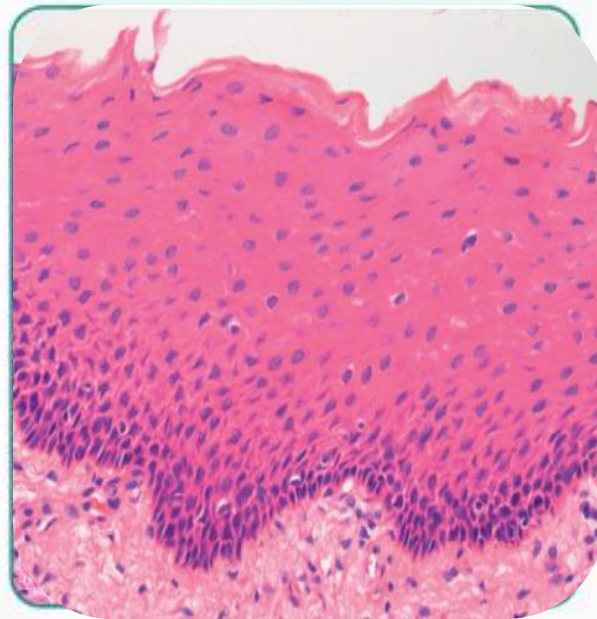
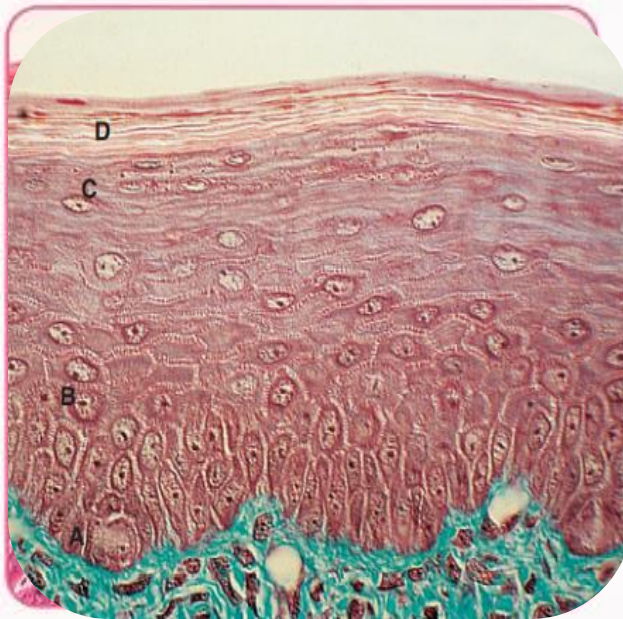
THE ACROBAT

LINING MUCOSA



THE FOOD CRITIC

SPECIALIZED MUCOSA



CLASSIFICATION OF ORAL MUCOSA

Masticatory Mucosa (60%)

keratinised

Found in high
compression/fr
iction

hard palate,
gingiva

Lining Mucosa (25%)

nonkeratinised

Areas subjected
to less stress

lip, cheek, floor of
mouth

Specialized Mucosa (15%)

gustatory mucosa
of tongue dorsum
(anterior 2/3,
partly keratinised)

vermilion zone of
lip.

Interesting fact: Posterior 1/3 of tongue dorsum = nonkeratinised, contains lingual tonsils.

FUNCTIONS

OF ORAL MUCOSA



PROTECTION

- Resists compressive & shearing forces
- Barrier against microorganisms
- Blocks toxins & antigens



Barrier for mechanical trauma & microbiological insults



SENSATION

- Richly innervated
- Touch
- Proprioception
- Pain
- Taste



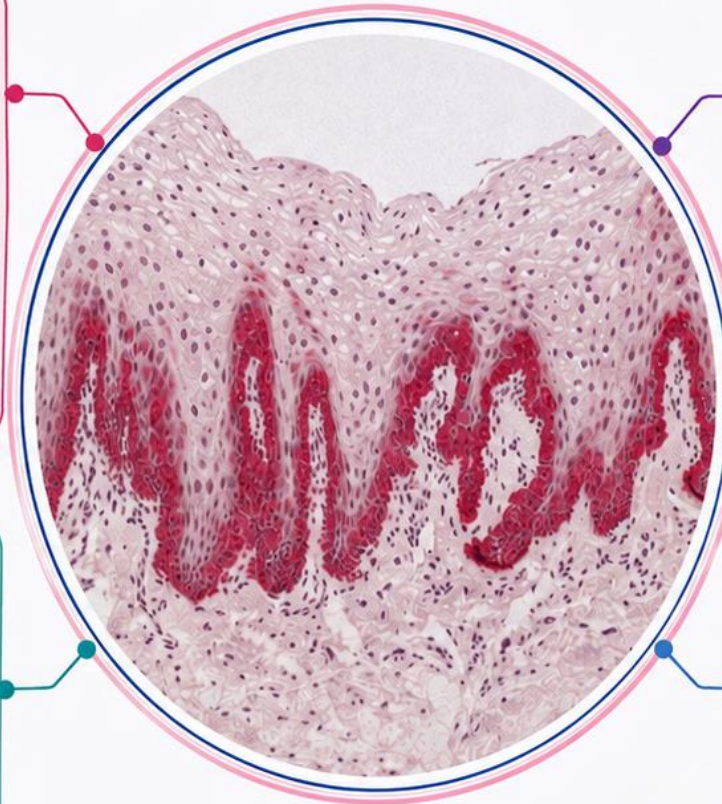
DEFENSE

- Immunological defense
- Humoral
- Cell-mediated



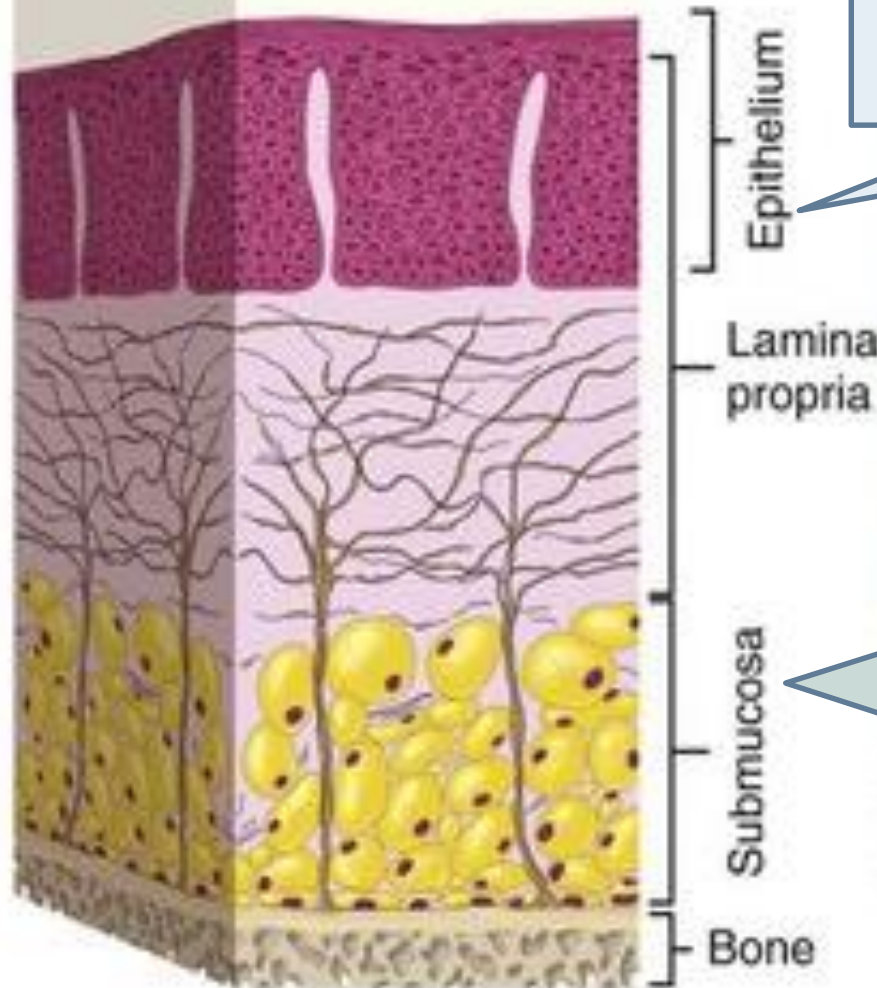
SECRETION

- Lubrication & buffering
- Secretes antibodies
- Mucous film retains water & electrolytes
- Acts as a protective barrier



Strong yet *flexible*, sensitive yet *protective* – oral mucosa is truly remarkable!

Oral mucosa



Stratified squamous epithelium
(keratinized usually eg., gingiva,
palate, dorsum of tongue)

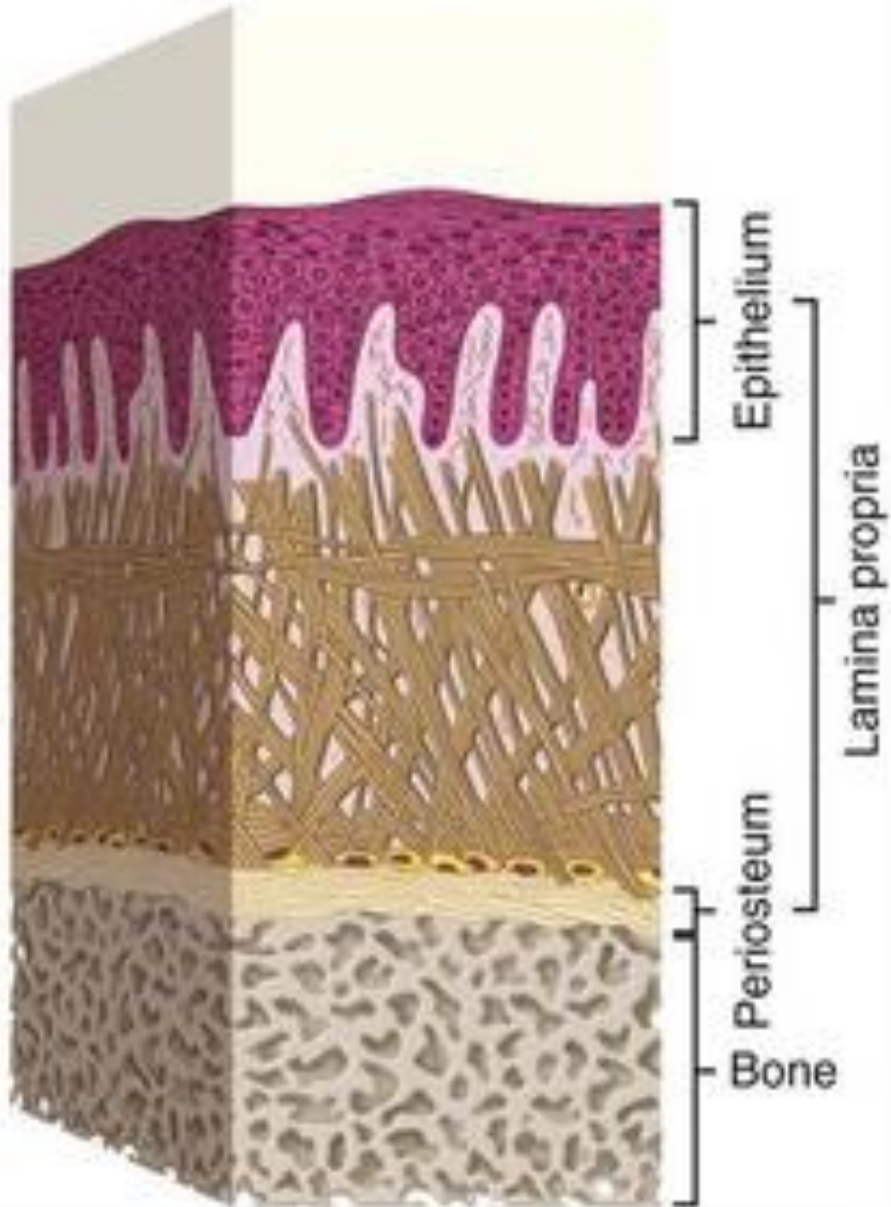
Origin of epithelium:
Ectoderm or endoderm

Lamina propria
Connective tissue

Submucosa
looser connective tissue with
fat deposits & glands.
larger nerves & blood vessels.
Lamina propria & submucosa's
connective tissue boundary =
indistinct

B

Oral mucoperiosteum



C

Masticatory mucosa has thick lamina propria which is usually firmly bound to bone (**mucoperiosteum**).

It is found in places of friction & compression

EPITHELIUM



LAYERS OF ORAL EPITHELIUM

**keratinised (cornified) layer
(stratum corneum).**

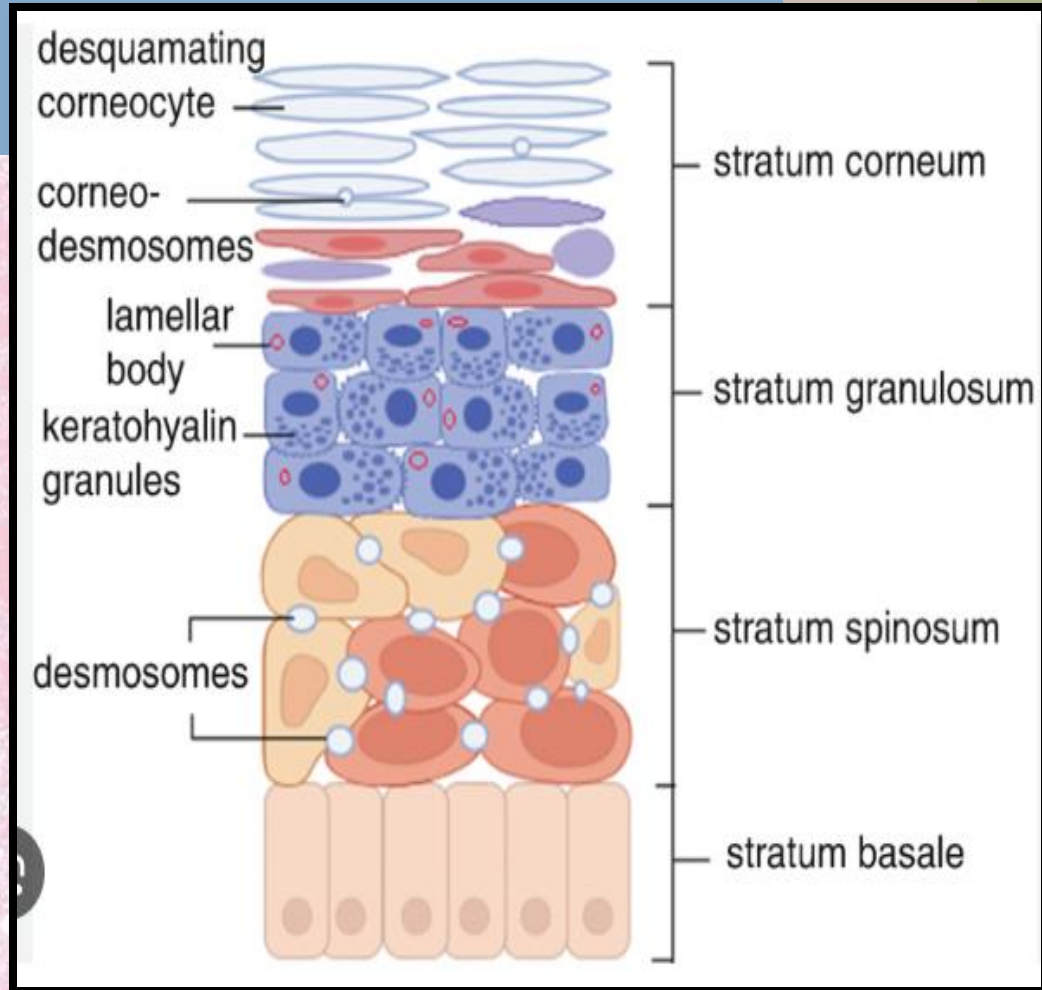
**granular layer (stratum
granulosum)**

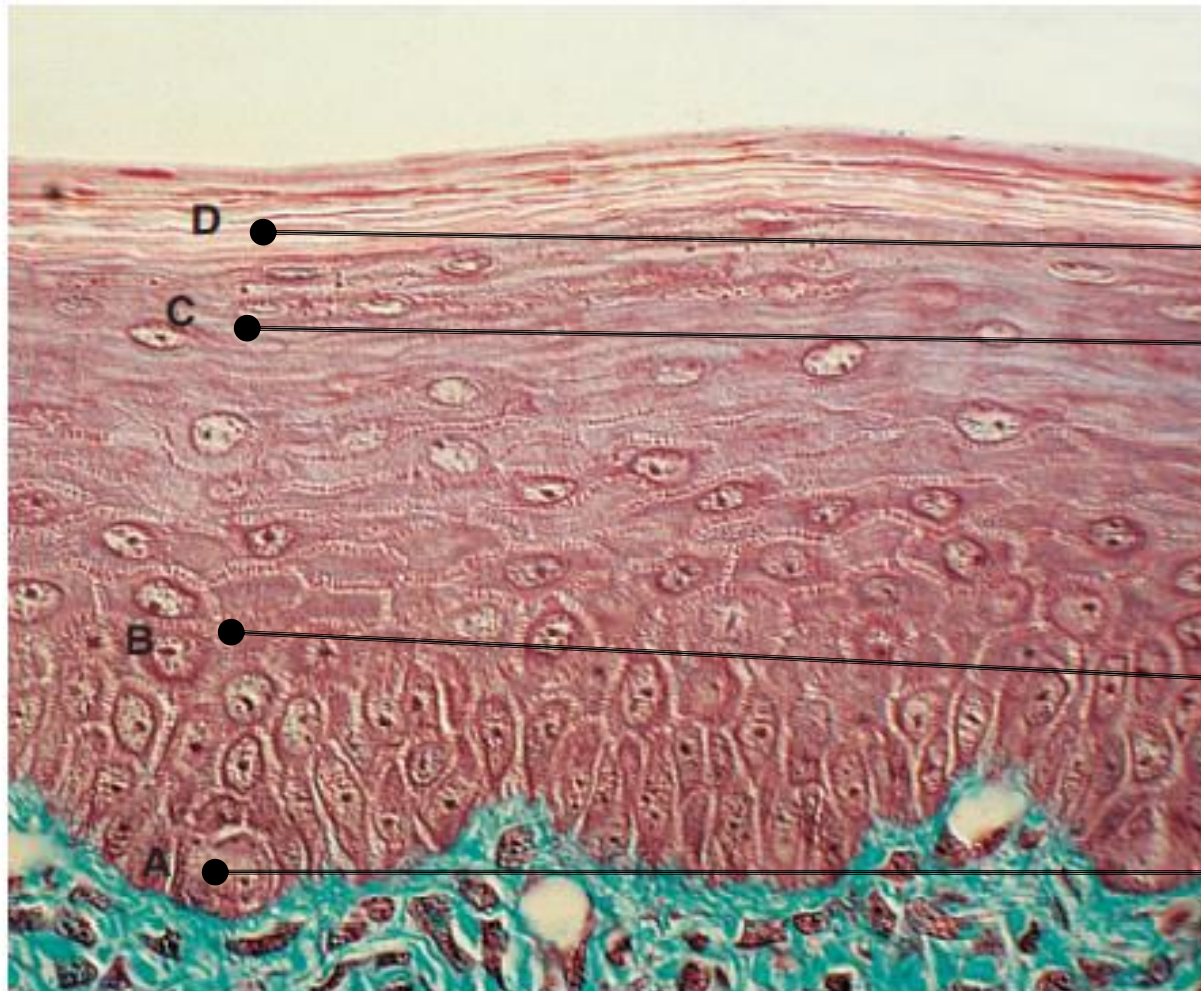
**prickle cell layer (stratum
spinosum)**

**basal layer (stratum
germinativum or stratum basale)**

Masticatory mucosa = all 4 layers (present)

Lining mucosa = outer 2 layers (granular and keratinised layers) absent & replaced by non-keratinizing superficial layer





Keratinised layer

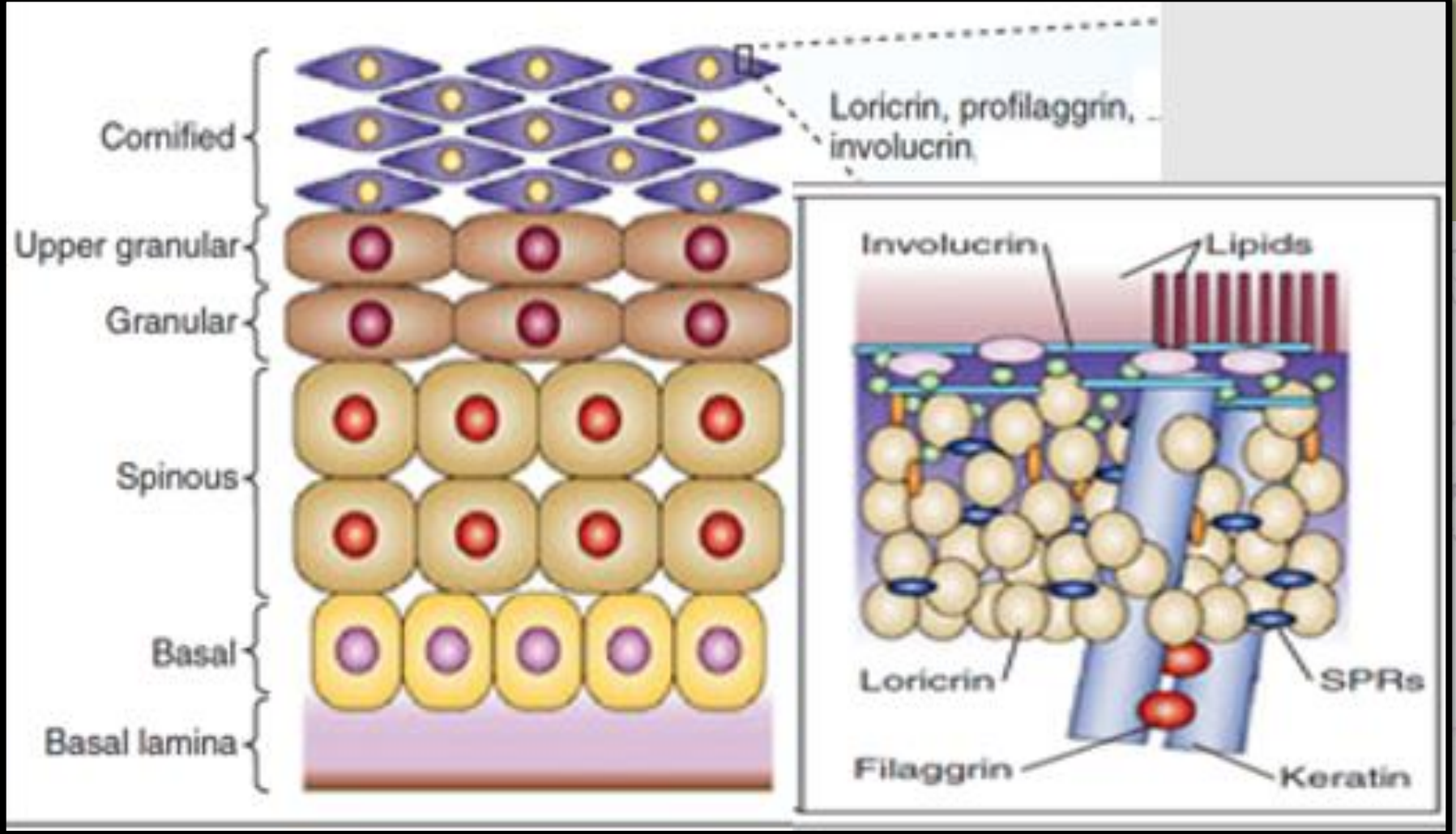
granular layer

Prickle cell layer

basal layer

Fig. 14.2 Section showing layers of keratinised oral epithelium. A = basal layer; B = prickle cell layer; C = granular layer; D = keratinised layer ($\times 180$).

End product of all layers : covering layer of keratinocytes filled with Cytokeratins



Proteins related to the cell membrane, such as loricrin, involucrin & filaggrin, are also synthesised as the cells differentiate.

Epithelial cells exhibit endocytosis & therefore have the capacity to internalise small particles.

TURNOVER TIME: ESTIMATED TIME NECESSARY TO REPLACE ALL EPITHELIAL CELLS

- **Progenitor population:** Divide & provide new cells (Proliferation)
- **Maturing population:** Undergo differentiation (maturation)

TURNOVER RATES OUTSIDE ORAL CAVITY

Skin	52 to 75 days
Gut	4 to 14 days

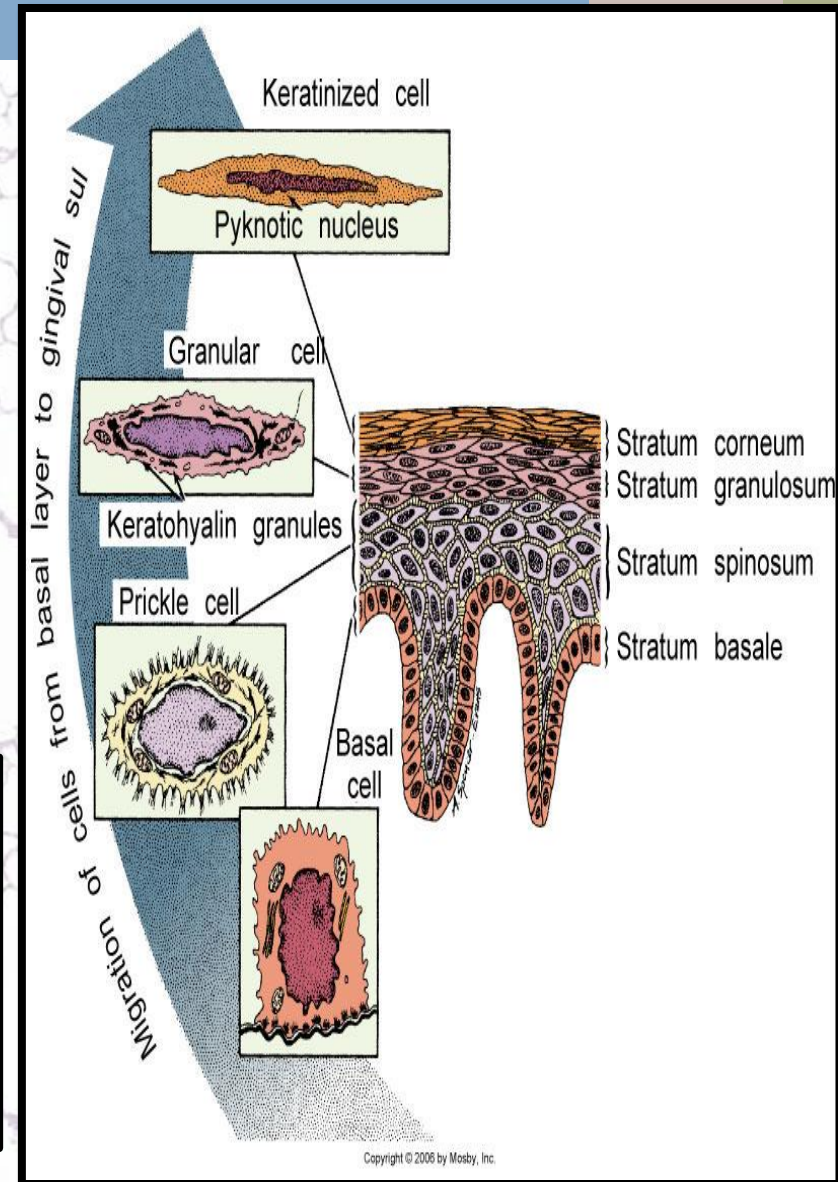
TURNOVER RATES: ORAL MUCOSA:

Junctional & sulcular epithelia: 5 days (fastest)

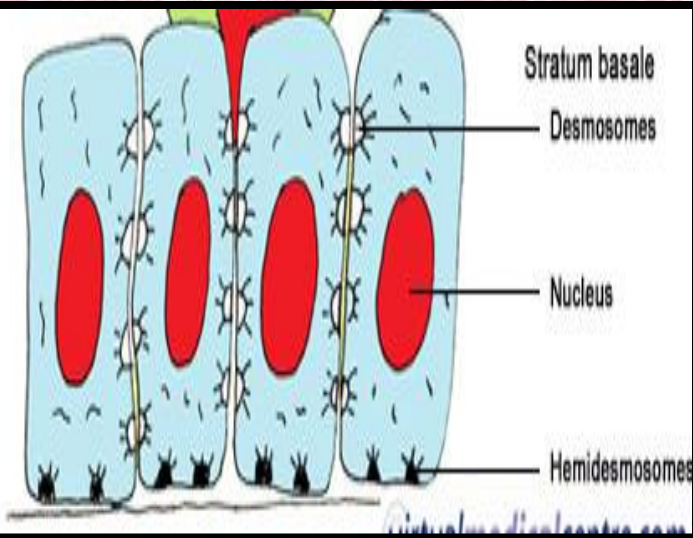
Junctional & sulcular epithelia: 2x faster than > nonmasticatory (lining mucosa)

lining mucosa little > masticatory mucosa

- Gingiva: 41 to 57 days
- Cheek: 25 days

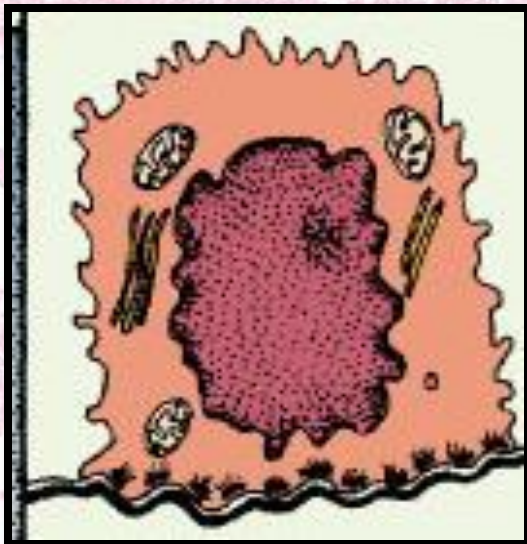


Stratum Basale



Single cell layer next to lamina propria → basal lamina demarcates from it

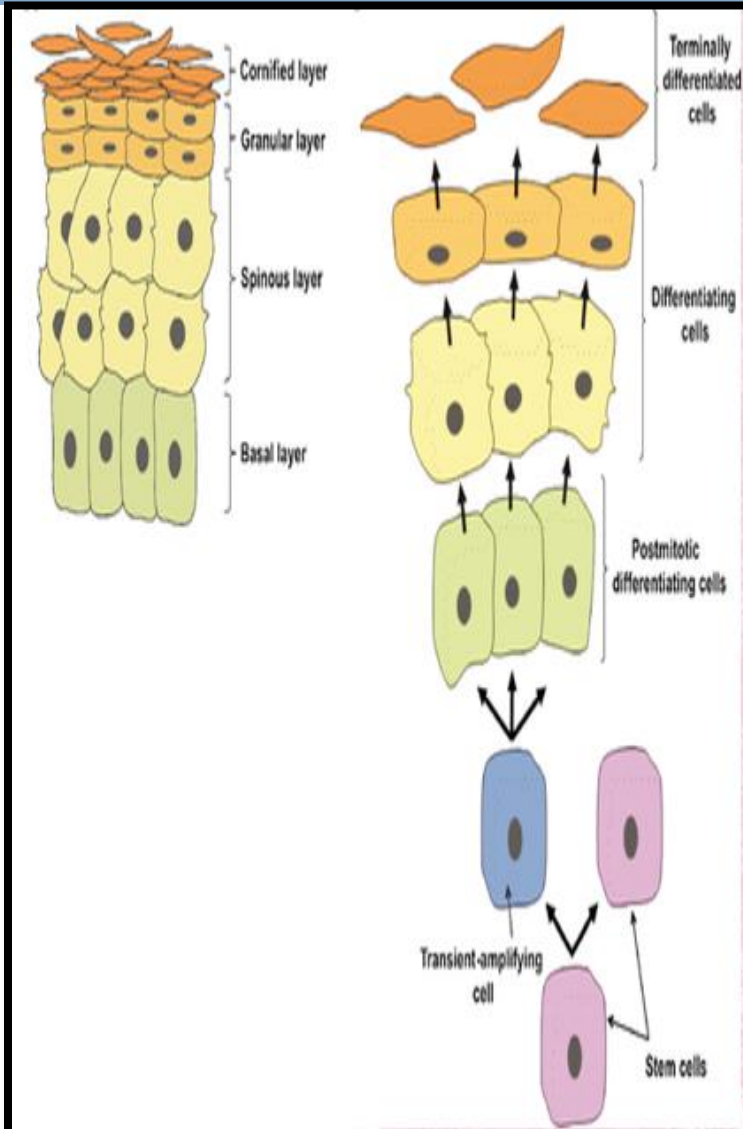
Cuboidal cells includes → stem cell population



Least differentiated cells of oral epithelium → limited organelles (ER, Golgi, mitochondria), some keratin tonofilaments (representing epithelial nature).

Cell contacts: desmosomes, hemidesmosomes, intermediate & gap junctions

Stratum Basale



stem cell population present mainly at base of rete ridges that project into lamina propria.

Stem cell mitosis $\rightarrow$ 2 daughter cells, at least one remains stem cell.

Stem cells $\rightarrow$ **transit-amplifying cells** $\rightarrow$ migrate up, differentiate $\rightarrow$ keratinocytes.

Not all dividing basal cells are true stem cells.

Stem cells express antiapoptotic proteins e.g. **Bcl2** (may help retain 'stemness').

Stratum spinosum

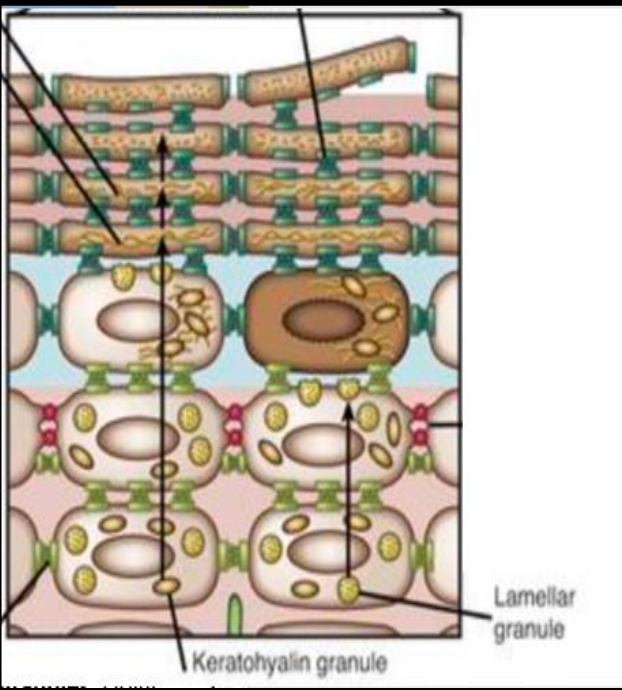
A: Prickle layer (upper part)

Round/ovoid cells, several cells thick; larger & rounder than basal cells

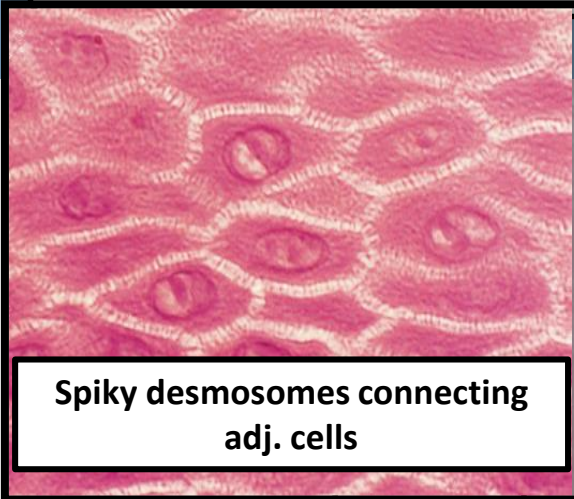
New CK types appear here (transition marker) → involved in tonofilaments formation, which become thicker and more conspicuous.

Upper prickle layer → small **membrane-coating granules** (~0.25 μm), phospholipid-rich, **lamellar structure** (only in keratinized epithelium)

Origin of membrane coating granules: Golgi apparatus
More superficial fate → come to lie close to cell membrane



Stratum spinosum

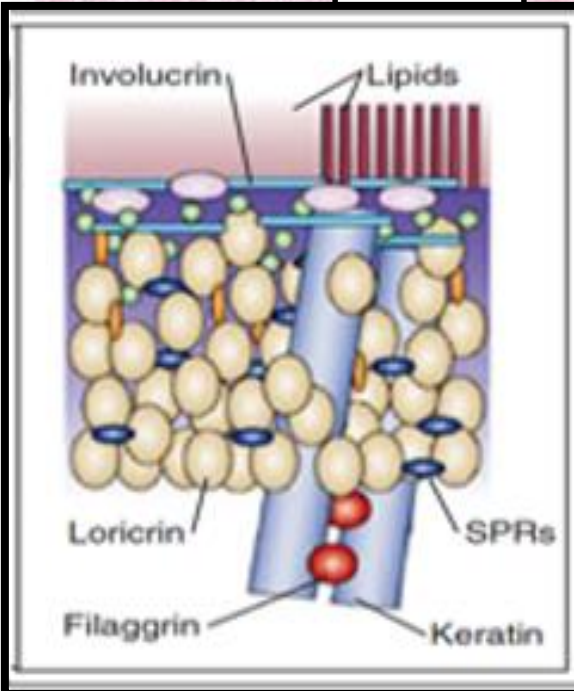


Desmosomes ↑ in number (~50% of intercellular space eventually) - shrinkage in histological prep separates cells at non-desmosomal points → classic 'spiny' / prickly appearance.

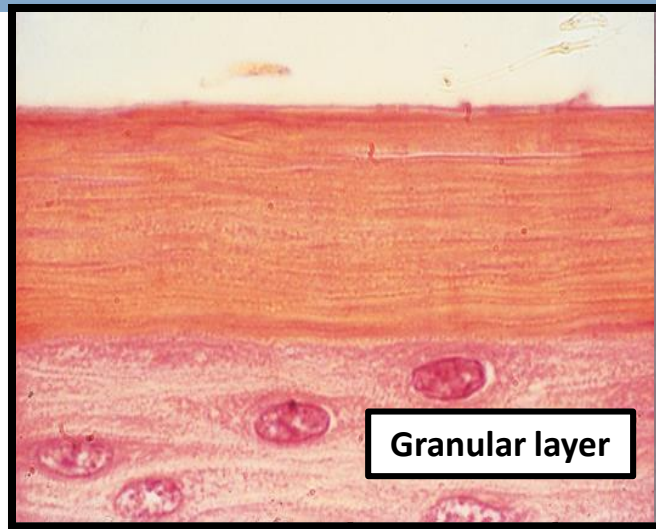
1st stages of maturation

Involucrin (the soluble precursor protein of the cornified envelope eventually found in the cornified layer) appears first in the prickle cell layer.

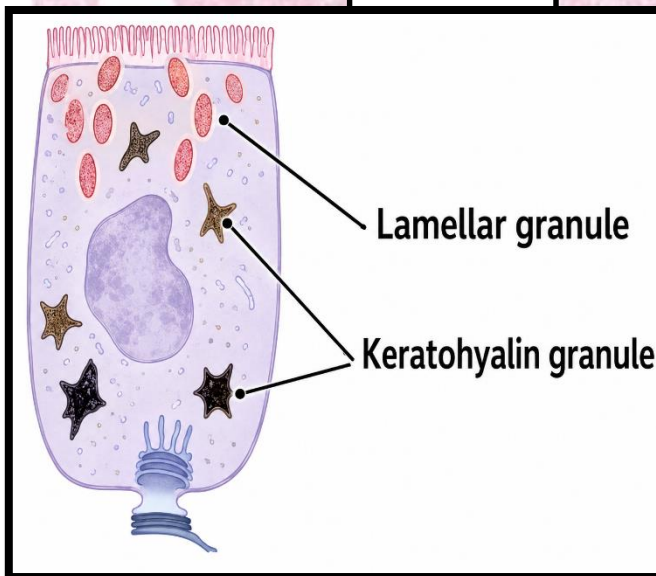
Parabasal = deepest part of prickle layer next to basal layer; may proliferate, resembles basal cells.



Stratum Granulosum



Granular layer



Further maturation; many organelles reduced/lost
— cytoplasm dominated by tonofilaments/
tonofibrils.

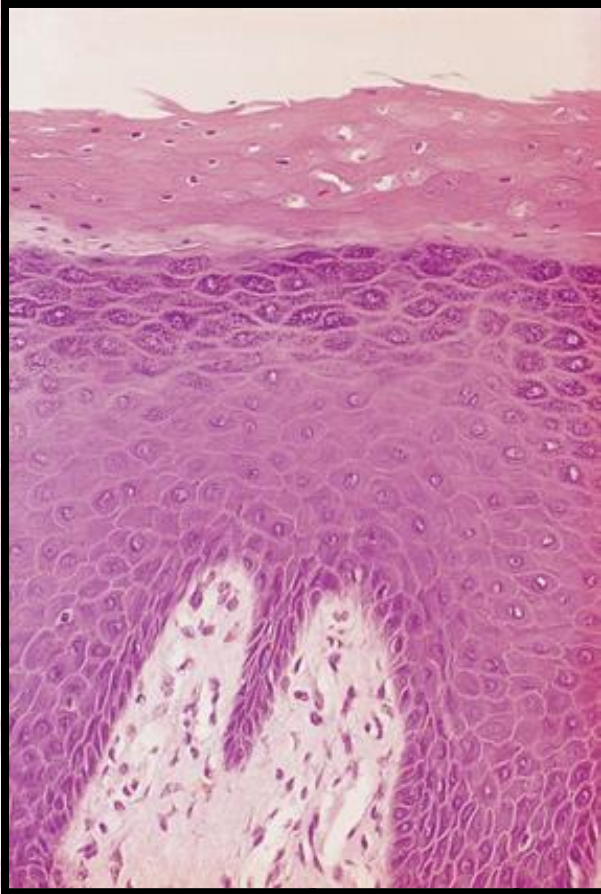
Cells larger & flatter.

Contains numerous **keratohyaline granules** (0.5–1.0 μm) — contain **profilaggrin** (precursor of **filaggrin**, binds keratin filaments into stable network).

Membrane-coating granules discharge lipid-rich contents into intercellular space → 'cements' with tight junctions → limits permeability & water loss.

Synthesis of **loricrin** and **involucrin** continues (form resistant cell envelope).

Stratum corneum (keratinized layer)

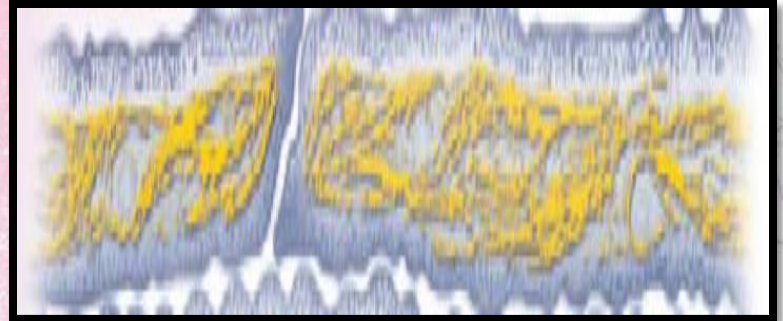


Final maturation stage: complete loss of organelles (nuclei, keratohyaline granules) — autolysis, likely via released proteases.

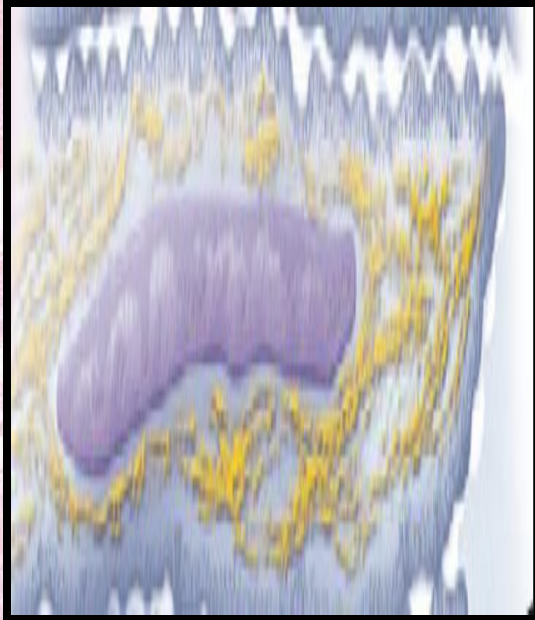
Cells filled entirely **with tonofilaments + filaggrin matrix protein** = collectively called **keratin**.

Keratin cross-linked by disulphide bonds → mechanical/chemical resistance.

parakeratinised oral epithelium (gingiva). Nuclei retained in heavily stained suprficial layers



Stratum corneum (keratinized layer)



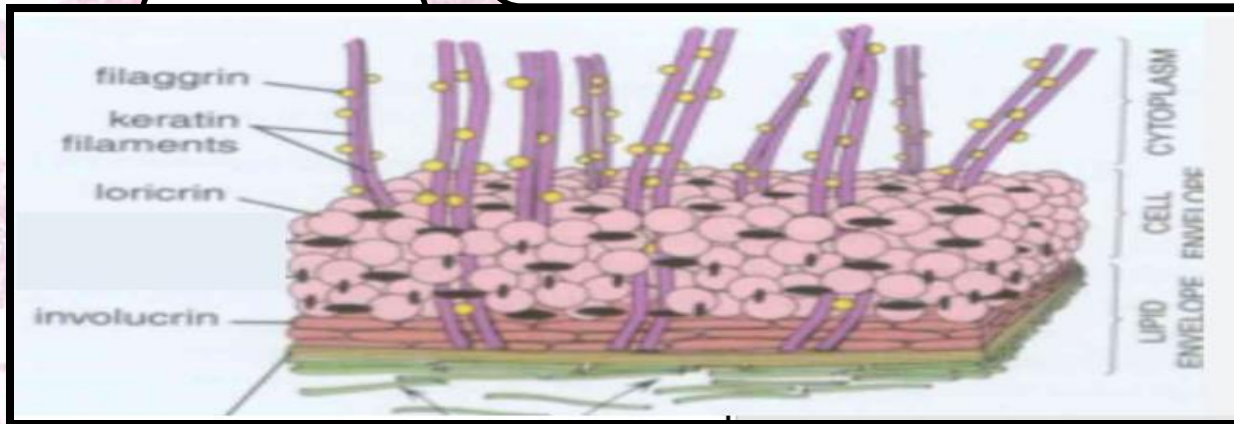
Involucrin cross-linked by transglutaminase enzyme → thin (10 nm) resistant cornified envelope beneath plasma membrane. Trigger = cell death + Ca^{2+} influx.

~75% of cornified envelope = loricrin;

Involucrin = ~5% acting as key lipid-binding site (forms water-insoluble barrier).

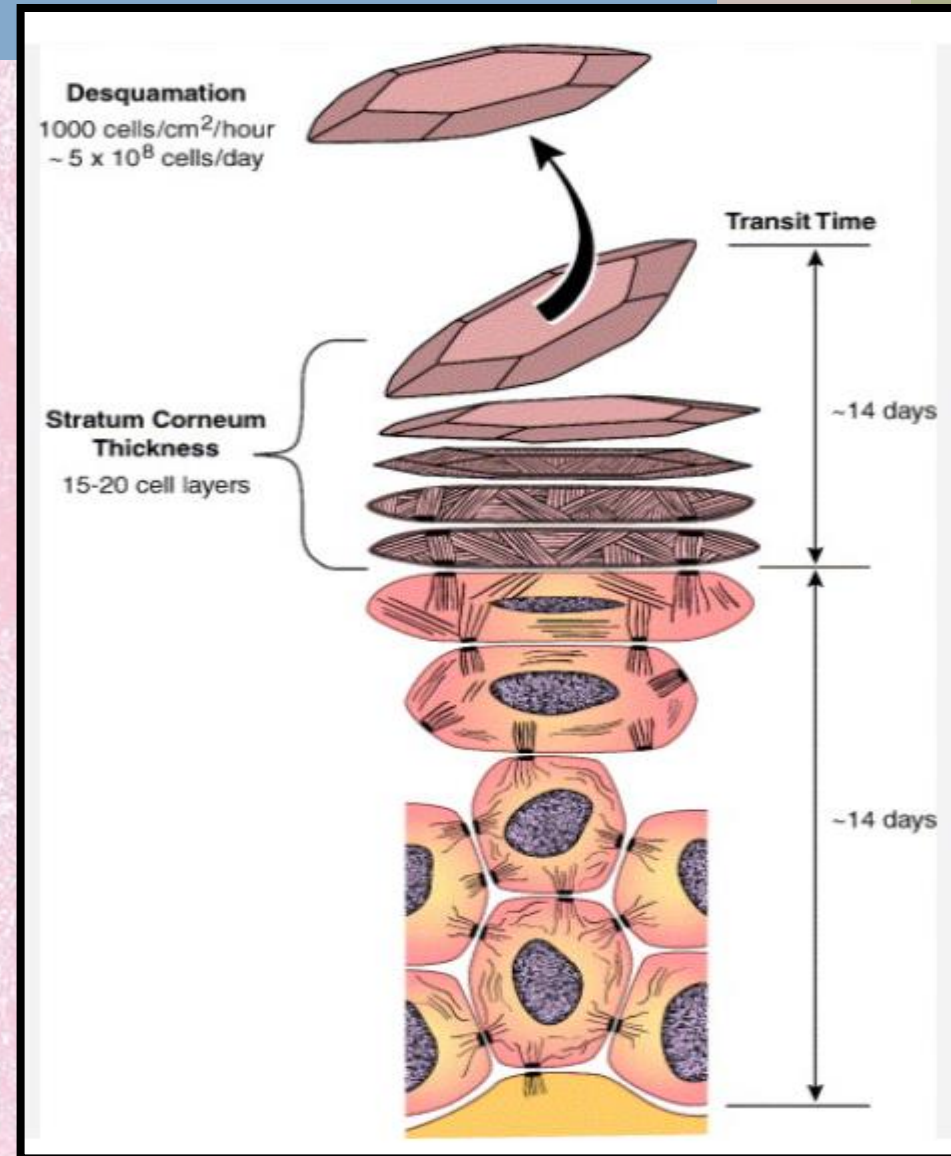
Parakeratinised = nuclei retained (small, shrunken) e.g. gingiva.

Orthokeratinised = no nuclei (more usual pattern).



DESQUAMATION

- **Superficial cells = epithelial squames, shed via desquamation (desmosomes weaken/disappear).**
- Keratinised layer thickness up to 20 cells – thicker in oral mucosa than most skin (except palms/soles).



NON-KERATINIZED EPITHELIUM

Granular and keratinized layers (absent); ends with superficial layer

Faster turnover rate

Keratin granules (Absent)

Tonofilament Organization: loosely arranged & less developed tonofilaments due to lack of filaggrin

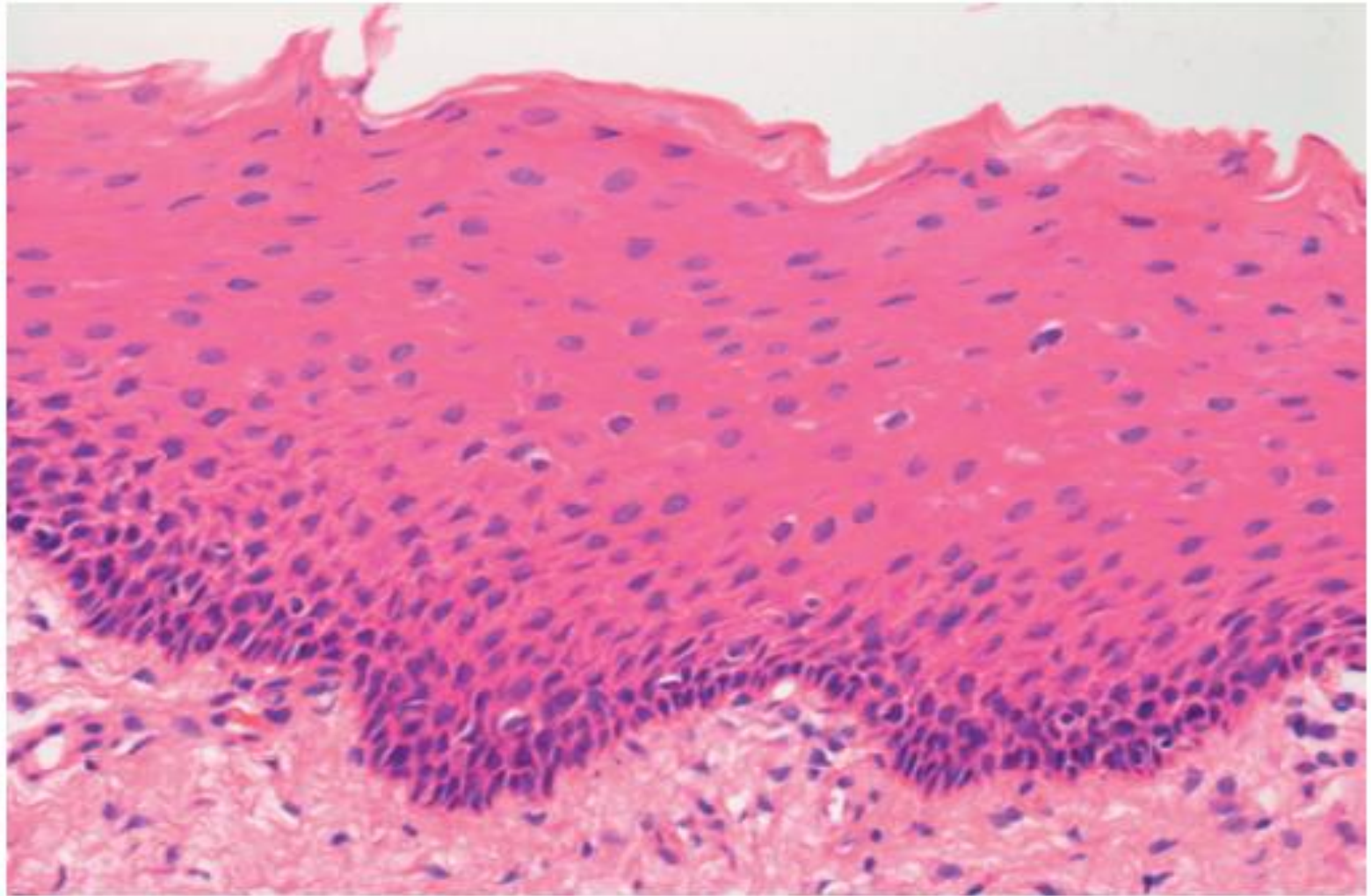
Organelles in lining epithelium > organelles in keratinized epithelium, but < basal layers

Surface cells retain prominent nuclei

Upper layers → not clearly demarcated

Smaller membrane-coating granules with less-defined lipid layers, resulting in increased permeability

Lacks filaggrin & loricrin but contains involucrin



NON-KERATINIZED EPITHELIUM

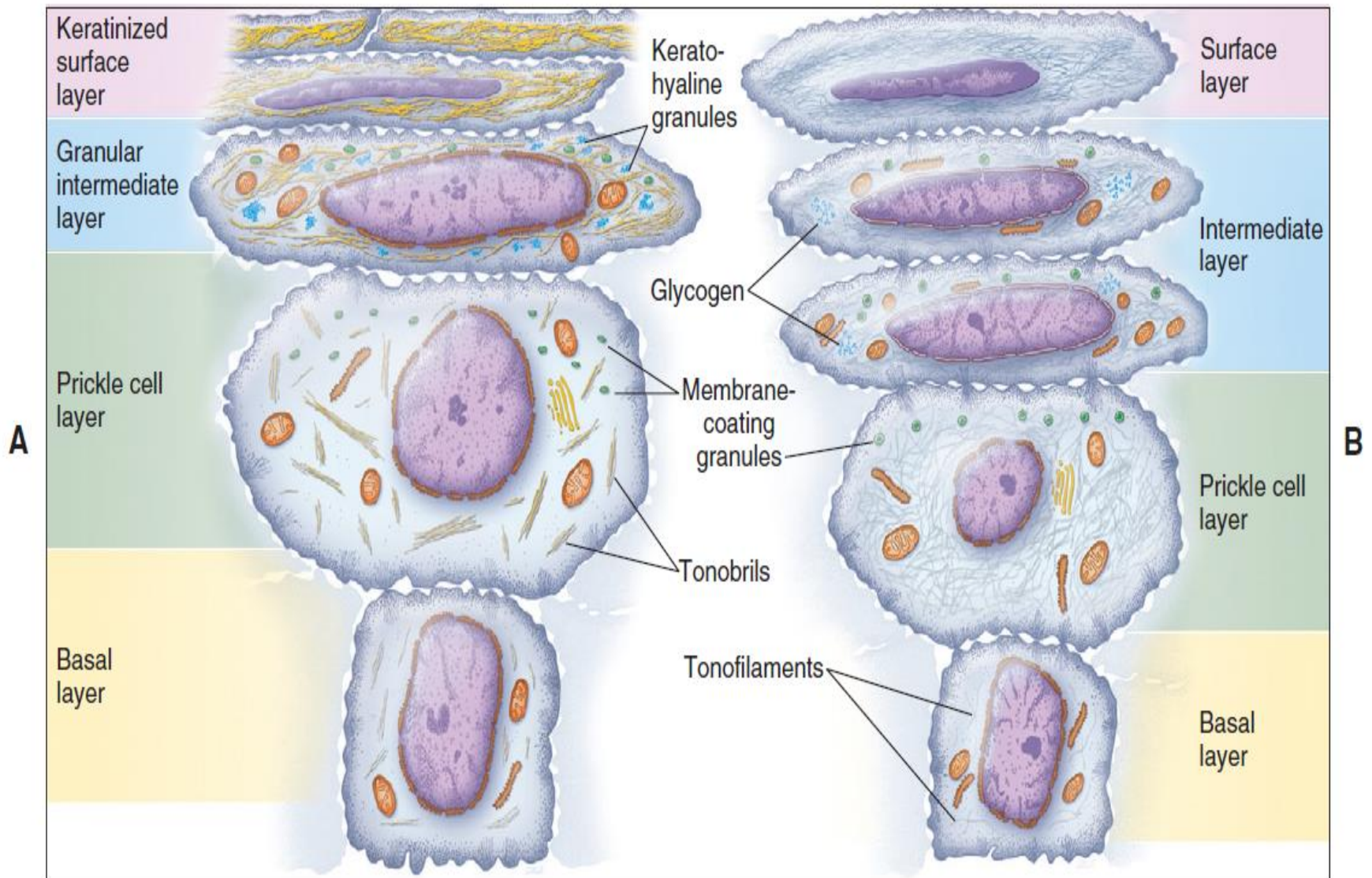
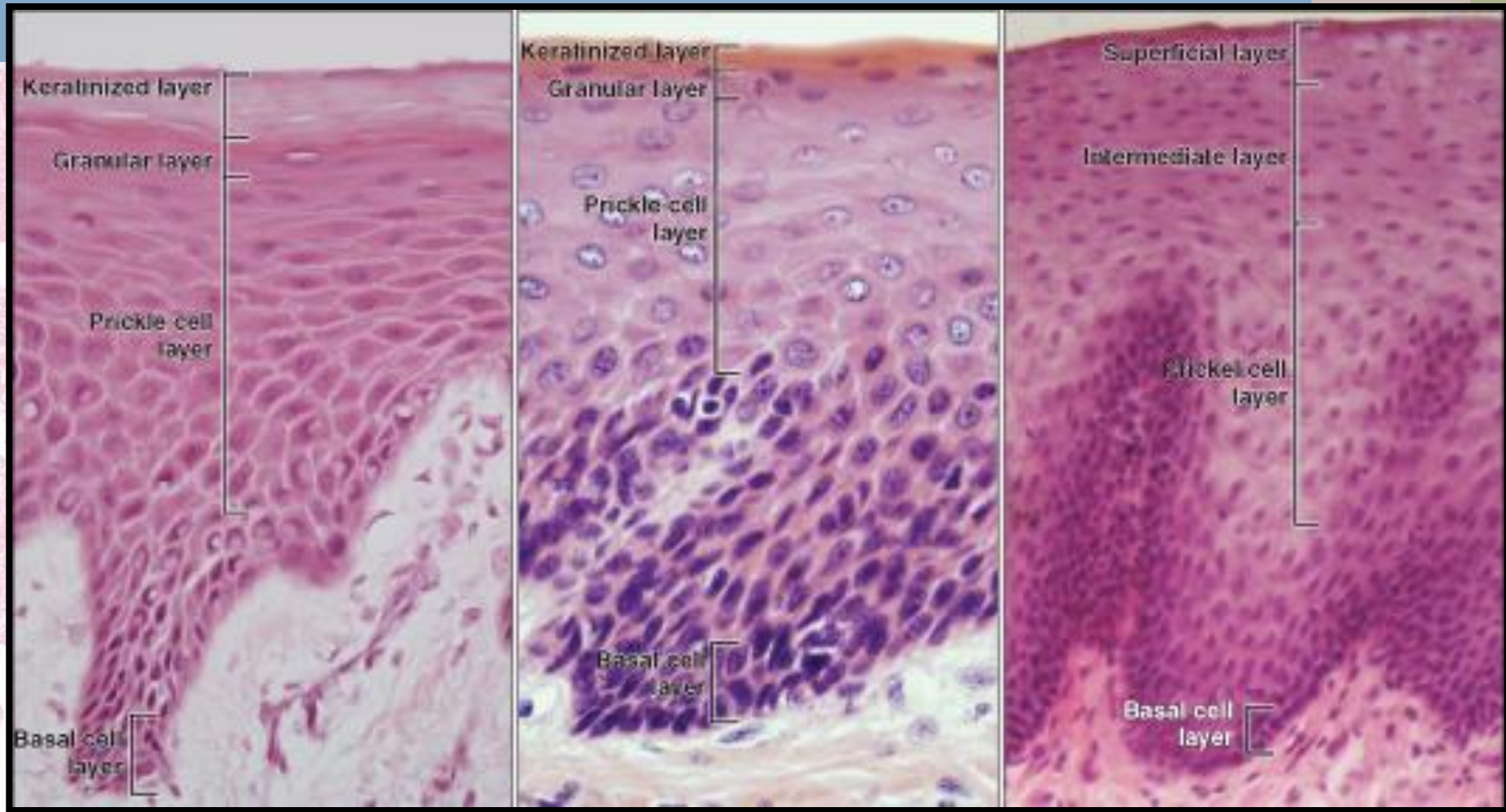


FIGURE 12-10 Principal structural features of epithelial cells in successive layers. **A**, Orthokeratinized oral epithelium. **B**, Nonkeratinized oral epithelium. (Adapted from Squier CA, Johnson NW, Hopps RM: *Human oral mucosa: development, structure, and function*, Oxford, UK, 1976, Blackwell Scientific. Note that cells are not drawn to scale.)

TABLE 12-1**Major Features of Maturation in Keratinized and Nonkeratinized Epithelium**

Keratinized Epithelium		Nonkeratinized Epithelium	
FEATURES	CELL LAYER	FEATURES	CELL LAYER
Cuboidal or columnar cells containing bundles of tonofibrils and other cell organelles; site of most cell divisions	Basal	Cuboidal or columnar cells containing separate tonofilaments and other cell organelles; site of most cell divisions	Basal
Larger ovoid cells containing conspicuous tonofibril bundles; membrane-coating granules appear in upper part of this layer	Prickle/ Spinosum	Larger ovoid cells containing dispersed tonofilaments; membrane-coating granules appear in upper part of layer; filaments become numerous	Prickle/ Spinosum
Flattened cells containing conspicuous keratohyaline granules associated with tonofibrils; membrane-coating granules fuse with cell membrane in upper part; internal membrane thickening also occurs	Granular	Slightly flattened cells containing many dispersed tonofilaments and glycogen	Intermediate
Extremely flattened and dehydrated cells in which all organelles have been lost; cells filled only with packed fibrillar material; when pyknotic nuclei are retained, parakeratinization occurs	Keratinized	Slightly flattened cells with dispersed filaments and glycogen; fewer organelles are present, but nuclei persist	Superficial

TYPES OF EPITHELIUM

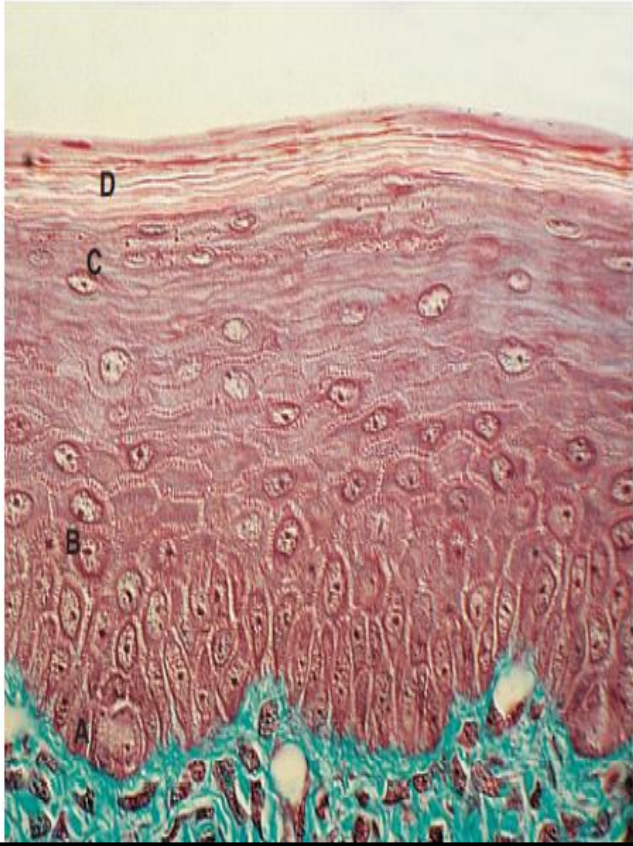


Orthokeratinized stratified squamous epithelium

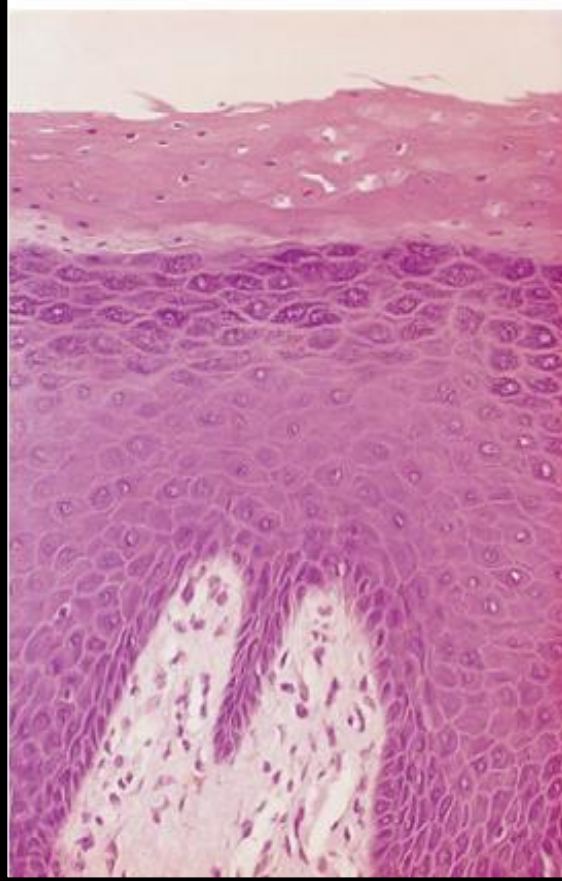
Parakeratinized stratified squamous epithelium

Non-keratinized (Lining) stratified squamous epithelium

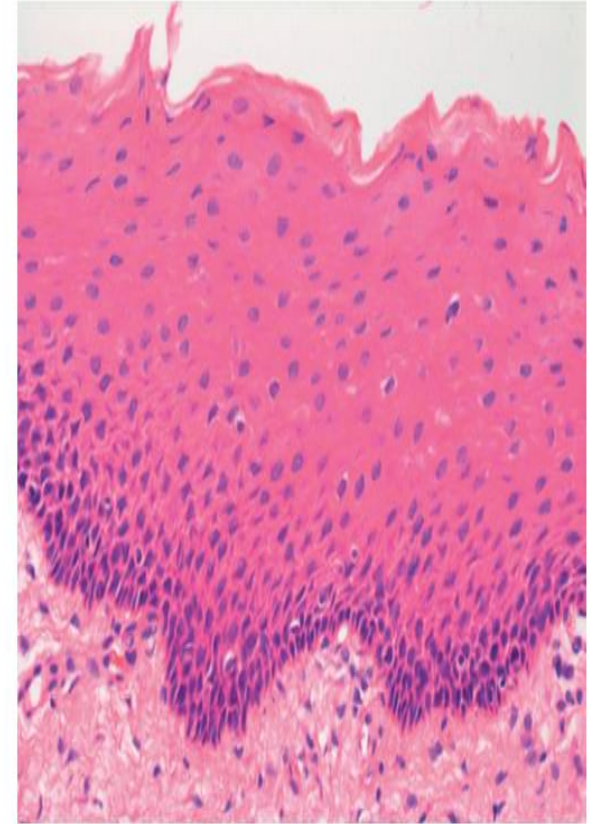
TYPES OF EPITHELIUM



Orthokeratinized stratified squamous epithelium



Parakeratinized stratified squamous epithelium



Non-keratinized (Lining) stratified squamous epithelium

Cell kinetics

**Derived from:
keratinocyte or
adjacent
fibroblasts**

**EPIDERMAL
GROWTH
FACTOR**

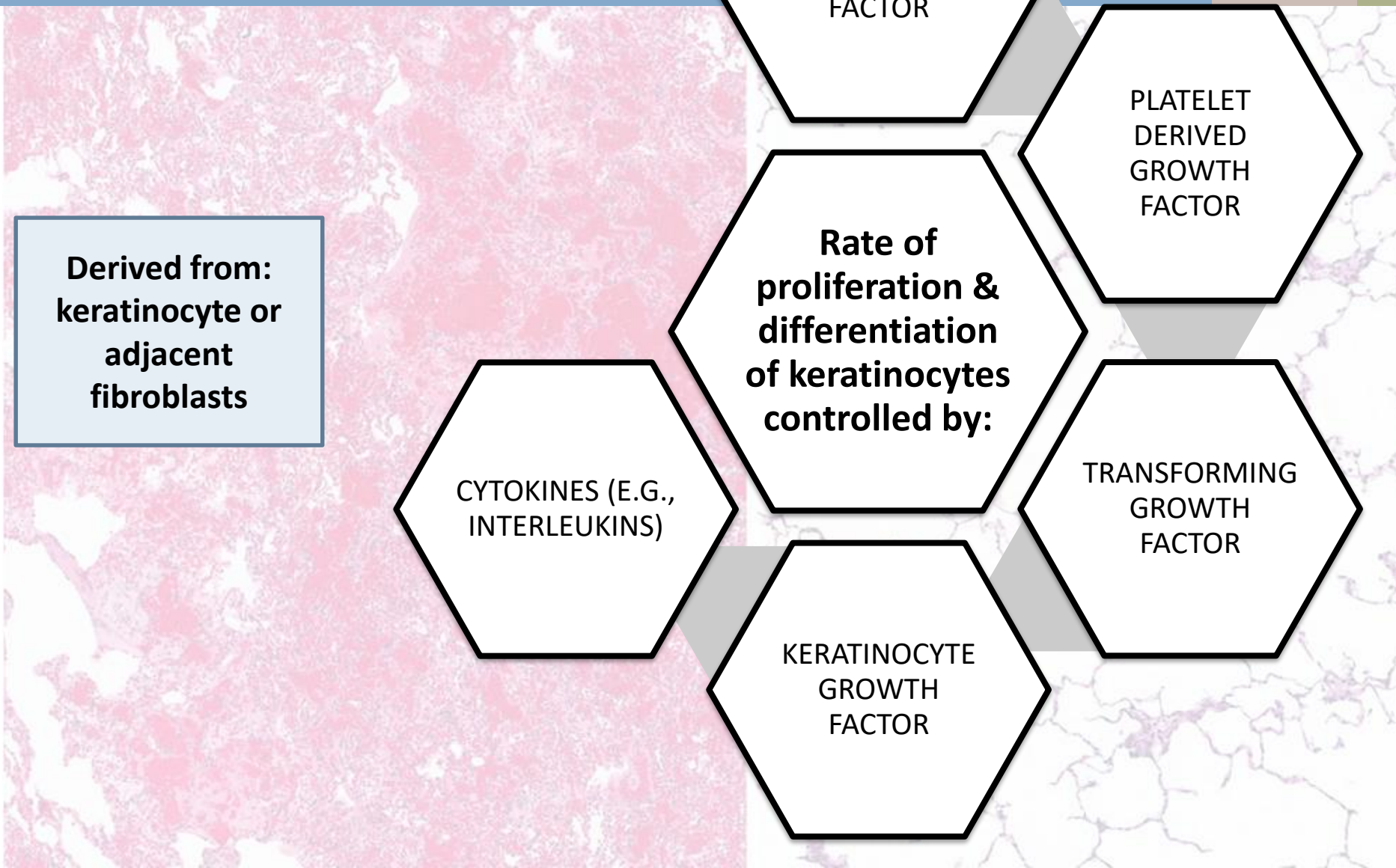
**PLATELET
DERIVED
GROWTH
FACTOR**

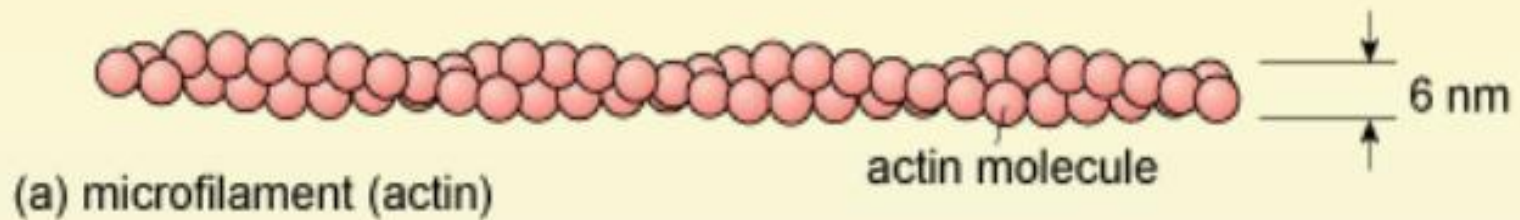
**Rate of
proliferation &
differentiation
of keratinocytes
controlled by:**

**CYTOKINES (E.G.,
INTERLEUKINS)**

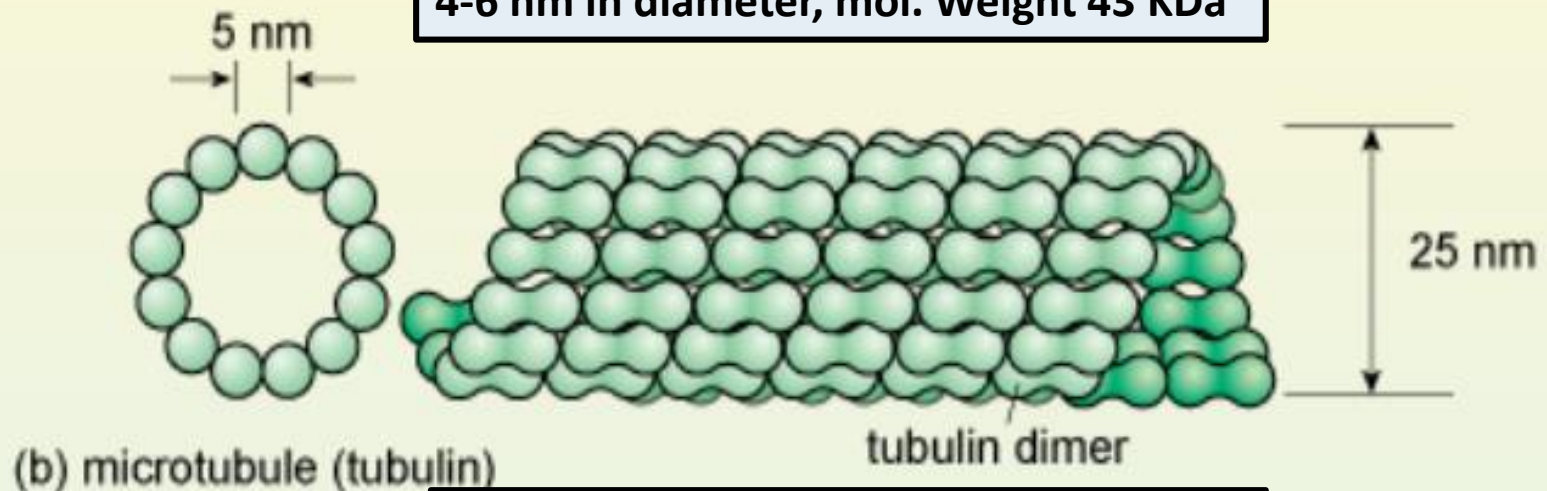
**TRANSFORMING
GROWTH
FACTOR**

**KERATINOCYTE
GROWTH
FACTOR**





4-6 nm in diameter, mol. Weight 43 KDa

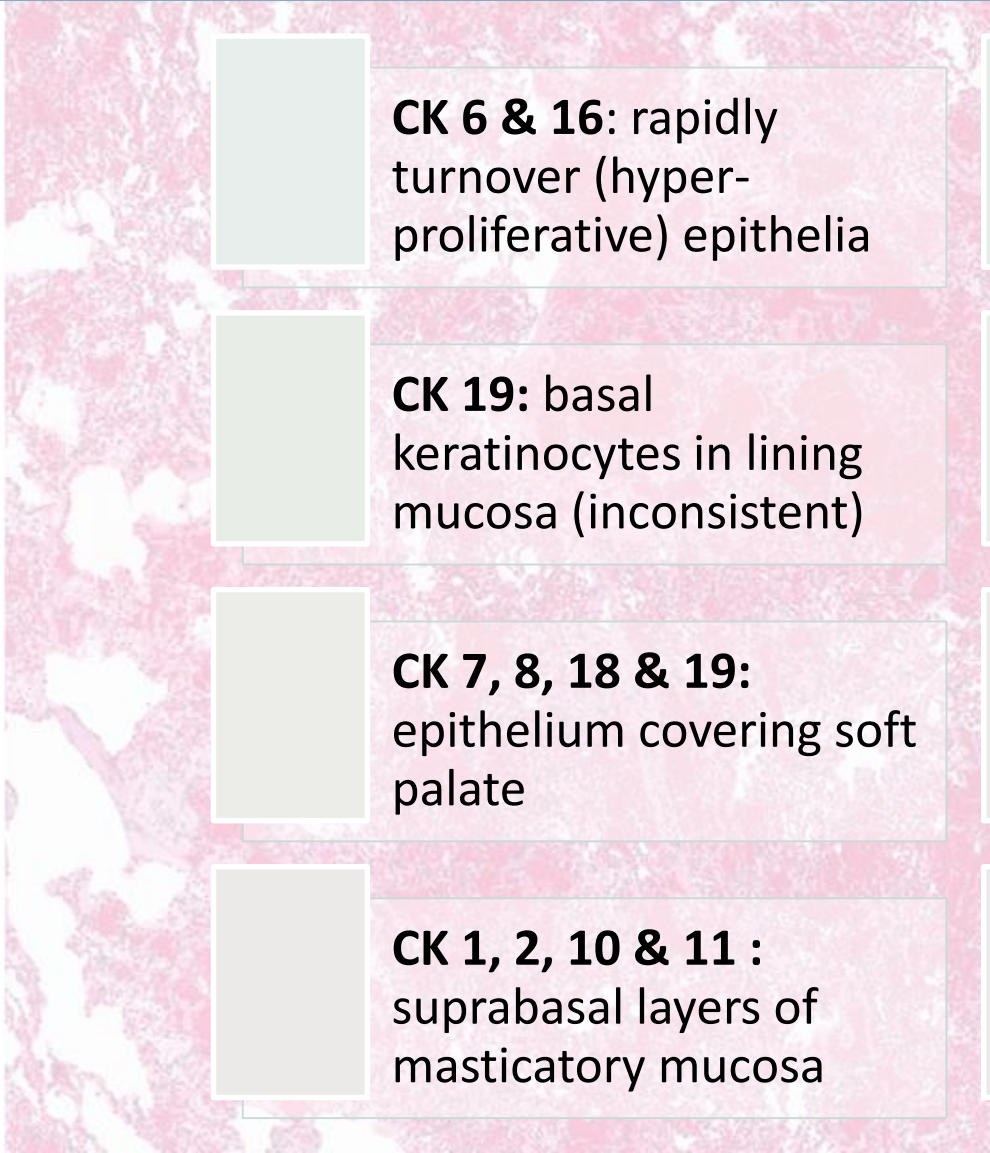


25 nm in diameter, mol. Weight 55 KDa



7-11 nm in diameter, mol. Weight 40-200 KDa

CYTOKERATINS & ITS CHARACTERISTICS

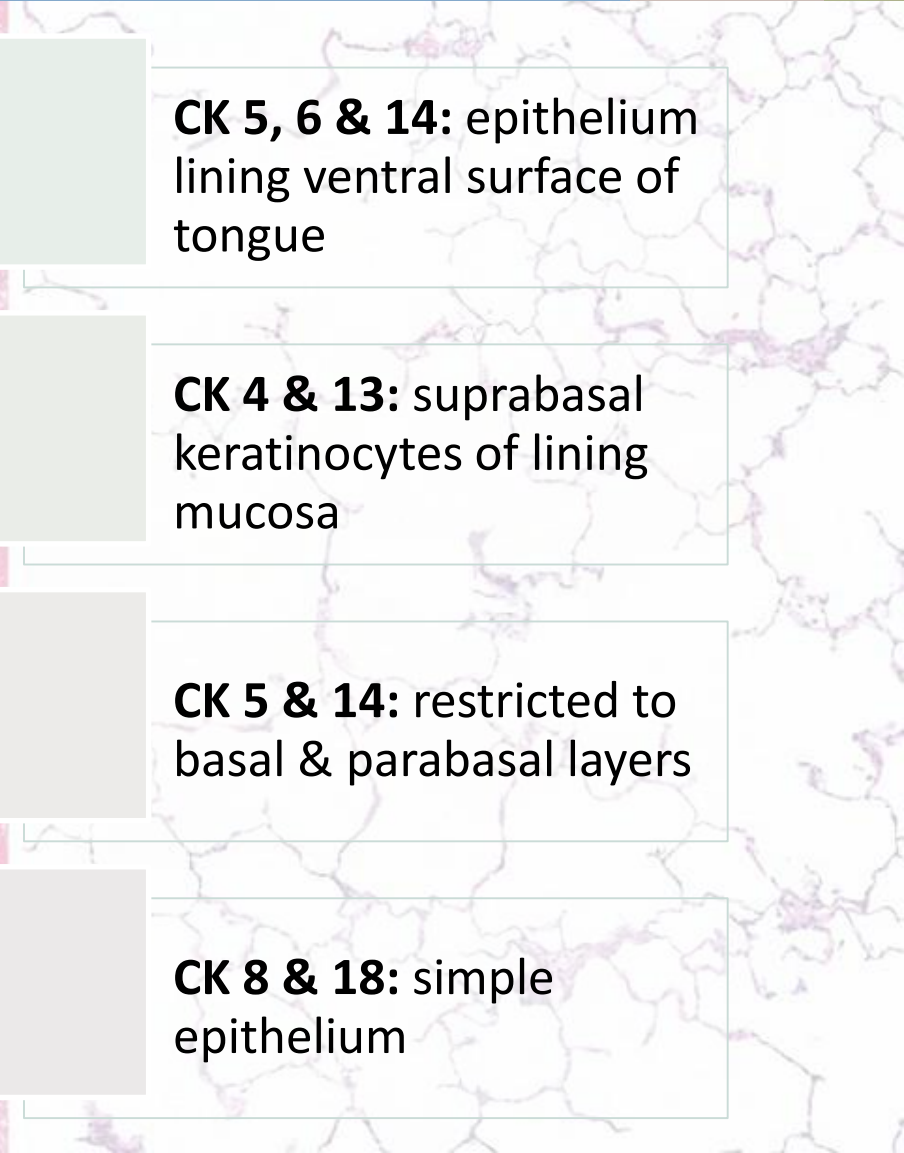


CK 6 & 16: rapidly turnover (hyper-proliferative) epithelia

CK 19: basal keratinocytes in lining mucosa (inconsistent)

CK 7, 8, 18 & 19: epithelium covering soft palate

CK 1, 2, 10 & 11 : suprabasal layers of masticatory mucosa



CK 5, 6 & 14: epithelium lining ventral surface of tongue

CK 4 & 13: suprabasal keratinocytes of lining mucosa

CK 5 & 14: restricted to basal & parabasal layers

CK 8 & 18: simple epithelium

Table 14.1 Distribution of cytokeratins in human stratified squamous epithelium (SSE)

	CK pair	
	Type II, basic, higher MW.	Type I, acidic, lower MW.
Suprabasal, keratinised SSE	1 and 2	10 and 11
Cornea	3	12
Suprabasal, nonkeratinised SSE	4	13
Basal keratinocytes (all SSE)	5	14
Hyperproliferative epithelia	6	16
Simple epithelia	8	18
Some simple epithelia, pilosebaceous tracts, basal cell carcinomas	5?	17
Basal keratinocytes in nonkeratinised SSE, some simple or 'plastic' epithelia (e.g. odontogenic)	5?	19

NONKERATINOCYTES

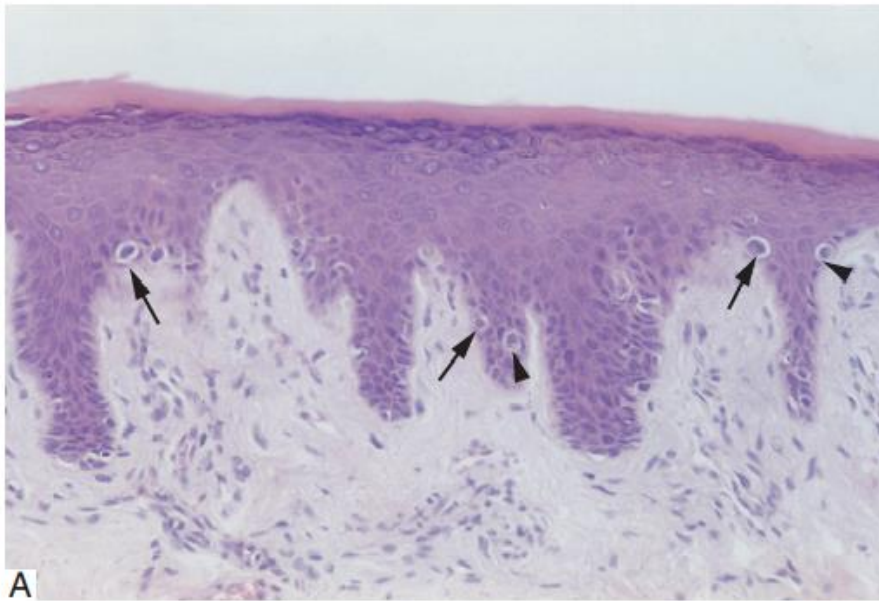
10% of cells in oral epithelium → non-keratinocytes which lack tonofilaments & desmosomes characteristic of keratinocytes (melanocytes & langerhan cells)

Cell names	Location	Function	Origin
Melanocytes	pigment-producing cells located in the basal layer	produce pigment melanin, using enzyme tyrosinase	Neural crest cells (present around 8 weeks IU) in skin
Langerhans cells	dendritic cells situated in the layers above the basal layer	• Part of immune system as → Antigen-presenting cell • Skin contact hypersensitivity reactions, • in antitumour immunity and in graft rejection; • propagators of human immunodeficiency virus (HIV)-1 transmission to T cells.	Bone marrow precursors (monocyte lineage) → from blood to lamina propria → penetrate BL to reach epithelium
Merkel cells	basal layer often closely apposed to nerve fibres.	Act as a receptor.	Neural Crest

NONKERATINOCYTES

- Other nonkeratinocytes : **inflammatory cells** that (migrated through epithelium).
- **Most common type of inflammatory cell → Lymphocytes.**
- **Others:** polymorphonuclear & leukocytes.
- Retention of lymphocytes within epithelial layer → **integrins binding (↑ in disease).**

1 Reason for masticatory epithelium permeability > nonkeratinised → ↑ # of inflammatory cells

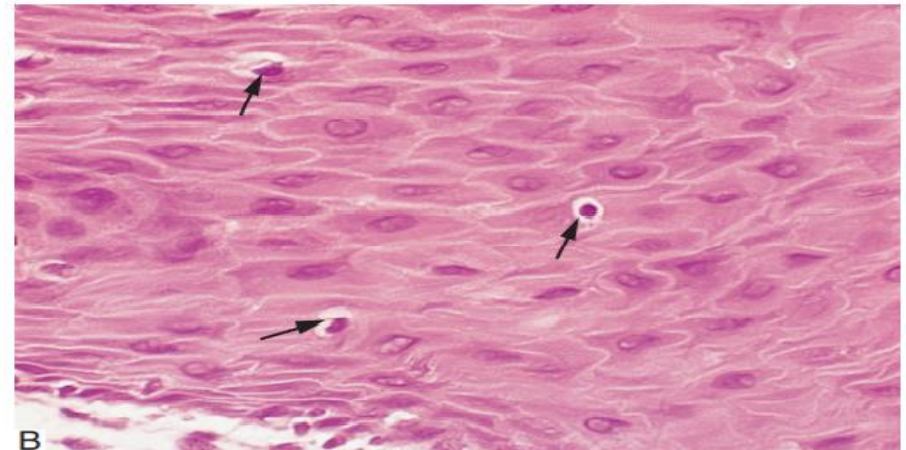


A

Melanin pigmentation not pronounced in → buccal mucosa, tongue, hard palate or lip

Fig. 14.19 (A) Section of the vermillion of lip showing clear cells (arrows) in the basal layer, representing mainly melanocytes ($\times 80$). (B) Clear cells in the upper layers of the epithelium (arrowed), mainly representing Langerhans cells (H & E; $\times 350$) Courtesy Prof. M M Smith.

Melanosomes → pigment packaged in small granules



B

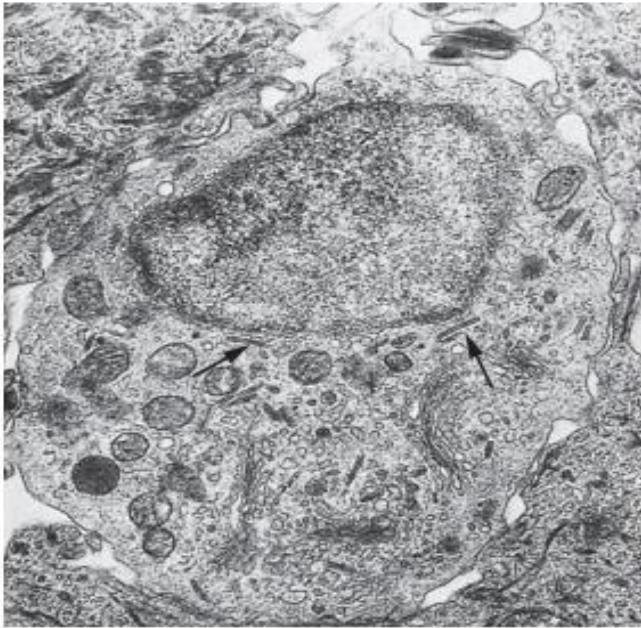


Fig. 14.24 Electron micrograph of Langerhans cell showing the characteristic Birbeck granules (arrows) ($\times 15,000$). Courtesy Dr N G El-Labban.

Birbeck Granules:

Ultrastructurally, the Langerhans cell contains characteristic, trilaminar, rod-shaped granules called Birbeck granules. These may be up to 50 nm long and 4 nm wide with a vesicular swelling at one end, resembling a tennis racquet.

- Most quick response (as APCs $\rightarrow$ bacterial structures) $\rightarrow$ through "toll" like receptors
- Dendritic cells express $\rightarrow$ class II molecules of major histocompatibility complex & Fc receptors $\rightarrow$ present antigenic material to T lymphocytes

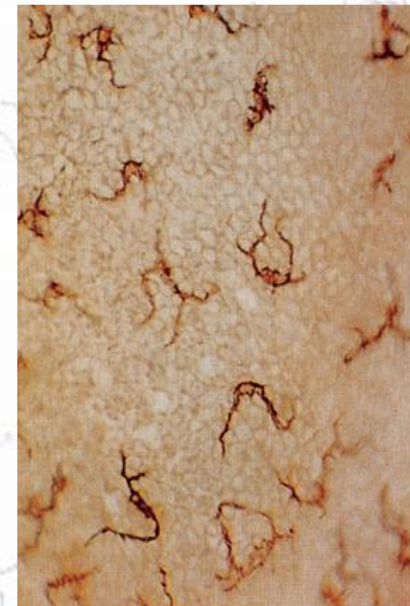


Fig. 14.23 Section of superficial layers of oral epithelium showing Langerhans cells (arrowed)

Merkel cells are common in masticatory epithelia (gingiva) but less frequent in lining mucosa such as the buccal mucosa.

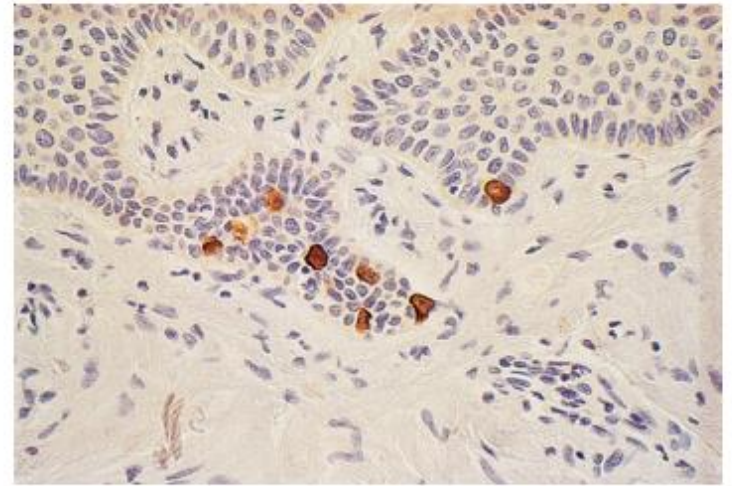


Fig. 14.25 Section showing Merkel cells in basal layer of epithelium staining positively (brown) for cytokeratin18 (streptavidin-biotinylated immunoperoxidase stain; $\times 200$). Courtesy Dr W A

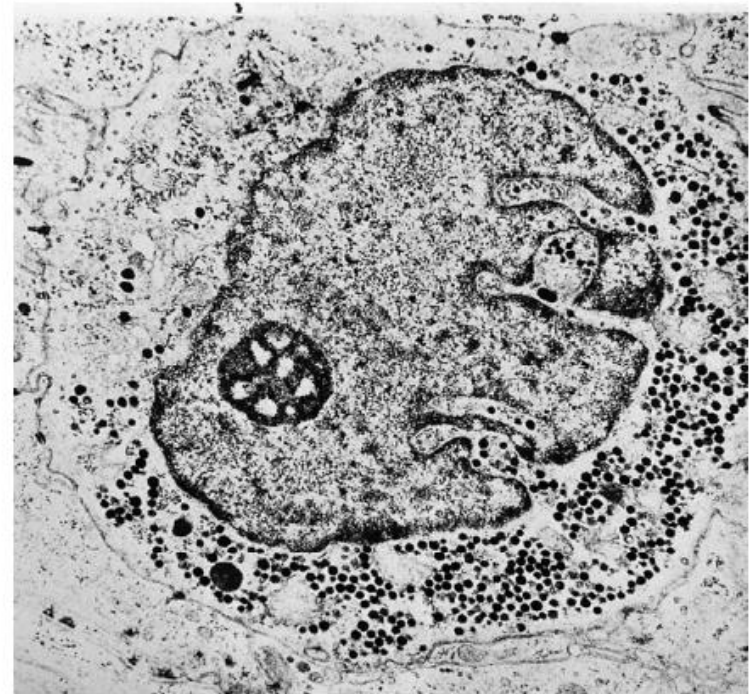
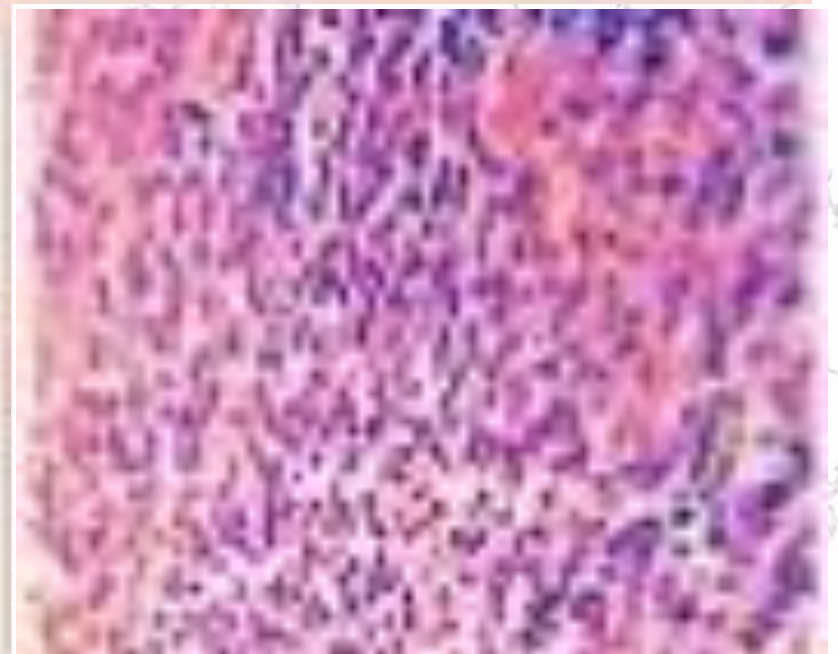
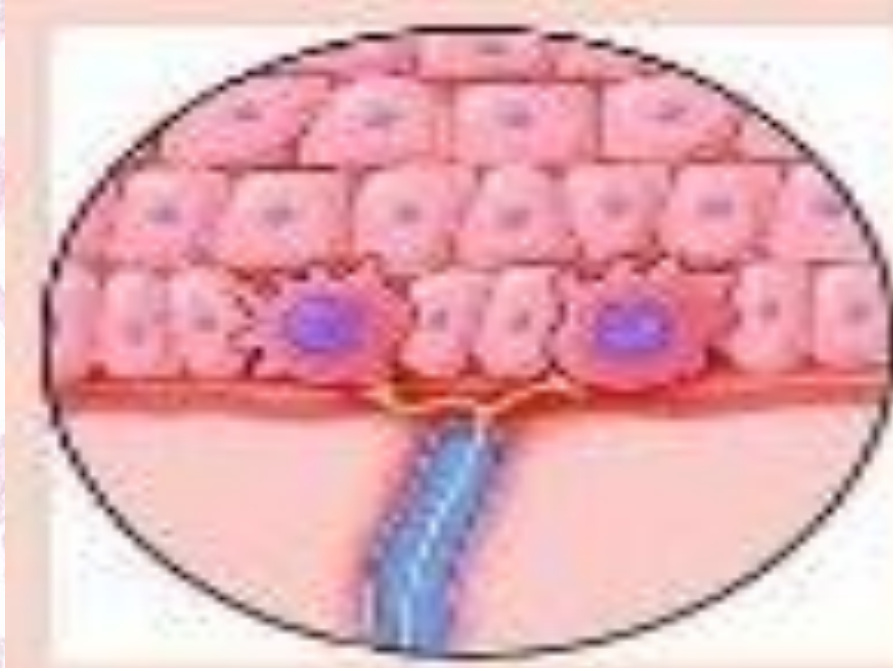
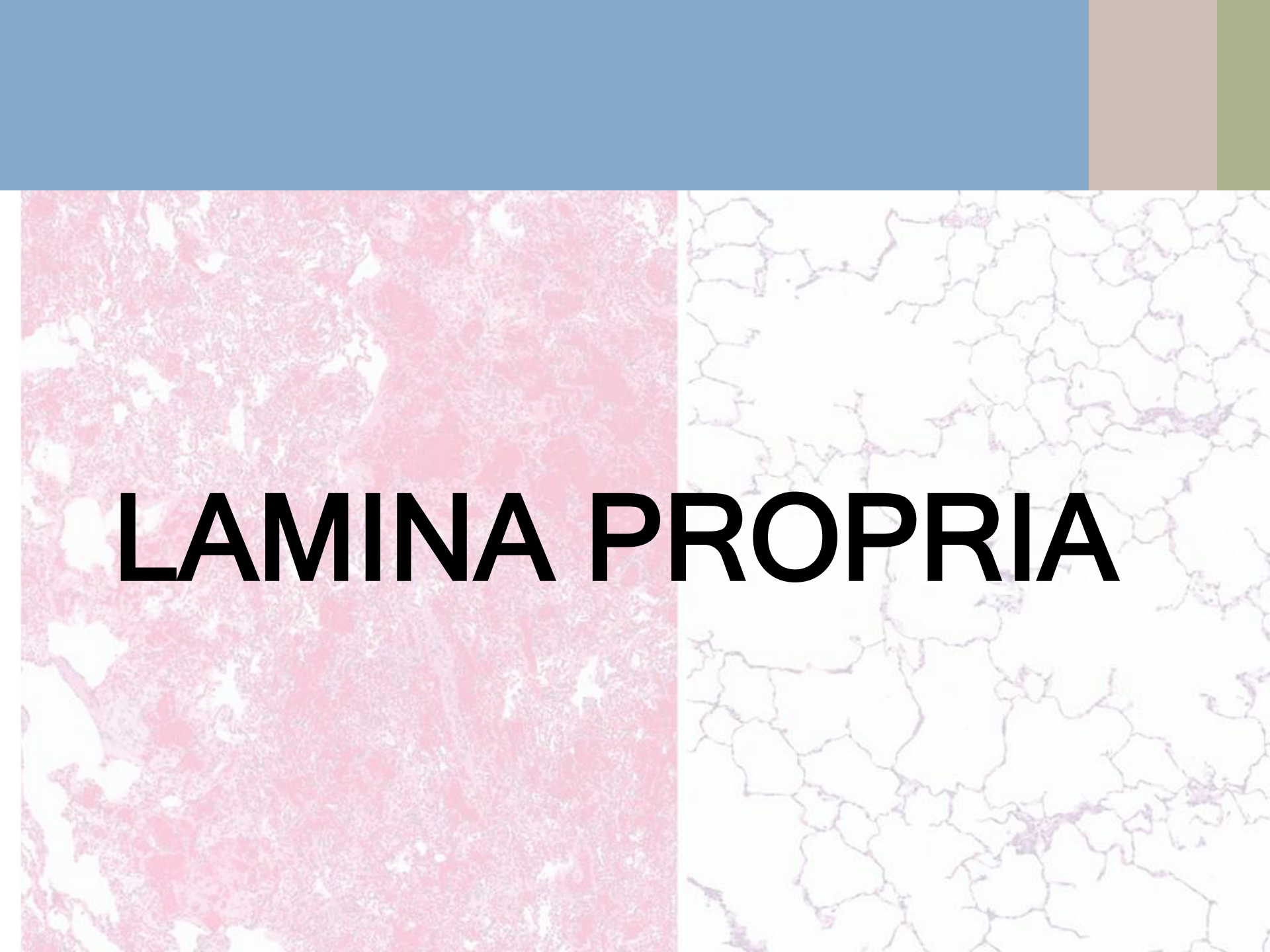


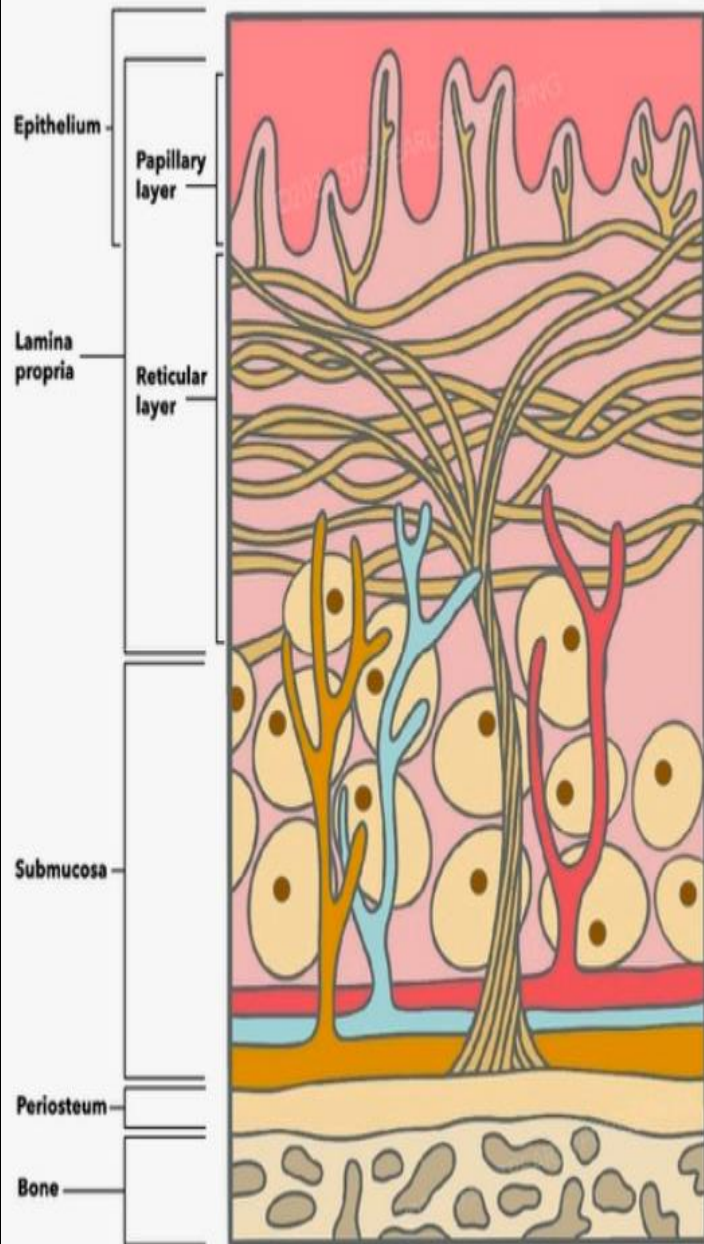
Fig. 14.26 Electron micrograph of Merkel cell. Note the indented nucleus and nuclear rodlet. The cytoplasm contains numerous characteristic dense granules 80–180 nm in diameter ($\times 16,000$). Courtesy Dr S-Y Chen.





LAMINA PROPRIA

Oral Mucosa



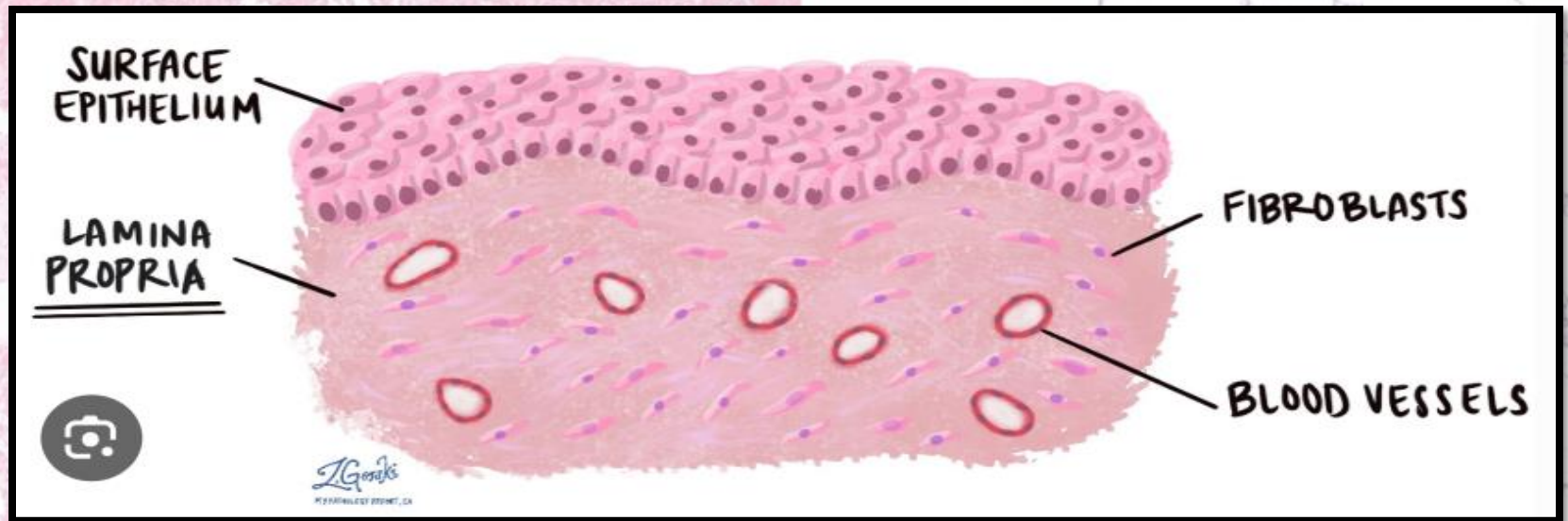
- **Superficial papillary layer** → b/w epithelial ridges (collagen fibres are thin & loosely arranged)
- **Deep reticular layer** → dominated by thick, parallel bundles of collagen fibres.

FUNCTIONS:

- mechanical support
- nutrition
- sensory → nerves
- defensive → blood vessels & salivary glands

PRINCIPLE CELLS: FIBROBLASTS

- Shape : spindle (variable)
- Synthetic organelles → consistent with → role in continuous production & secretion of extracellular fibres & ground substance for lamina propria



STEM CELLS

- A population of stem cells exists in the lamina propria, characterised by → expression of embryonic & pluripotency markers → Sox2 & Notch signalling pathways.
- They exhibit high proliferation rates *in vitro*.
- They have the capacity to differentiate into several lineages, such as fat, bone & cartilage cells.
- They also demonstrate → immunoprivilege feature → may be important in tissue engineering.

EXTRACELLULAR MATRIX

collagen fibres:

- **Type I (90%)
Type III (8%)**
- **Types IV and VII → (associated with the presence of basement membranes)**
- **types V and VI → (small amounts)**

Elastin Fibres

- **# dependent on site**

Oxytalan Fibres

- **Also described**

**Ground substance of lamina propria →
hydrated gel of proteoglycans and glycoproteins.**

Macrophages

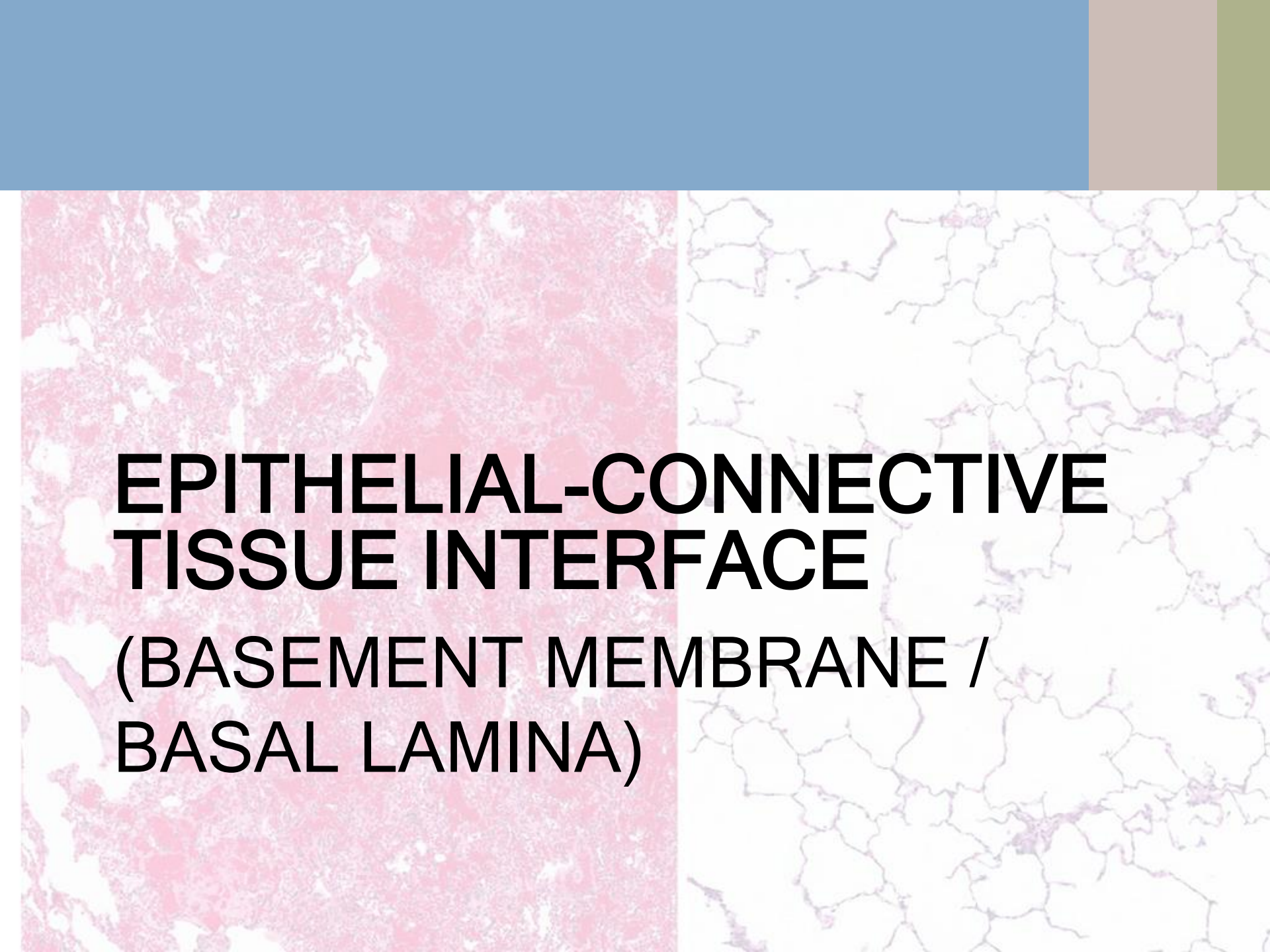
- Known as → histiocytes (in fixed inactive stages) indistinguishable from fibroblasts.
- smaller, darker nucleus than fibroblasts
- contain lysosomes but little ER.
- Function: phagocytic role + antigen-presenting cells.

Mast Cells

- mononuclear, spherical or elliptical in shape
- contain histamine & heparin intracellular granules
- Role in vascular homeostasis
- Inflammation
- in cell-mediated immunity
- are responsible for anaphylactic (type 1) hypersensitivity.

Lymphocytes

- small numbers
- in healthy mucosa, but increase dramatically in inflammation.

The background of the slide is composed of three distinct visual elements. At the top, there is a horizontal bar with a blue-to-brown gradient. Below this, the slide is split into two vertical panels. The left panel shows a pink-stained histological section, likely of an epithelial layer, with a dense cellular structure. The right panel shows a light-colored histological section, likely of adipose tissue, characterized by large, clear, polygonal cells. The text is overlaid on the left panel.

EPITHELIAL-CONNECTIVE TISSUE INTERFACE

**(BASEMENT MEMBRANE /
BASAL LAMINA)**

BASEMENT MEMBRANE

complex arrangement
linking surface epithelium
to lamina propria

In light microscope, a layer
1–2 μm thick is seen on the
lamina propria side of the
junction.

Appears thicker under LM
b/c of inclusion of some
subepithelial collagen fibres
with similar staining
properties as basal lamina

Ultrastructurally, possesses
specialized attachment
plaques, the
hemidesmosomes, (in
length b/c of increased
density)

Cytoplasmic keratin
filaments insert into
hemidesmosome.

BASAL LAMINA

Under the electron microscope, the layer appears much thinner and is termed the basal lamina.

All the major products of the basal lamina appear to be synthesised by the epithelial cells.

located immediately adjacent to the cell membrane.

In routine electron microscopy, the basal lamina is seen to consist of 2 layers:

- **lamina lucida:** Electron-lucent (20–40 nm thick) lies immediately under epithelium (preparation artefact)
- **lamina densa:** Thicker (20–120 nm).

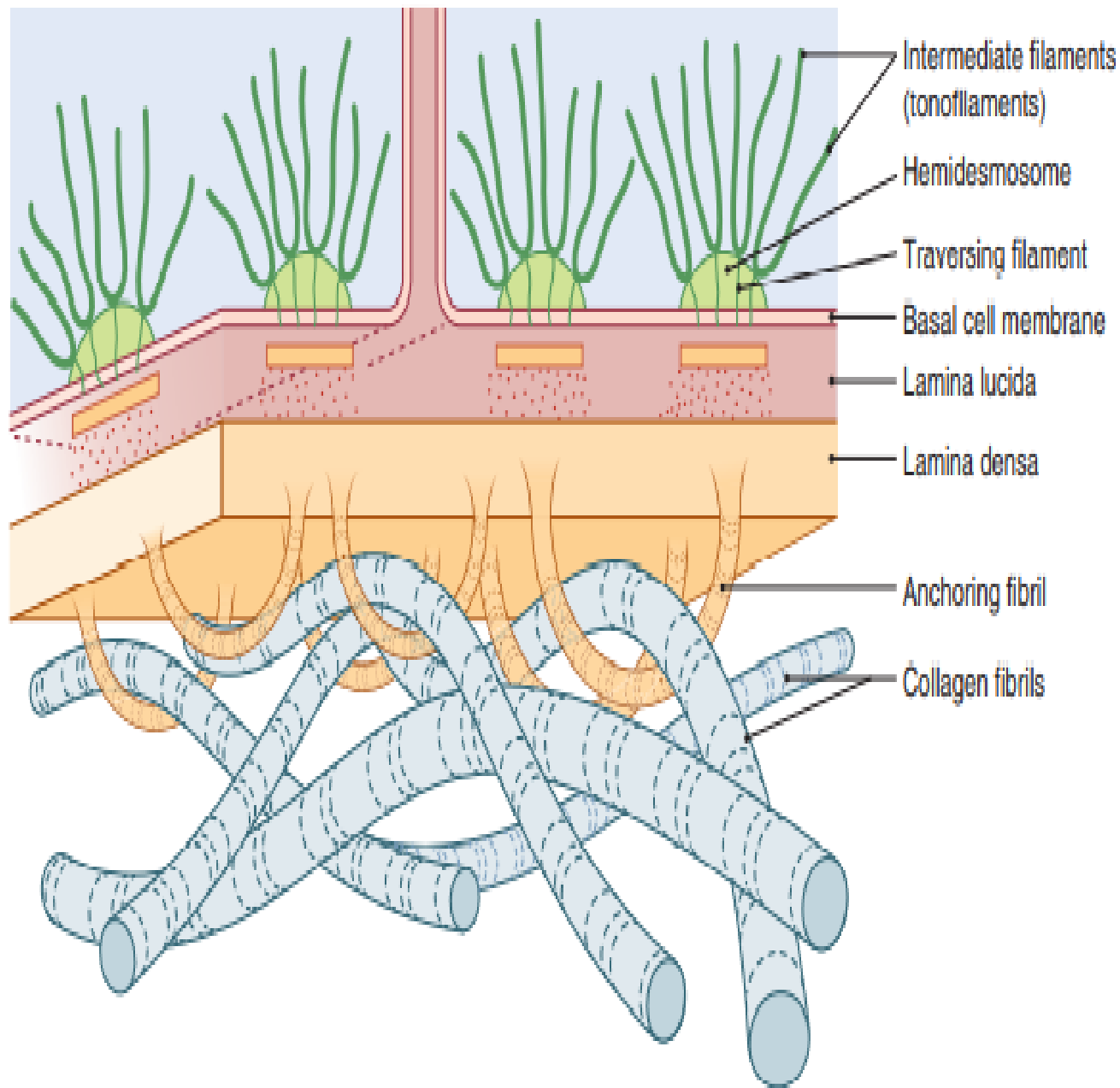


Fig. 14.29 Electron micrograph showing appearance of basal lamina ($\times 24,000$). Courtesy Prof. C A Squier.

Composition

Collagen:

type IV
(nonfibrillar)
collagen
(Network)

Proteoglycan & Glycoprotein molecules:

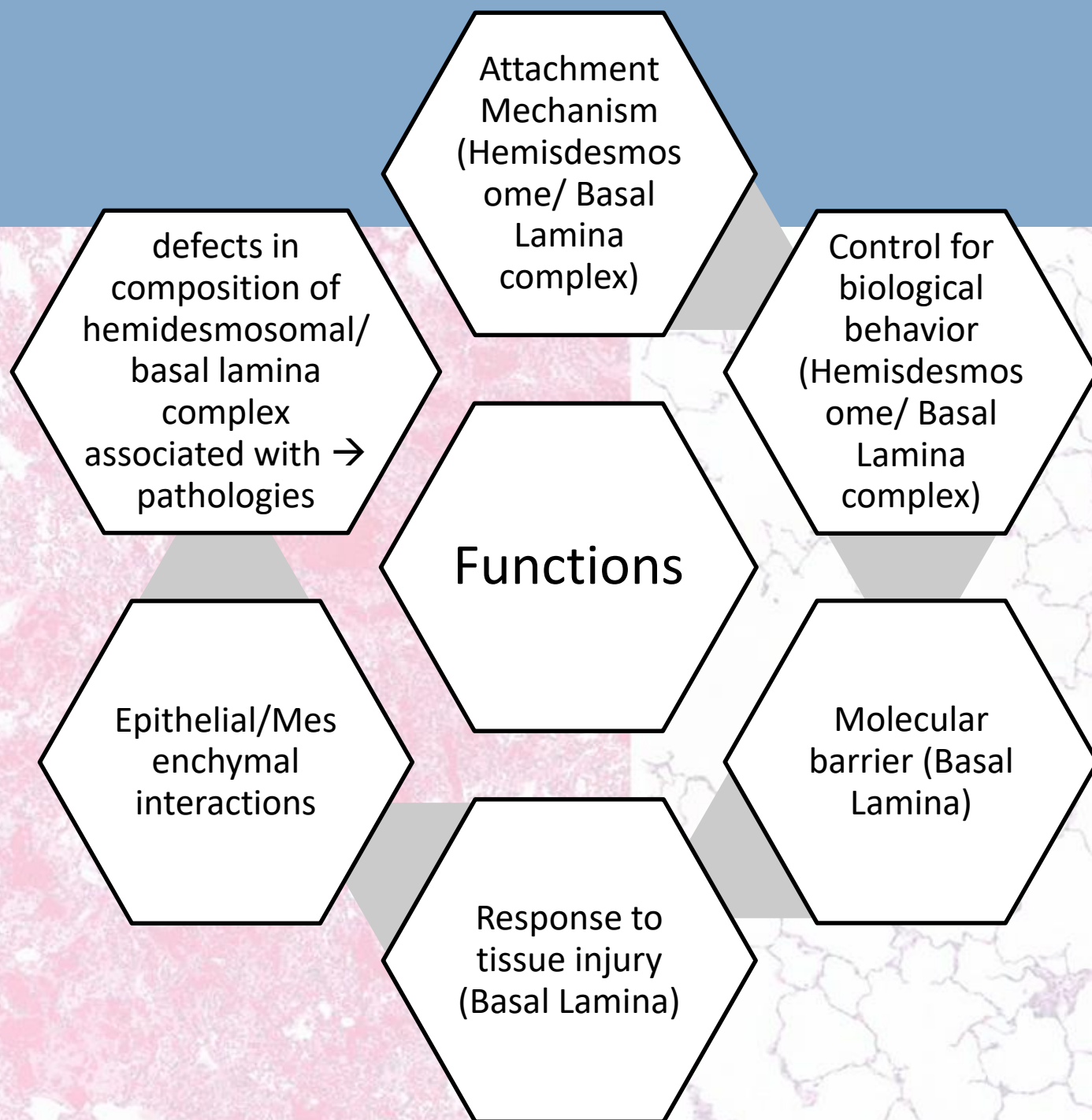
fibronectin,
laminin &
perlecan
(heparin
sulphate
proteoglycan).

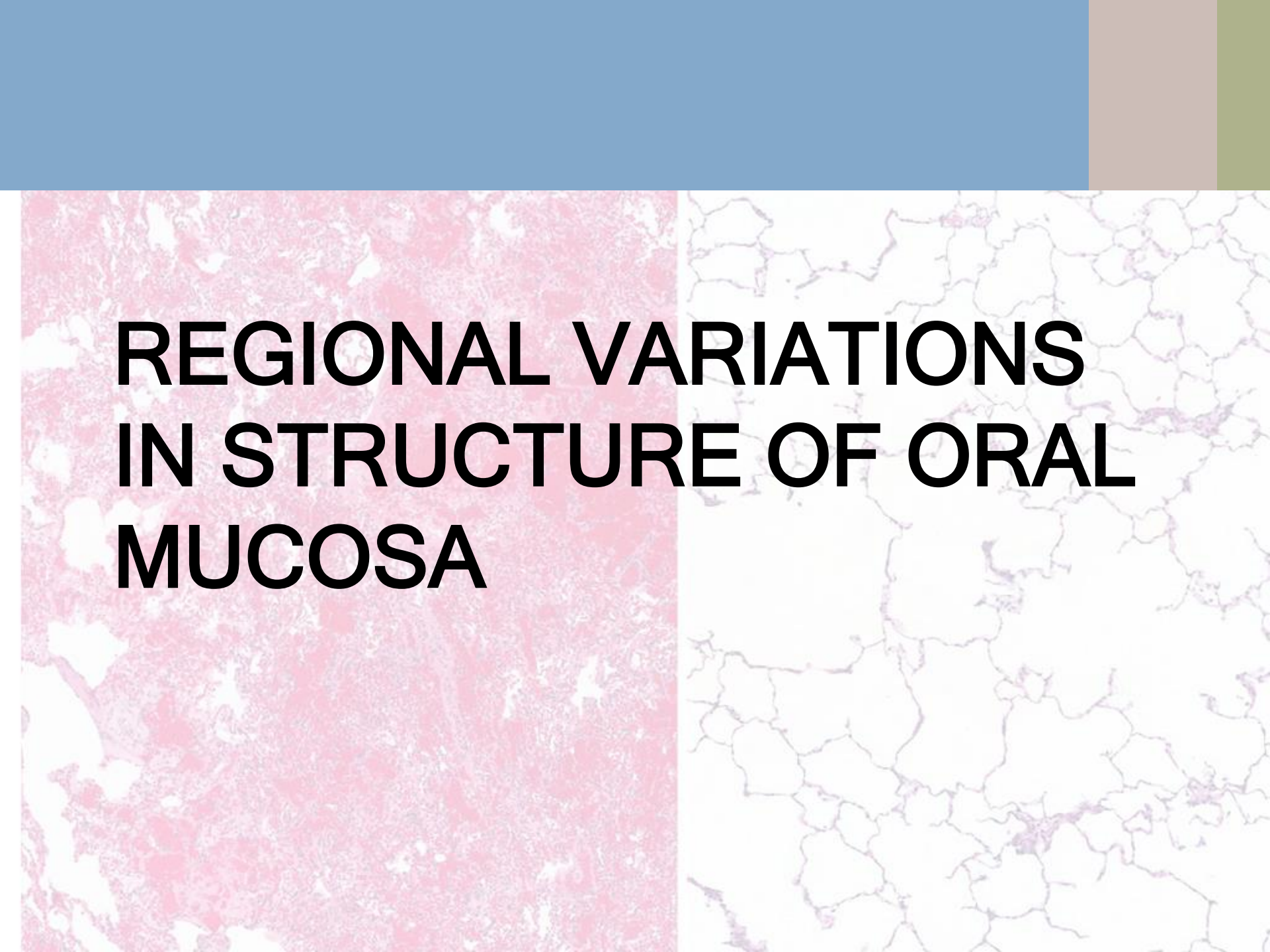
Transmembrane molecules:

integrins &
bullous
pemphigoid
antigens (BPAG-1
and BPA-2).
(strengthen link
b/w cell & basal
lamina)

Other Collagen types:

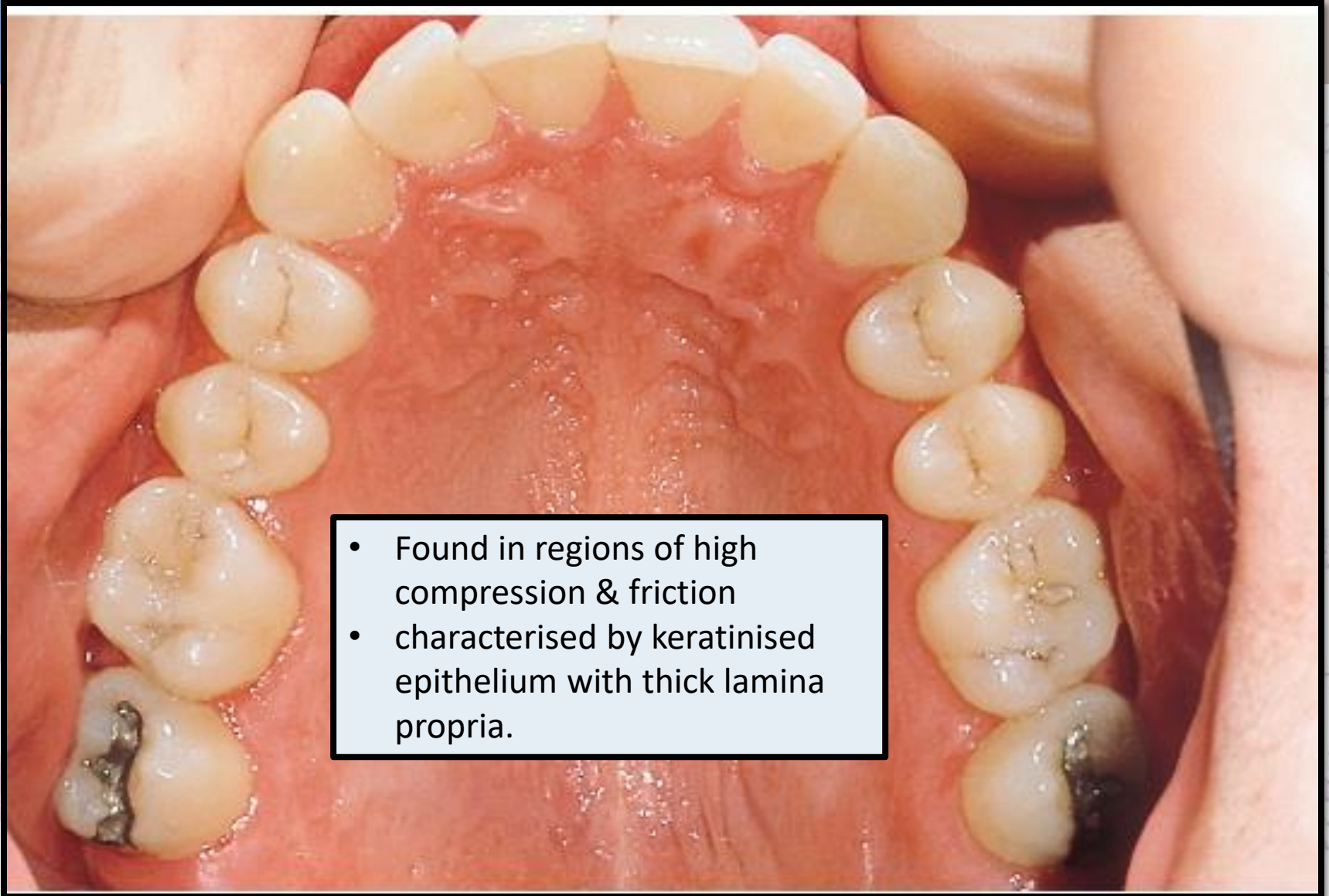
Looping fibrils of
type VII collagen
binding to type
IV collagen of
basal lamina &
interdigitating
with type I & type
III collagen of
ECM.
(strengthening
bonds)



The background of the slide features two microscopic images of oral mucosa. The left image shows a dense, pink-stained tissue with a granular texture. The right image shows a lighter, more cellular tissue with a network of purple-stained fibers. The title text is overlaid on the left image.

REGIONAL VARIATIONS IN STRUCTURE OF ORAL MUCOSA

Masticatory Mucosa (25% of oral mucosa)



- Found in regions of high compression & friction
- characterised by keratinised epithelium with thick lamina propria.

Eg: gingiva & palate except side of palate

Lining Mucosa 60% of oral mucosa



- mobile, distensible & not subject to ↑ levels of friction
- Nonkeratinised (loose lamina propria)
- Collagen fibres loosely arranged as network (allow free movement)
- Elastic fibres (allow recoil to prevent the mucosa being chewed)
- has a submucosa.

The lips, cheeks, buccal and labial sulci, alveolar mucosa, floor of the mouth, ventral surface of tongue & soft palate

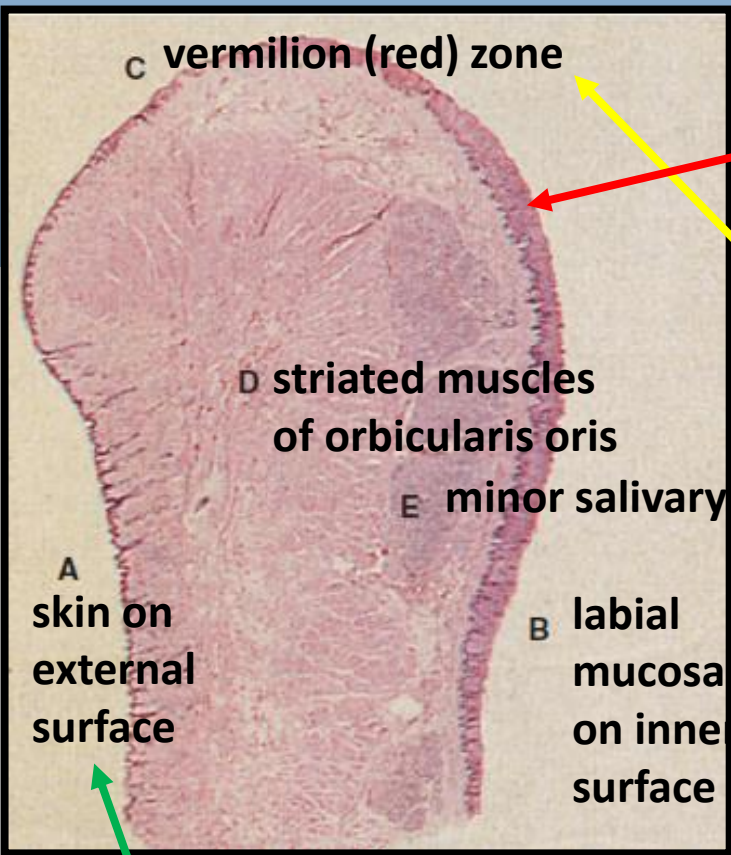
Specialized Mucosa 15% of oral mucosa

- specialised gustatory mucosa of the dorsum of the tongue
- Where vermillion zone forms a transition b/w skin & oral mucosa.

Table 14.2 Principal features and regional variations of the oral mucosa

Region	Epithelium		Lamina propria		Submucosa		Type of mucosa
	Thickness	Keratinisation	Papillae	Fibre types	Density	Attachment	
Labial and buccal mucosa	Thick	Nonkeratinised	Short and irregular	Collagen and some elastin fibres	Dense	Firmly to underlying muscle	Lining
Vermilion (red) zone of lip	Thin	Keratinised	Long and narrow	Collagen and some elastin fibres	Dense	Firmly to underlying muscle	Specialised
Alveolar mucosa	Thin	Nonkeratinised	Short or absent	Many elastin fibres	Loose	Loose attachment to periosteum	Lining
Attached gingiva	Thick	Keratinised and parakeratinised	Long and narrow	Dense collagen firmly attached to underlying periosteum	No distinct submucosa		Masticatory
Floor of mouth	Thin	Nonkeratinised	Short and broad	Collagen and some elastin fibres	Loose	Loose attachment to underlying muscle	Lining
Ventral surface of tongue	Thin	Nonkeratinised	Short and numerous	Collagen and some elastin fibres	Not very distinct layer; attached to underlying muscle		Lining
Dorsum of tongue (anterior two-thirds)	Thick	Primarily keratinised	Long	Collagen and some elastin fibres	Not very distinct layer; attached to underlying muscle		Specialised gustatory
Dorsum of tongue (posterior one-third)	Variable	Generally nonkeratinised	Short or absent	Collagen and some elastin fibres	Not very distinct layer; attached to underlying muscle		Lining gustatory
Hard palate	Thick	Keratinised	Long	Dense collagen in submucosa laterally, but lamina propria firmly bound to periosteum without submucosa in midline			Masticatory
Soft palate	Thick	Nonkeratinised	Short	Many elastin fibres	Loose	Loose attachment to underlying tissues	Lining

LIP



Intermediate zone:

- junction b/w vermilion & labial mucosa
- lacks granular layer a thick parakeratinised layer
- **suckling pad → thickened in infants**

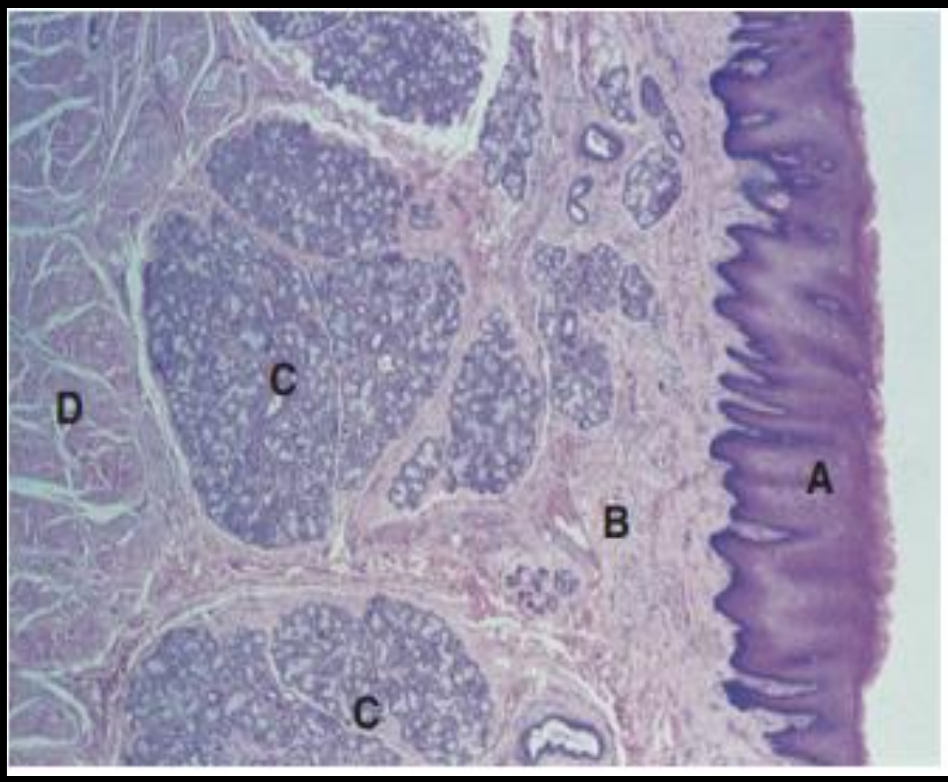
Vermillion zone (red/transitional zone):

- No skin appendages
- Sebaceous glands → present occasionally at mouth angles
- No mucous glands, needs salivary lubrication.
- Epithelium keratinised (thin & translucent)
- **Red colour (a uniquely human feature) → Long, narrow connective tissue papillae with capillary loops close to surface + epithelial translucency**

Skin (outer surface):

- keratinised epidermis on dermis
- sweat glands
- sebaceous glands
- (hair follicle → produces hair keratin).

Epithelial thickness increases progressively from skin to mucosal aspect.



Section of Labial mucosa

A = nonkeratinised oral epithelium;

B = lamina propria;

C = minor salivary gland in submucosa;

D = fibres of orbicularis oris

Labial mucosa (inner surface):

- Relatively thick, nonkeratinised epithelium; wide lamina propria,
- short/irregular papillae; submucosa with many minor mucous salivary glands
- dense connective tissue strands bind mucosa to orbicularis oris
- Striated muscle (in core)
- Minor mucous salivary glands (in substantial amounts → submucosa)
- Epithelial thickness ↑ (from skin to mucosal aspect)

CHEEK

Lining mucosa

nonkeratinised
epithelium; dense
lamina propria with
short irregular papillae.

present with many minor
mucous salivary glands
overlying buccinator muscle
fibres.

Main ducts of minor salivary
glands penetrate epithelium to
drain into vestibule.

Submucosa

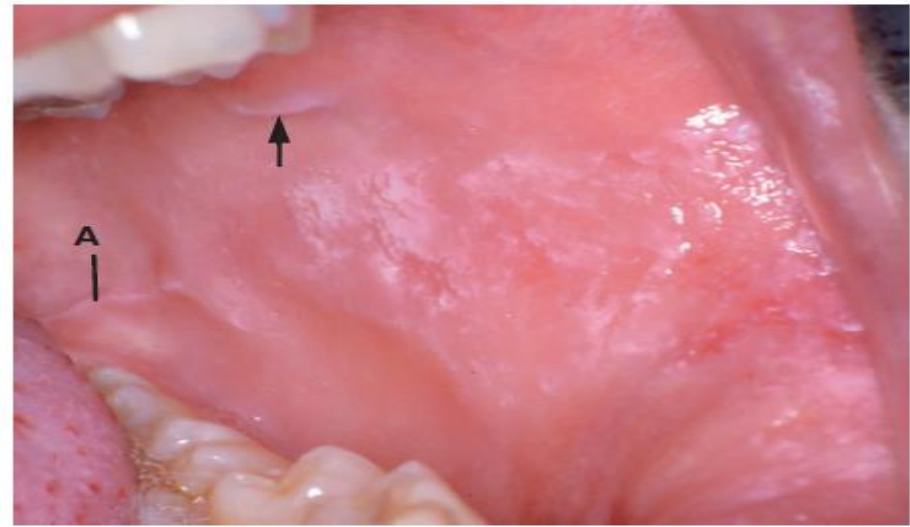


Fig. 1.24 View of buccal mucosa showing a linea alba adjacent to the molar teeth (A) at the level of the occlusal plane. In front of this line, the white patches on the cheek represent more diverse cheek chewing. Arrow shows the parotid papilla. Courtesy Prof. P R Morgan.

Linea alba →

- white line along occlusal plane where epithelium becomes keratinised (friction).
- result of low-grade, intermittent trauma due to folds of cheek mucosa being trapped b/w teeth

FORDYCE SPOTS

- Ectopic sebaceous glands without any associated hair follicles,
- occur singly or in clusters
- may also be seen on margin of lips
- ↑ with age.
- more obvious after puberty (male), menopause (female);
- Yellow patches — must differentiate from pathology.

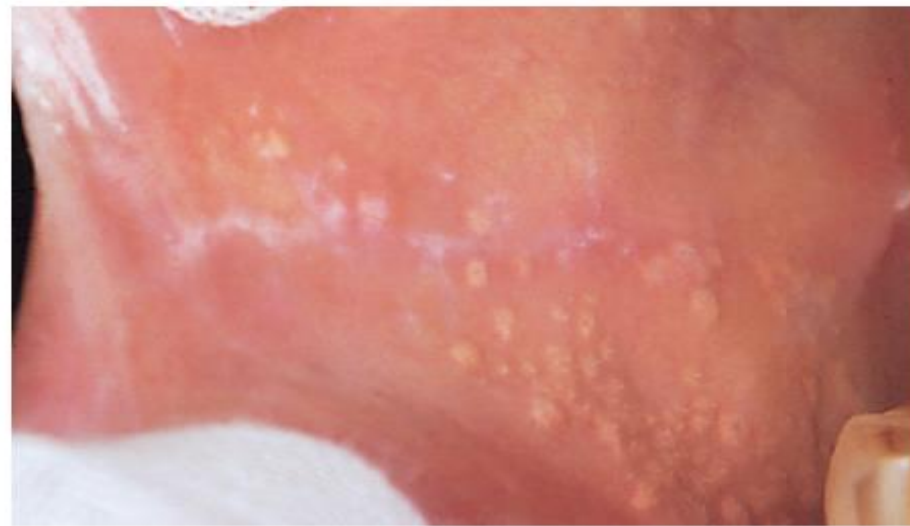


Fig. 1.8 Inner surface of the cheek, showing **Fordyce** spots as yellowish patches. (From Cawson RA, Odell EW (eds) 2008 Cawson's essentials of oral pathology and oral medicine, 8th edn. Churchill Livingstone, Edinburgh, with permission.)



Fig. 1.21 **Fordyce** spots appearing as yellow spots on the vermilion (red zone) of the lip. The black spots below represent hair follicles on the surface of the adjacent skin of the chin. Courtesy Prof. P R Morgan.

GINGIVA & ALVEOLAR MUCOSA

Attached gingiva

- main component → directly bound down to underlying alveolar bone & tooth -> mucoperiosteum
- surface stippling in healthy state, which corresponds to sites of intersecting epithelial ridges
- Width varies regionally

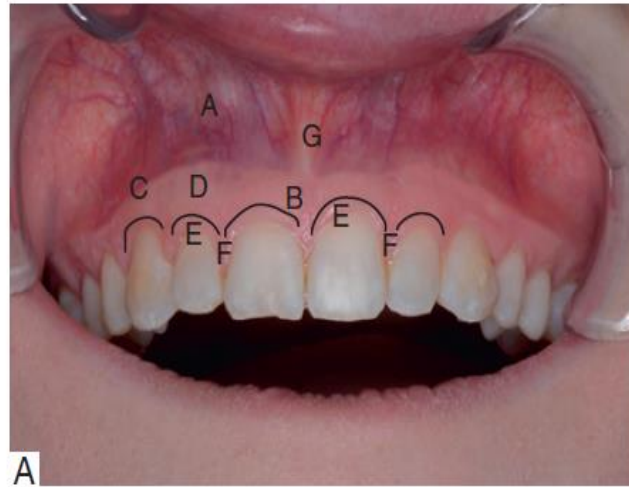


Fig. 14.37 *In vivo* appearance of maxillary gingiva (A) and mandibular gingiva (B). A = alveolar mucosa; B = free gingival groove indicated by curved black line; C = mucogingival junction; D = attached gingiva; E = free gingiva; F = interdental papilla.

Free gingiva

- Coronal to attached gingiva, narrow rim of mucosa that is not bound down to underlying hard tissue.
- Smooth
- Width varies regionally

interdental papilla

- Part of gingiva that fills space b/w teeth.
- shape in 3 dimensions,
- Histological appearance, depend on the shape & nature of contact b/w adj. teeth.

Free gingival groove

- junction b/w free & attached gingiva
- coronal limit → gingival margin
- 40% of teeth
- Follows contours of CEJ
- produced by the principal collagen fibres bundles running from cervical cementum to gingiva, or
- may correspond to heavy epithelial ridge.
- When present, free gingival groove lies approx. at CEJ level

Mucogingival junction:

- Apical demarcation Apically b/w attached gingiva from alveolar mucosa
- It normally lies 3–5 mm below level of alveolar crest.
- height of gingiva above this junction is about 4–6 mm.
- Histologically, demarcation b/w keratinised & nonkeratinised epithelium of junction) is well-defined

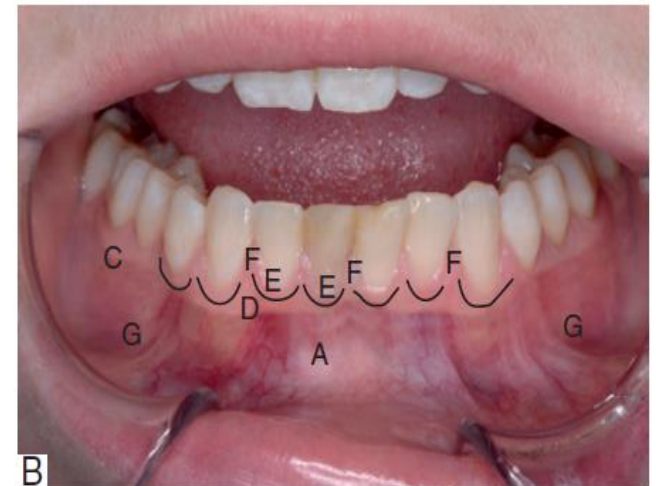
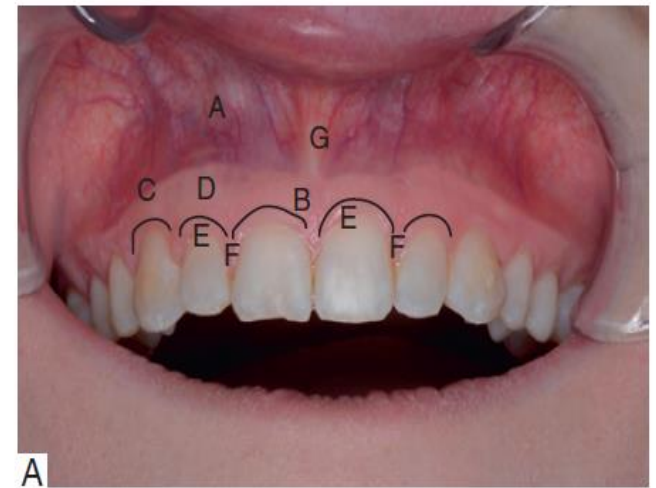


Fig. 14.37 *In vivo* appearance of maxillary gingiva (A) and mandibular gingiva (B). A = alveolar mucosa; B = free gingival groove indicated by curved black line; C = mucogingival junction; D = attached gingiva; E = free gingiva; F = interdental papilla.

ALVEOLAR MUCOSA

Loosely attached to the lower part of the alveolar bone

The d/f in appearance b/w alveolar mucosa & attached gingiva → related to differences in keratinisation & translucency.

The alveolar mucosa epithelium is translucent, thin, nonkeratinised epithelium overlying a lamina propria that shows poorly developed dermal papillae

Small blood vessels are clearly seen lying superficially.

The extensive submucosa houses many minor mucous salivary glands

near the vestibular sulcus, the submucosa → is loosely attached with numerous elastin fibres, allowing for free movement.

Where alveolar mucosa adjoins the attached gingiva, is thicker than elsewhere (& commonly thicker than the adjacent keratinized epithelium).



Fig. 14.38 Section of alveolar mucosa. A = nonkeratinised oral epithelium; B = lamina propria; C = minor salivary gland in submucosa (H & E; ×15).

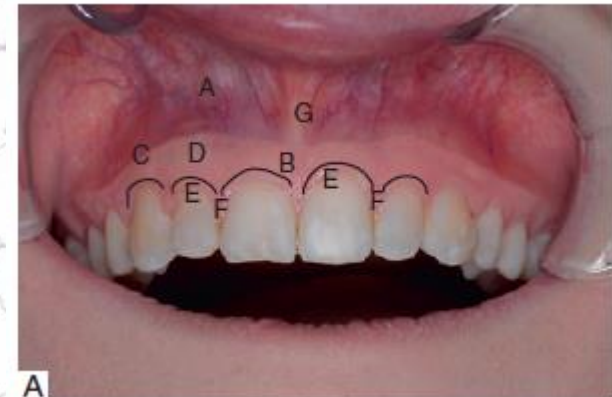
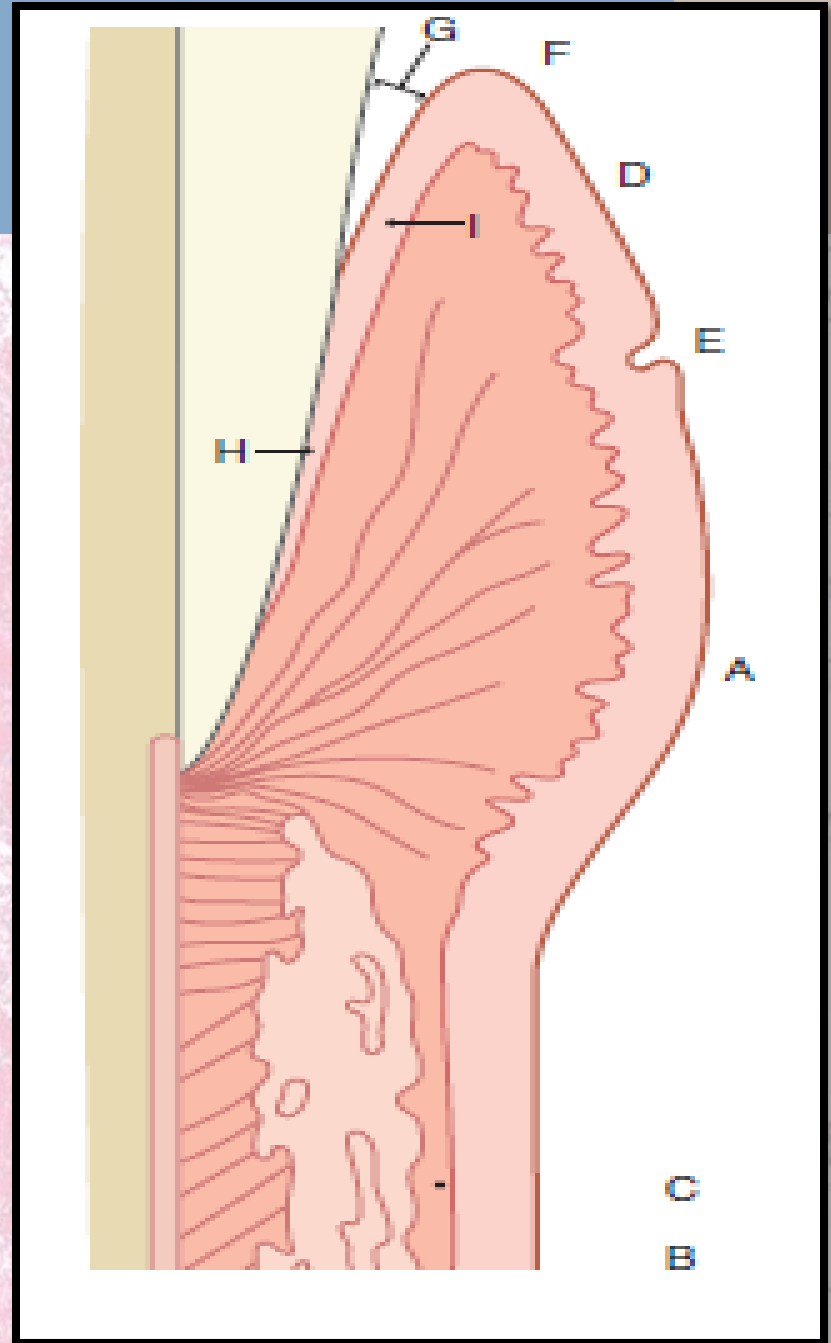
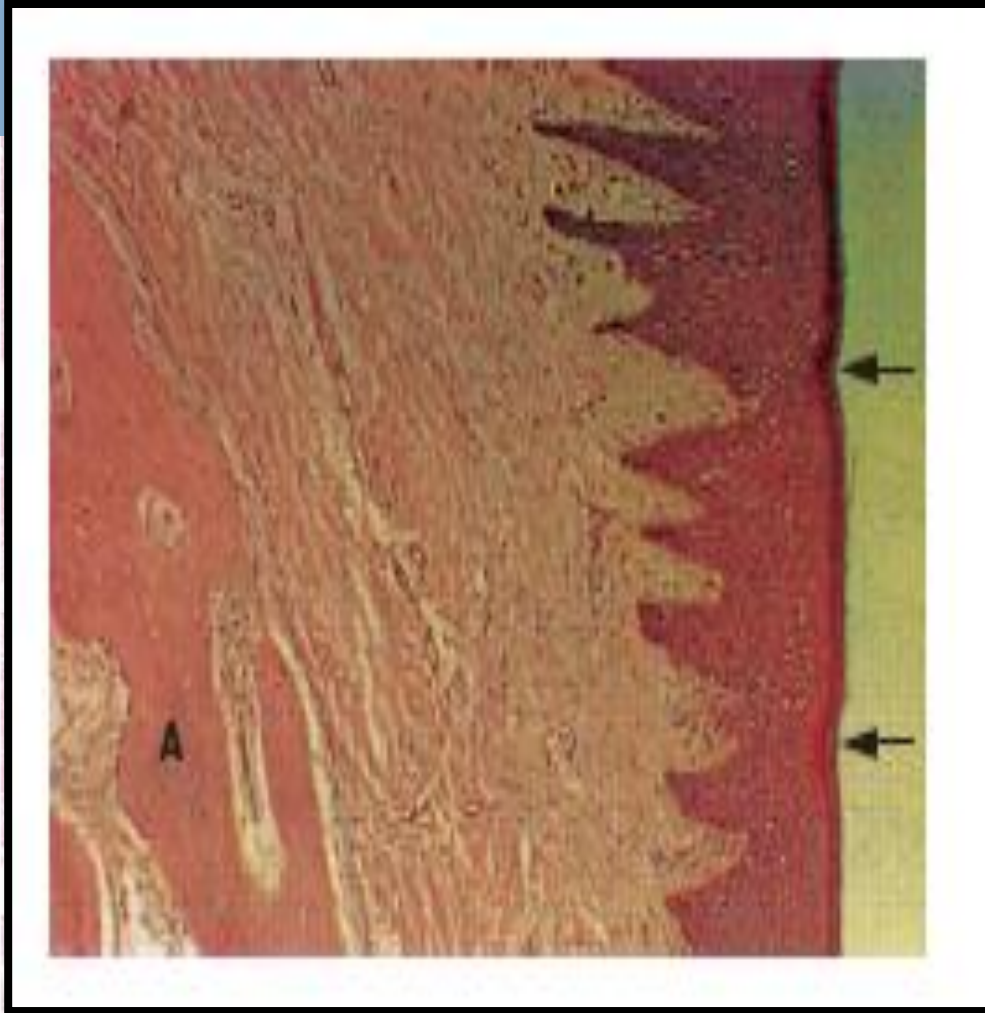


Fig. 14.36 Diagrammatic representation of gingiva.

A = attached gingiva;
B = alveolar mucosa;
C = submucosa associated with alveolar mucosa;
D = free gingiva;
E = free gingival groove;
F = gingival margin;
G = gingival sulcus;
H = junctional epithelium;
I = crevicular epithelium.



ATTACHED GINGIVA



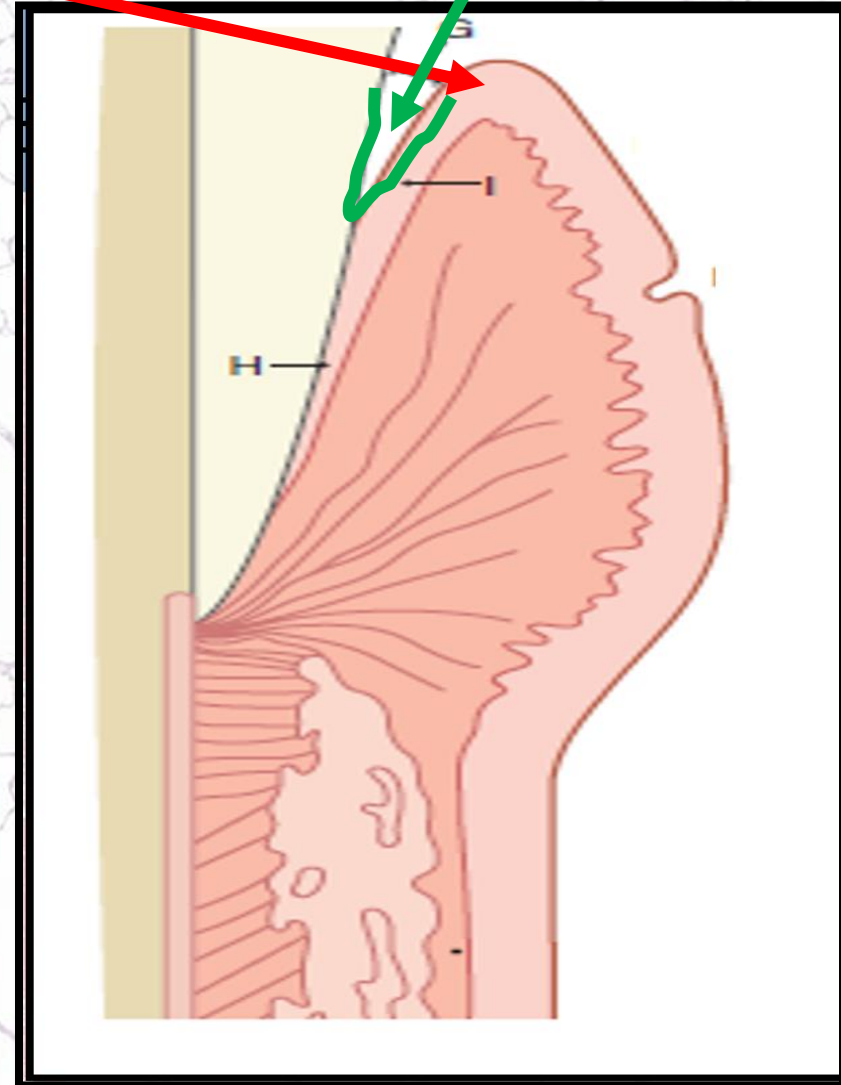
- Masticatory mucosa on external surface
- degree & extent of keratinization vary considerably b/w & within individuals.
- Orthokeratined normally (no inflamed state)
- parakeratinised (75%)
- Nonkeratinised (10%)
- Papillation (variable), aligned in rows

Fig. 14.40 Section of the attached gingiva. The epithelium is a keratinised stratified squamous epithelium. The lamina propria is dense and relatively avascular and the interface with the epithelium is highly folded. The lamina propria is directly attached to the underlying alveolar bone (A), forming a mucoperiosteum. Arrows indicate surface stippling (H & E; $\times 75$).

FREE GINGIVA & GINGIVAL SULCUS

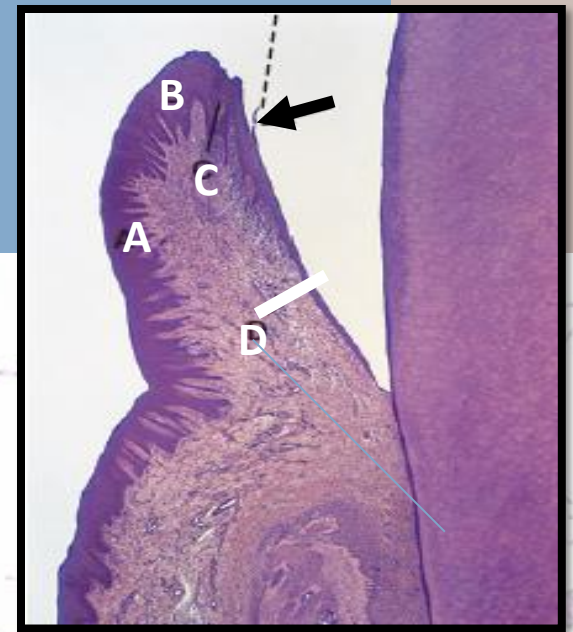
- Mucosa indistinguishable from attached gingiva, demarcation only by free gingival groove (or a line of stipples).
- The gingival crevice → bounded by gingival margin above & junctional epithelium below.

- In clinically healthy mouths, the clinical crevice is 0.5–2.0 mm deep.
- Crevices deeper than 3 mm (measured clinically) are generally accepted as diseased and are described as 'periodontal pockets'



CREVICULAR (SULCULAR) EPITHELIUM

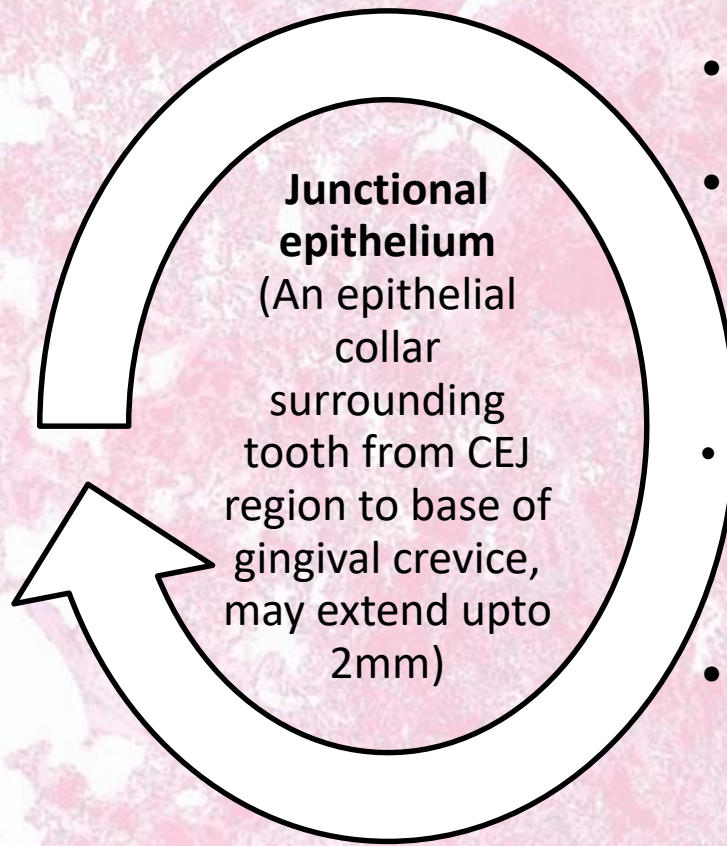
- Epithelium on inner surface of gingiva
→ sulcular epithelium & junctional epithelium
- Nonkeratinised (both), thin & lack granular layer.
- form a gingival cuff at the site where oral mucosa meets tooth.
- Histologically, crevicular epithelium may be distinguished from junctional epithelium by having a more folded interface with the underlying CT.
- an enamel cuticle may be seen at the interface between the 2 epithelia
- crevicular epithelium merges with junctional but no distinct boundary
- Externally, base of crevice corresponds approx. with free gingival groove, when present.



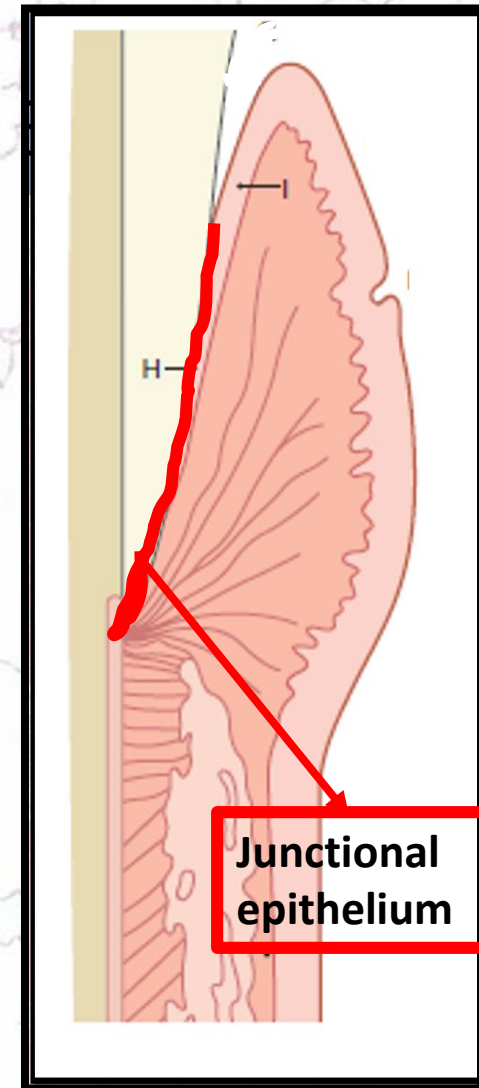
A = Region of attached gingiva covered by masticatory epithelium;
B = region of free gingiva covered externally by masticatory epithelium;
C = nonkeratinised crevicular epithelium;
D = nonkeratinised junctional epithelium.

Note the tag of enamel cuticle (arrow) that helps to delineate the oral sulcular epithelium from the junctional epithelium

JUNCTIONAL EPITHELIUM



- Coronal cell thickness → 15–30 cells (up to 100 μm);
- Apical cell thickness → 1–3 cells
- 2 zones: single-layer cuboidal **stratum germinativum** + **several flattened layers (stratum spinosum)**, without any granulosum or corneum.
- ↑ turnover rate of any oral mucosa (~5–6 days); cells exfoliate coronally into crevice,
- Origin → odontogenic as CK profile resemble, **reduced enamel epithelium** (probably stratum intermedium



PRINCIPLE FEATURES OF JUNCTIONAL EPITHELIUM

unique → 2 basal lamina with differences

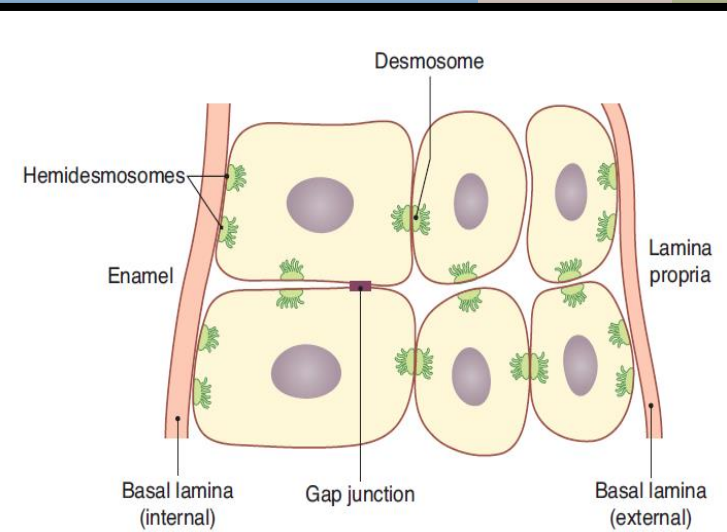
Internal Basal Lamina → Basal lamina in contact with tooth

lack type IV collagen, anchoring fibrils & laminin) with
↑ turnover

Ultrastructure → 2 zones (100–140 nm). : an electron-lucent zone adj. to cell (preparation artefact) & an electron-dense layer against the tooth surface. (Lamina Lucida + Lamina Densa → pattern unclear)

Attachment Apparatus of junctional epithelium to tooth → hemidesmosomes (thickenings of inner leaflet of plasma membrane + basal lamina (**epithelial attachment**))

Opp. to attachment plaque at enamel surface → peripheral dense line comparable to lamina lucida of basal lamina b/w epithelium & connective tissue.



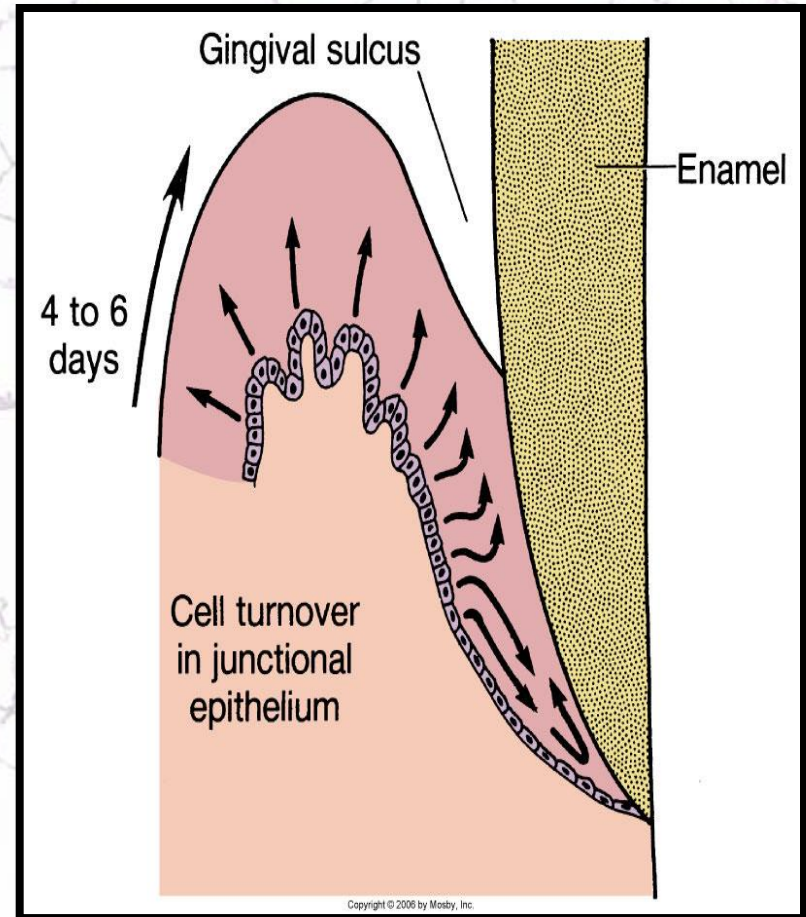
External basal lamina → On the other surface of the junctional epithelium in contact with lamina propria is the normal basal lamina

Cells of st. germinativum rest on this typical lamina propria, with many capillaries & ↑ cellularity than other parts of gingiva & smooth interface with connective tissue

PRINCIPLE FEATURES OF JUNCTIONAL EPITHELIUM

TURNOVER RATE OF JUNCTIONAL EPITHELIUM

- Continual breakdown & reformation of lamina densa, hemidesmosomes & desmosomes → allows cells to alter their relationship as they migrate through the JE
- Turnover rate → dependent on demands placed upon tissue & appears to be directly related to degree of inflammation.



PRINCIPLE FEATURES OF JUNCTIONAL EPITHELIUM

PERMEABILITY OF JUNCTIONAL EPITHELIUM

- ↓ desmosomes & larger intercellular spaces (5% of the volume) → ↑ permeability (crevicular fluid & defence cells can pass across JE.)
- ↑ permeability → also b/c of lack of membrane-coating granules

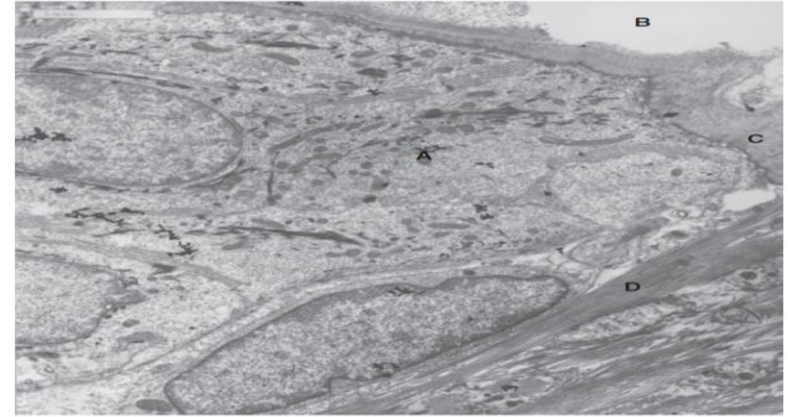
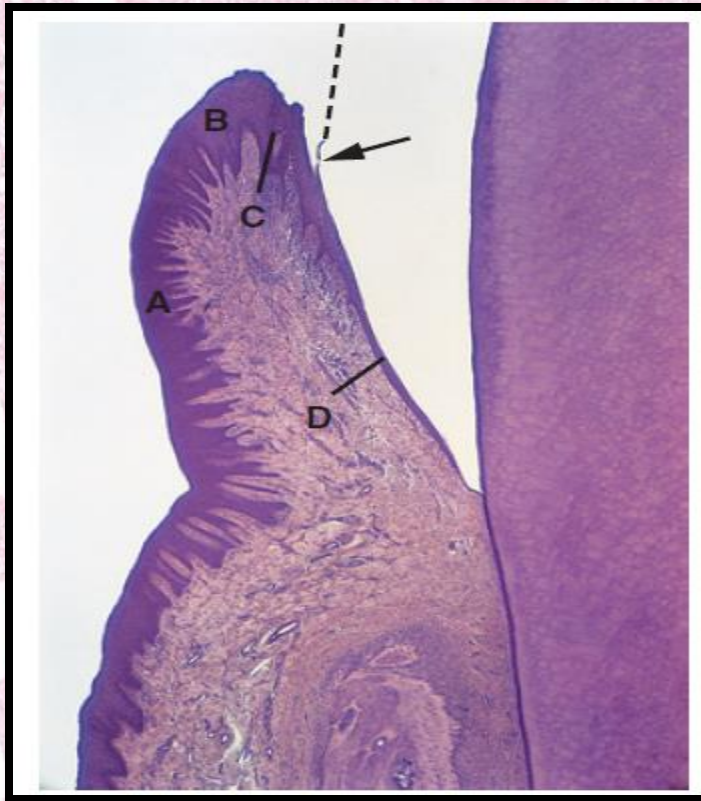


Fig. 14.49 Electron micrograph of demineralised section showing the apical limit of the junctional epithelial cells (A) at the cement-enamel junction. B = enamel space; C = cementum; D = collagen fibre of gingiva ($\times 2,500$). Courtesy Prof. M M Smith.

NEUTROPHILS IN JE

- Healthy gingival tissue may exhibit neutrophils in intercellular spaces, indicative of its protective role.
- Transmigration of neutrophils from the adjacent capillaries:
INTERCELLULAR ADHESION MOLECULE (ICAM)-1 by cells of junctional epithelium, together with underlying fibroblasts & endothelial cells



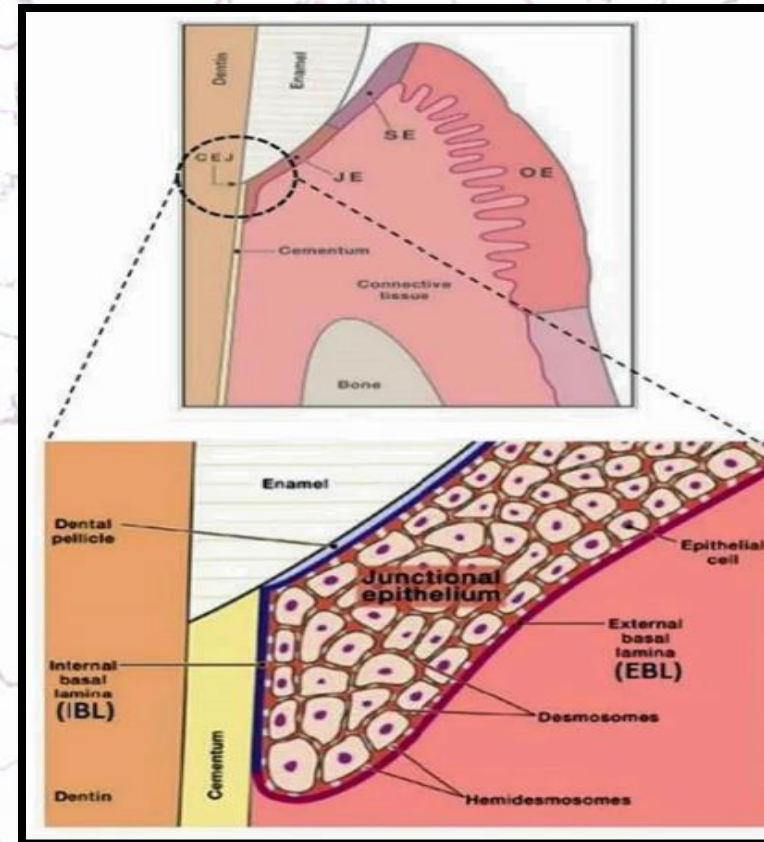
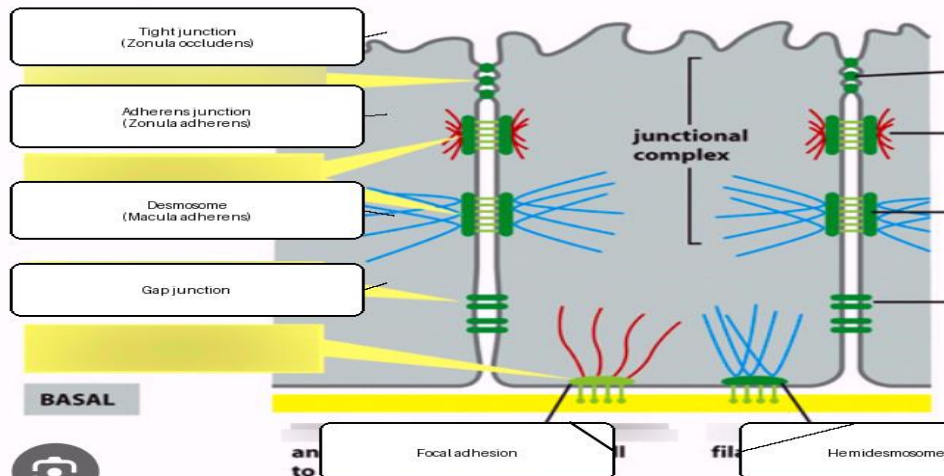
ENAMEL CUTICLE:

- not visible in decalcified preparations
- nonmineralised structure interposed b/w junctional epithelium & underlying hard tissue.
- Extent varies & is not always present.

PRINCIPLE FEATURES OF JUNCTIONAL EPITHELIUM

Synthesis, transport & secretion of Basal Lamina components by JE cells: cytoplasm contains numerous free ribosomes, cisternae of RER & a prominent Golgi apparatus.

Absent membrane-coating granules, Keratohyaline granules but few cytokeratin filaments are present

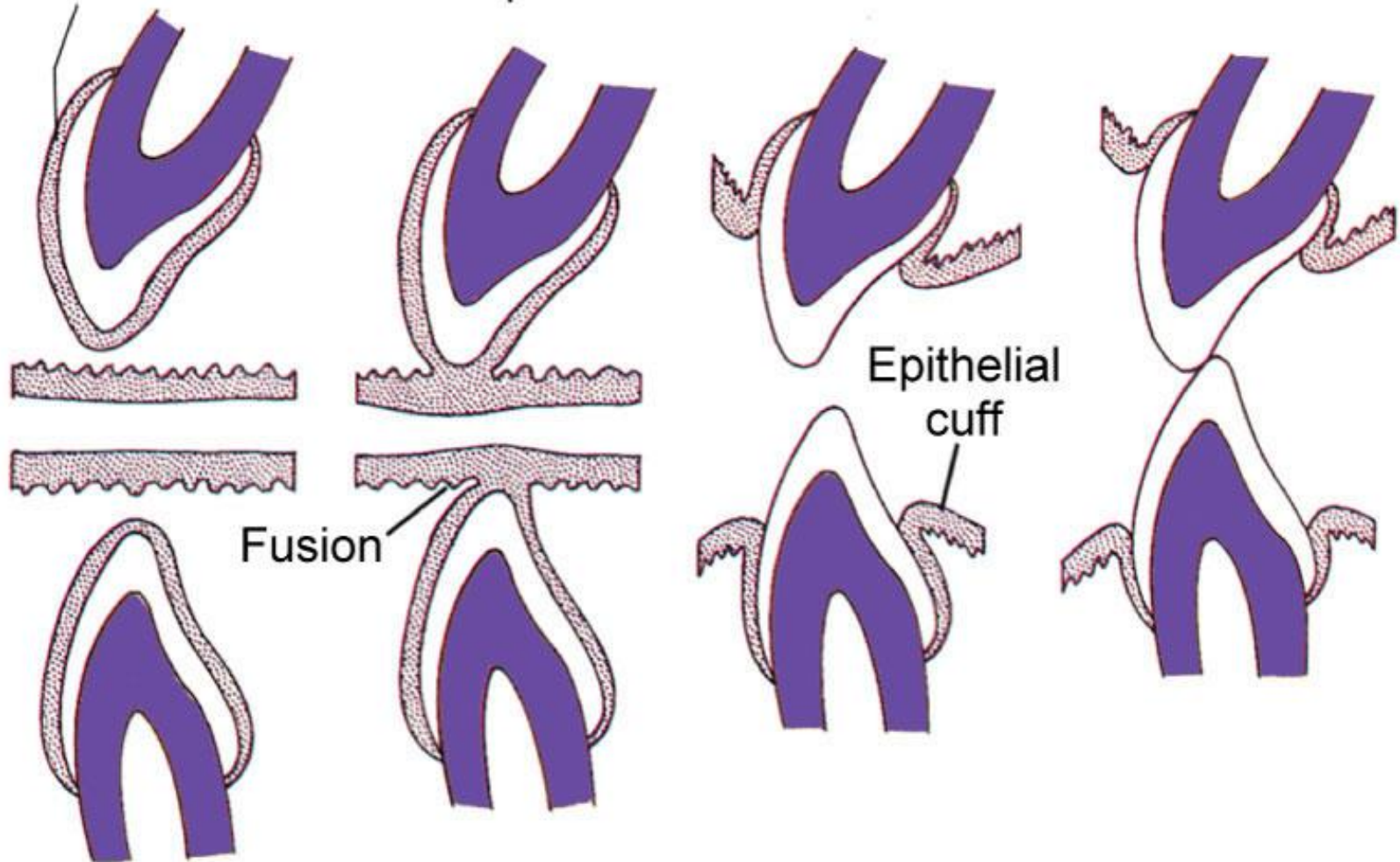


Cell contacts: desmosomes & gap junctions; tight junctions (rare)

VARIATION OF JUNCTIONAL EPITHELIUM LENGTH WITH STAGE OF ERUPTION

- Length of JE on enamel varies with eruption stage:
- newly erupted tooth = most enamel covered;
- Occlusal plane stage = $\sim 1/4$ of enamel surface covered;
- eventually JE lies near CEJ.
- With root exposure, JE may proliferate apically onto cementum (via basal lamina) – could represent physiological passive eruption OR disease-related change.
- Following surgical gingivectomy, a new JE rapidly reforms with all original characteristics – proliferating (keratinised) gingival epithelial cells are modified by underlying CT to adopt the JE phenotype.

Reduced enamel epithelium



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VASCULATURE & INNERVATION OF JUNCTIONAL EPITHELIUM

Crevicular plexus:

- Rich blood supply arranged as complex anastomosing network, major vessels representing post-capillary venules (In lamina propria associated with JE)
- separated from surrounding looping vessels that lie within dermal papillae of attached gingiva → obvious source of GCF.
- The vessels of plexus → sensitive to stimulation & are likely to vasodilate under slightest of insults.
- In response to plaque, they may become more permeable, ↑ production of crevicular fluid.
- The JE has a rich plexus of nerves close to its basal layer, with many endings penetrating into the epithelium itself.

DENTOGINGIVAL JUNCTION

- The dentogingival junction seals underlying CT of periodontium from oral environment.
- The strength of the seal is thought to be dependent not only upon the attachment of the JE to tooth but also upon the pressure exerted by the fibres & tissue fluid of the underlying CT.
- The weakness of dentogingival junction derives from its situation, the enamel being a nonshedding surface, allowing persistent bacterial colonisation.
- The epithelium is therefore exposed to toxic products emanating from the consequent bacterial accumulation.

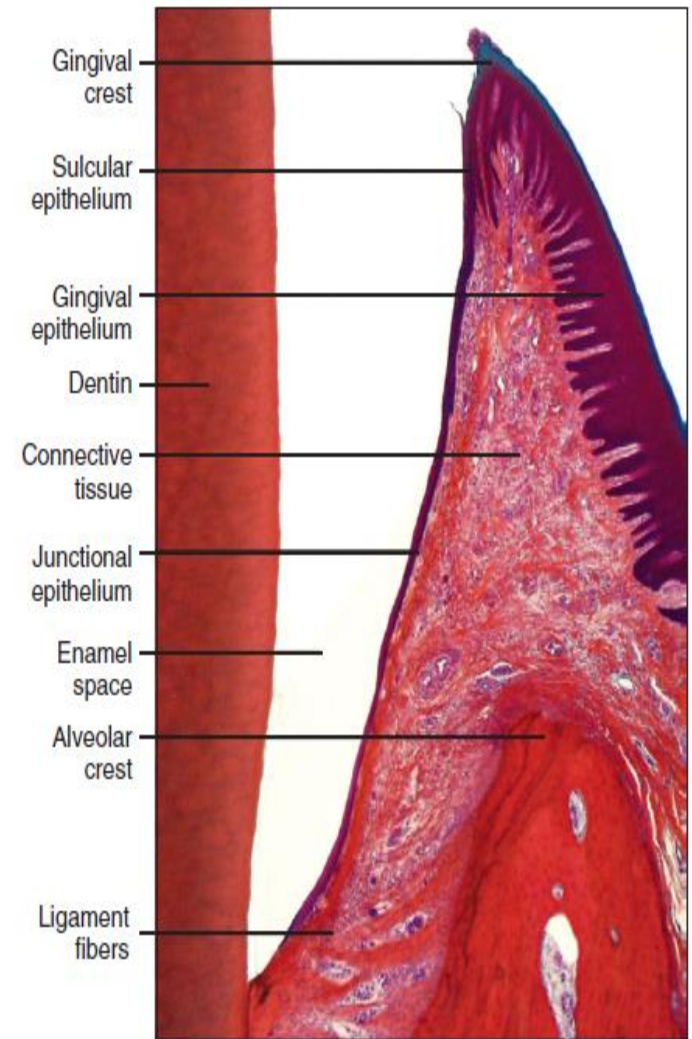


FIGURE 12-33 The dentogingival junction. Together, the sulcular and junctional epithelium constitute the dentogingival junction. The junctional epithelium attaches to the tooth surface. Decalcification of the specimen has removed the tooth enamel, leaving an enamel space. The sulcular epithelium is short in this preparation because the tooth is not fully erupted.

Epidermal
growth factor

**Factors only in
Junctional
Epithelium as
compared to
Crevicular
Epithelium**

characteristic
expression of
basal markers
CK5, 14, & 19
throughout ALL
layers
(nondifferentiating
tissue) —
unlike crevicular
epithelium
(CK4).

Integrin ($\alpha 6 \beta 4$) →
transmembrane receptor
glycoprotein, component of
hemidesmosomes (both
basal laminae of junctional
epithelium as compared to
single basal lamina of
crevicular epithelium).

CK19 distribution &
 $\alpha 6$ component

Gingival crevicular fluid (GCF)

JE is **permeable** — tissue fluid & cells pass readily from connective tissue into crevice.

Healthy tissue flow rate: **0.05–0.2 $\mu\text{L}/\text{min}$** ; the fluid doesn't solely response to pathological stimuli.

transudate/exudate of periodontal tissues, reaching crevice via intercellular routes (osmosis); basement membrane filters out large components.

Composition (health) similar to plasma: Na, K, Ca, Mg (Ca higher than saliva — aids pellicle protein precipitation, bacterial attachment, calculus formation); immunoglobulins IgA, IgG, IgM; fibrinogen; complement; protease inhibitors; PMNs; epithelial squames; plaque bacteria & products.

In disease: inflammatory products $\rightarrow$ $\uparrow$ cytokines, metalloproteinases (protective reaction may also damage host tissue).

GCF composition indicates periodontal health status — e.g. after orthodontic loading, $\uparrow$ in cathepsin B, prostaglandins, ILs, TNF, EGF (bone/PDL remodelling markers). Potential biomarker source for periodontal disease severity.

INTERDENTAL GINGIVA

part of gingiva b/w adjacent teeth.

The shape & arrangement → depend on shape of contact b/w teeth (although free & attached gingivae are always present).

Occupies space b/w teeth & conforms to its shape.

wedge-shaped appearance → from B & L aspects

Col epithelium → continuous with JE on each side (similarly nonkeratinised & initially derived from Reduced enamel epithelium, but it is as difficult to to keep plaque-free with inflammatory cells infiltrating the underlying lamina propria.

In spaced teeth → col (absent) with very flat gingiva, covered by a keratinised epithelium.

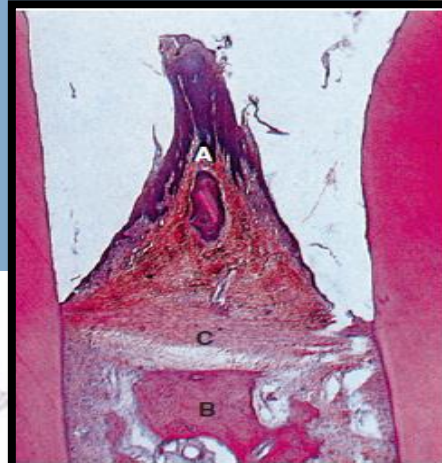
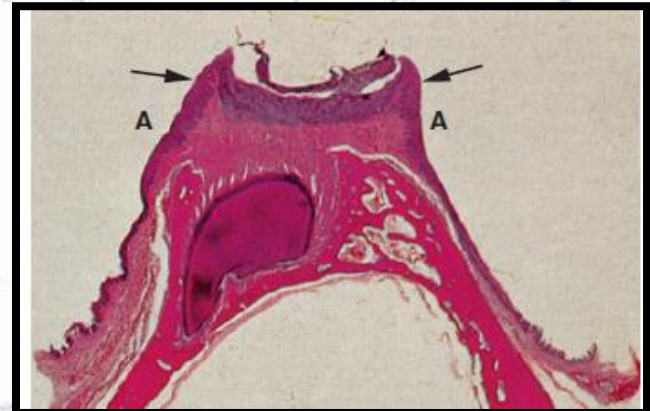


Fig. 14.52 Demineralised section showing the interdentals gingiva (A) between two cheek teeth in the anteroposterior plane. B = alveolar crest; C = transseptal group of gingival fibres (H & E; $\times 20$).

B/w anterior teeth (which contact only at a small point), → similarly 'pointed' when viewed in a BL plane.



In post. teeth, (broader contact area) → typical wedge shape appearance from B or L side but across its BL plane there are 2 peaks on the B & L aspects with a curved depression in b/w (the interdental col), which follows the contour around the contact point.

GINGIVAL FIBRES

- collagen fibres → type I collagen (dense principal bundles to support free gingiva, bind the attached gingiva to the alveolar bone & tooth to resist masticatory loads, & linking of teeth to one another.
- named according to orientation of attachments
- The turnover rate of gingival collagen appears to be faster than in other parts of the oral cavity.

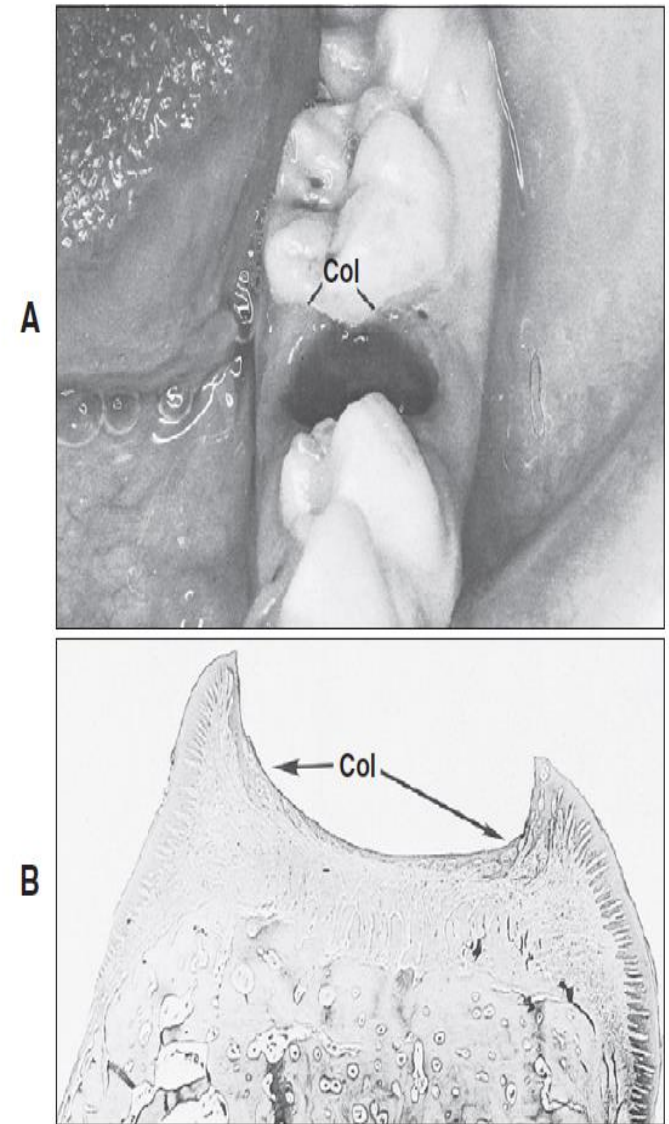


FIGURE 12-36 Dental col. **A**, Clinical appearance. **B**, Histologic section. The distinction between the keratinized gingival epithelium and the epithelium of the col is evident.

Dentogingival fibres → from root surface above alveolar crest & radiate to insert into lamina propria of gingiva.

superficial fibres: lie beneath crevicular epithelium

middle group: lies almost horizontally & deepest group courses b/w gingiva & alveolar bone.

Longitudinal fibres → extend for long distances within the free gingiva some possibly for the whole length of the arch.

C = circular fibres;

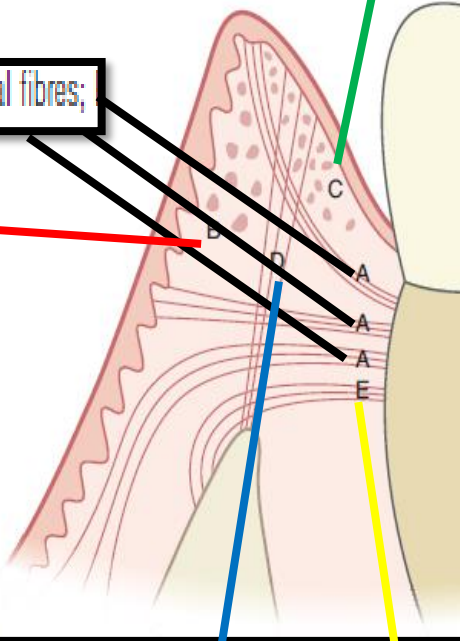
A = dentogingival fibres;

B = longitudinal fibres;

Circular fibres → encircle each tooth within the marginal and interdental gingiva.

Some attach to cementum, some to alveolar bone.

Some cross interdentally to join the fibre group of the adjacent tooth.



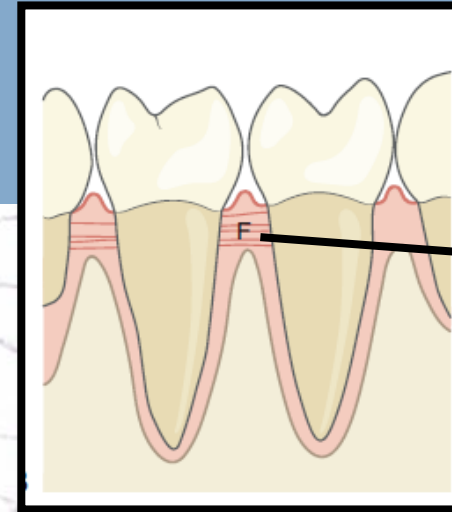
D = alveologingival fibres;

E = dentoperiosteal fibres;

Alveologingival fibres → run from the crest of the alveolar bone & interdental septum, radiating coronally into the overlying lamina propria of the gingiva.

Dentoperiosteal fibres → occur only in labial/buccal and lingual gingiva. They arise from cementum and pass over the alveolar crest to insert into the periosteum.

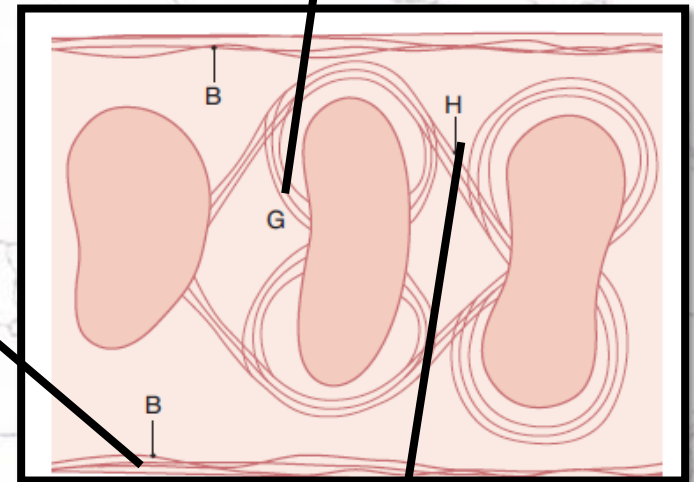
Transseptal fibres → pass horizontally from root of 1 tooth, above alveolar crest, to be inserted into root of adj. tooth. Provide an anatomical basis for linking all teeth in dentition. They have been implicated in mechanism of mesial drift.



F = transseptal fibres

Semicircular fibres → emanate from cementum near the CEJ, cross the free marginal gingiva and insert into a similar position on the opposite side of the tooth.

G = semicircular fibres;

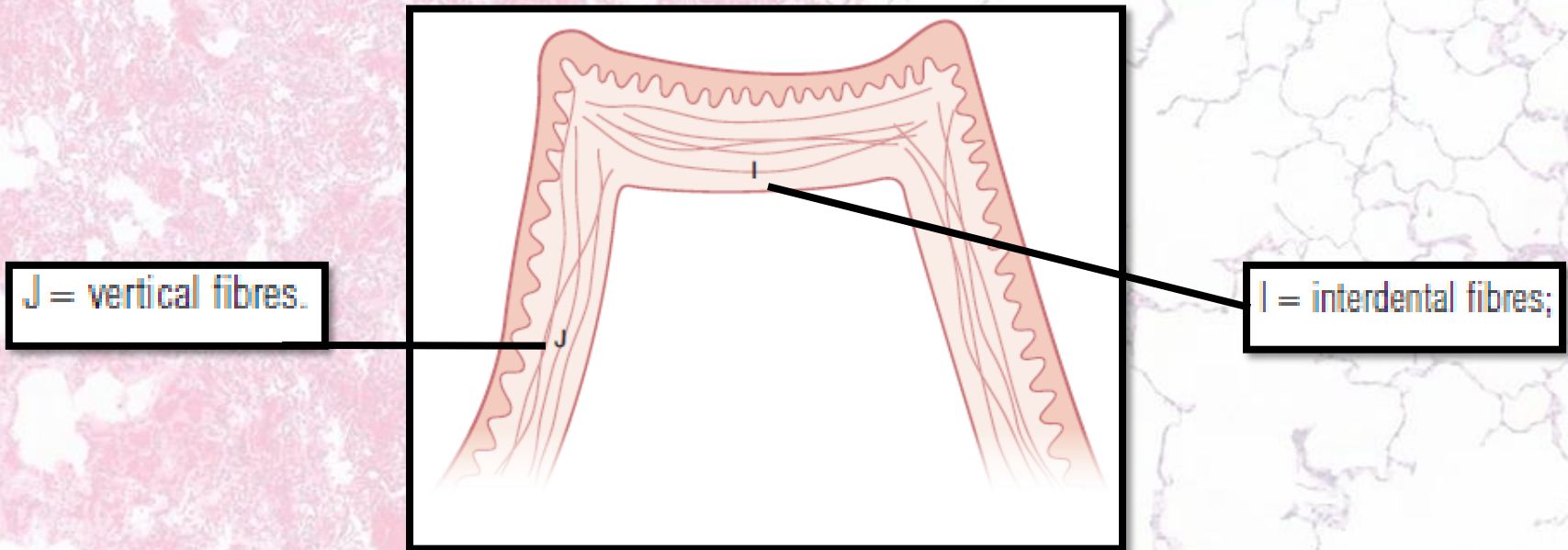


B = longitudinal fibres;

Transgingival fibres → reinforce the circular and semicircular fibres. The fibres arise from the cervical cementum and extend into the marginal gingiva of the adjacent tooth, merging with the circular fibres.

H = transgingival fibres;

Interdental fibres → pass through the coronal portion of the interdental gingiva in the buccolingual direction, connecting buccal and lingual papillae.



Vertical fibres a→ rise in alveolar mucosa or attached gingiva and pass coronally towards the marginal gingiva and interdental papilla.

FEATURES DISTINGUISHING LAMINA PROPRIA OF GINGIVA FROM PDL

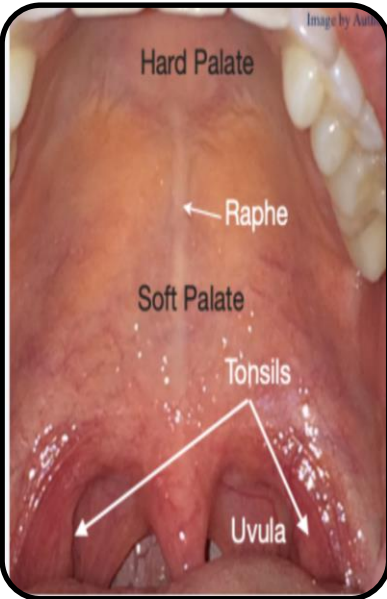
- Fibroblasts lack alkaline phosphatase,
- Fibroblasts have less contractile proteins
- Fibroblasts can release $\uparrow$ prostaglandin in response to histamine.
- The ECM has $\downarrow$ ground substance
- The ECM has less type III collagen,
- The ECM is hyaluronan-rich
- The ECM has has a $\downarrow$ turnover rate.

VASCULATURE OF GINGIVA

- The vasculature of lamina propria → very rich
- Forms two plexuses → one beneath oral gingival epithelium & other beneath oral sulcular epithelium.
- Plexuses allow tissues to respond very quickly to stimuli.
- Each dermal papilla of lamina propria beneath oral gingival epithelium possesses an ascending arterial loop & a descending venous loop, b/w which lies a terminal capillary loop.

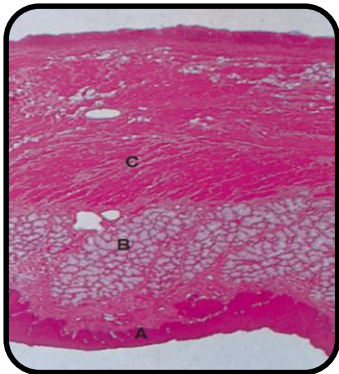
PALATE

Hard palate



- Typical masticatory mucosa — partly keratinised, partly parakeratinised.
- Central region: NO submucosa → dense lamina propria bound directly to bone (mucoperiosteum), as in attached gingiva.
- Where palate joins alveolus → submucosa present, contains main neurovascular bundles + minor mucous glands (mostly post.) + adipose tissue (mostly ant.).
- **Nasal surface** of hard palate = **respiratory mucosa** (ciliated columnar epithelium with goblet cells)
- Vascular submucosa beneath with mixed mucous/serous glands.

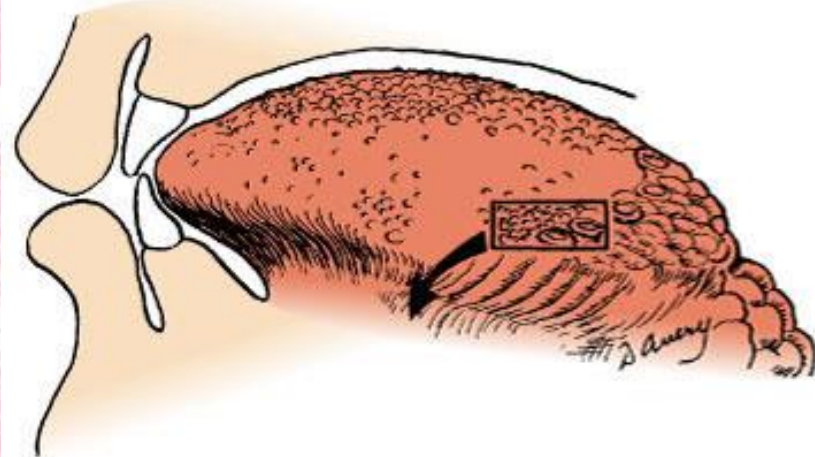
Soft palate



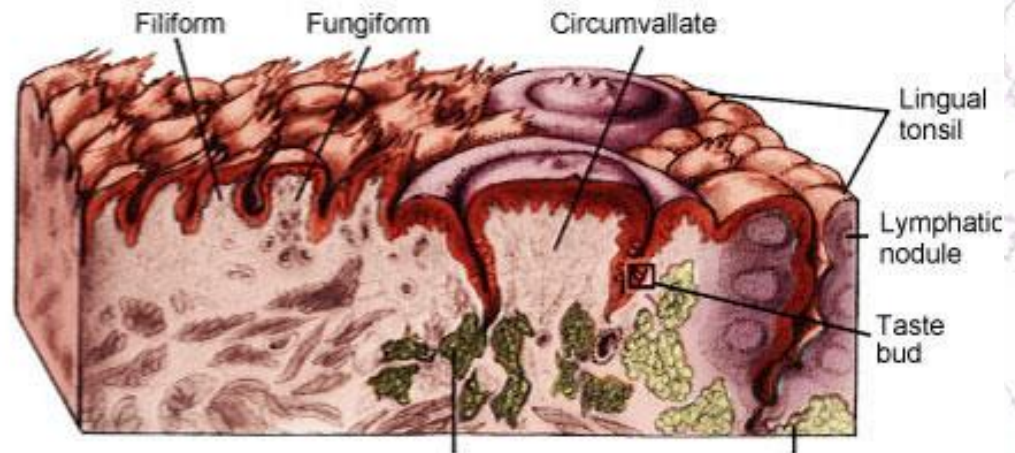
- Oral surface = **nonkeratinised lining mucosa**; short, broad papillae; lamina propria rich in elastin, relatively thin collagen bundles.
- Broad submucosa with many small mucous glands, attaching into palatal muscles.
- Nasal surface of soft palate = respiratory mucosa (ciliated columnar epithelium).

TONGUE & FLOOR OF MOUTH

- Ventral tongue surface + floor of mouth = typical **lining mucosa**: thin, nonkeratinised, short papillae.
- Submucosa extensive on floor of mouth but indistinct/absent on ventral tongue (mucosa binds directly to muscle).
- Thin epithelium + vascular connective tissue → route for rapid drug absorption (sublingual route).
- **Sulcus terminalis** divides tongue into anterior 2/3 (palatal surface) and posterior 1/3 (pharyngeal surface).



A



Von Ebner's glands Mucous glands and ducts

B

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PAPPILLAE ON ANT. 2/3RD OF TONGUE

Filiform papillae

most numerous; dorsum classed as masticatory mucosa b/c of these.

Keratinised, hairlike tufts; nonkeratinised b/w papillae.

Central lamina propria core with secondary papillae.

Highly abrasive during mastication (compress bolus against palate).

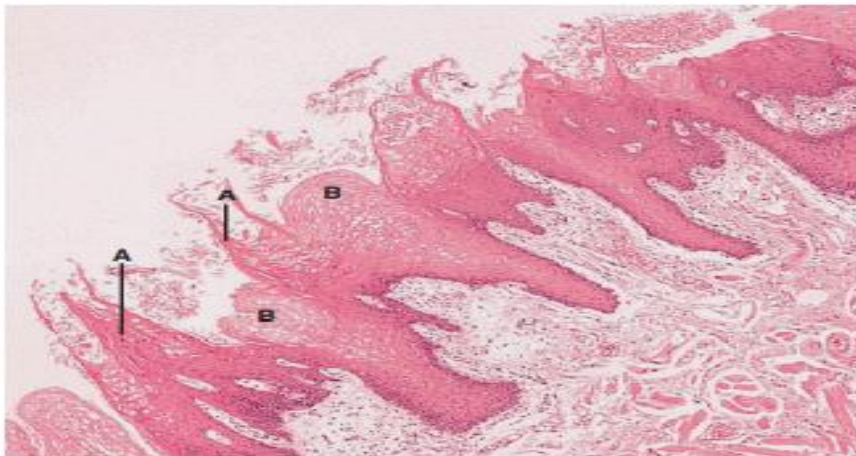


Fig. 14.66 Section showing dorsum of anterior two-thirds of tongue covered by keratinised filiform papillae (A) with nonkeratinised regions between (B) (H & E; $\times 35$). Courtesy Prof. P R Morgan.

Foliate papillae

one or two longitudinal clefts on posterior lateral tongue;

taste buds present in nonkeratinised epithelium.

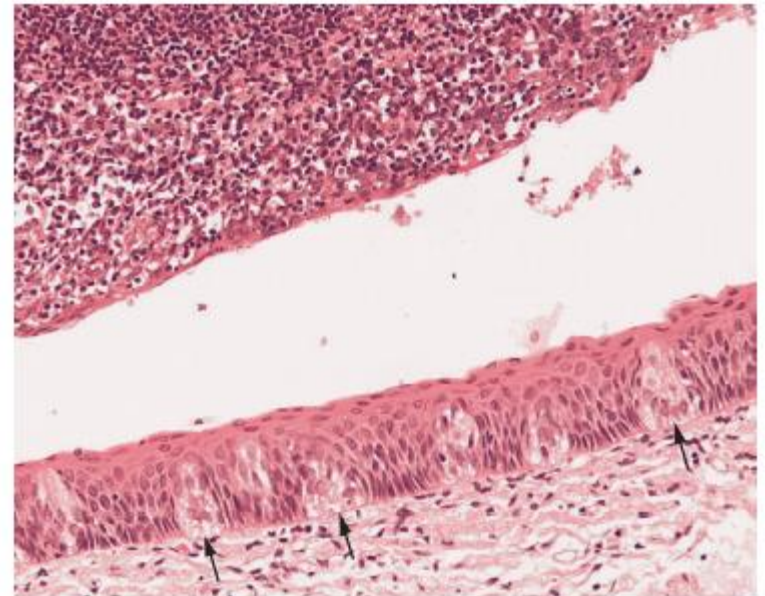


Fig. 14.70 Section of foliate papilla showing taste buds (arrowed). Note the adjacent lymphoid material characteristic of the posterior part of the tongue ($\times 200$).

PAPPILLAE ON ANT. 2/3RD OF TONGUE

Fungiform papillae

isolated, mushroom-shaped, 150–400 μm , scattered among filiform papillae;

thin epithelium (may/may not be keratinised);

vascular lamina propria core; may bear taste buds.

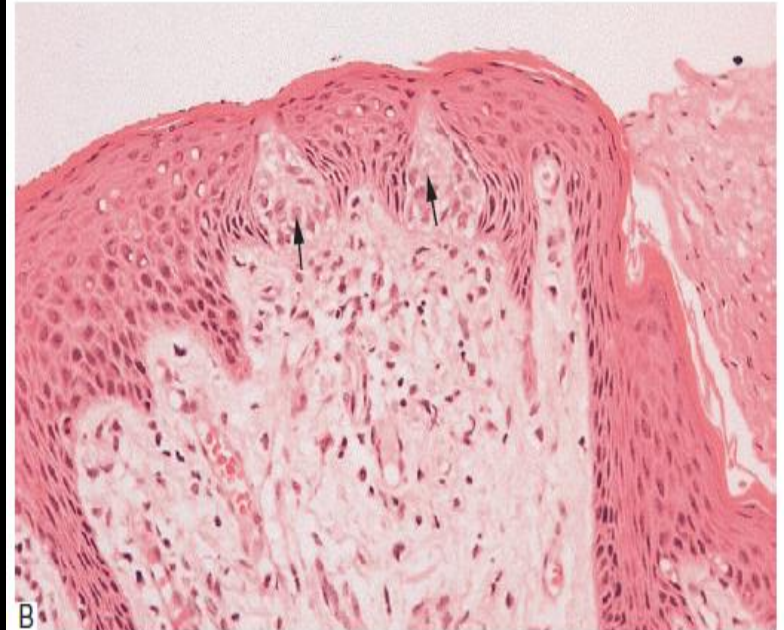
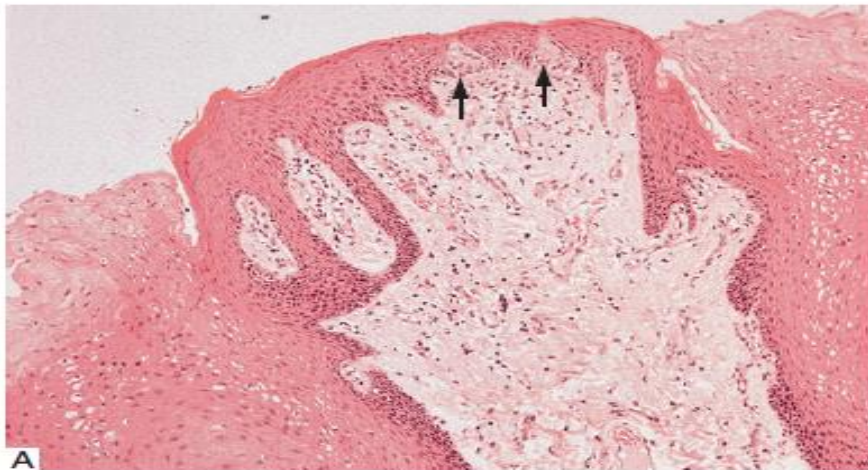


Fig. 14.68 (A) Section of fungiform papilla on dorsal surface of anterior part of tongue showing taste buds (arrowed). The papilla is keratinised (H & E; $\times 120$). (B) High-power view of surface of fungiform papilla seen in (A), showing taste buds (arrows) ($\times 240$). Courtesy Prof. P R Morgan.

PAPPILLAE ON ANT. 2/3RD OF TONGUE

Circumvallate papillae

large, rounded, surrounded by a trench, do NOT project above tongue surface;

nonkeratinised covering epithelium; taste buds predominate on internal trench wall;

small serous glands of **von Ebner** empty into trench base;

unencapsulated mucous glands also occur in tongue muscle (esp. posteriorly).

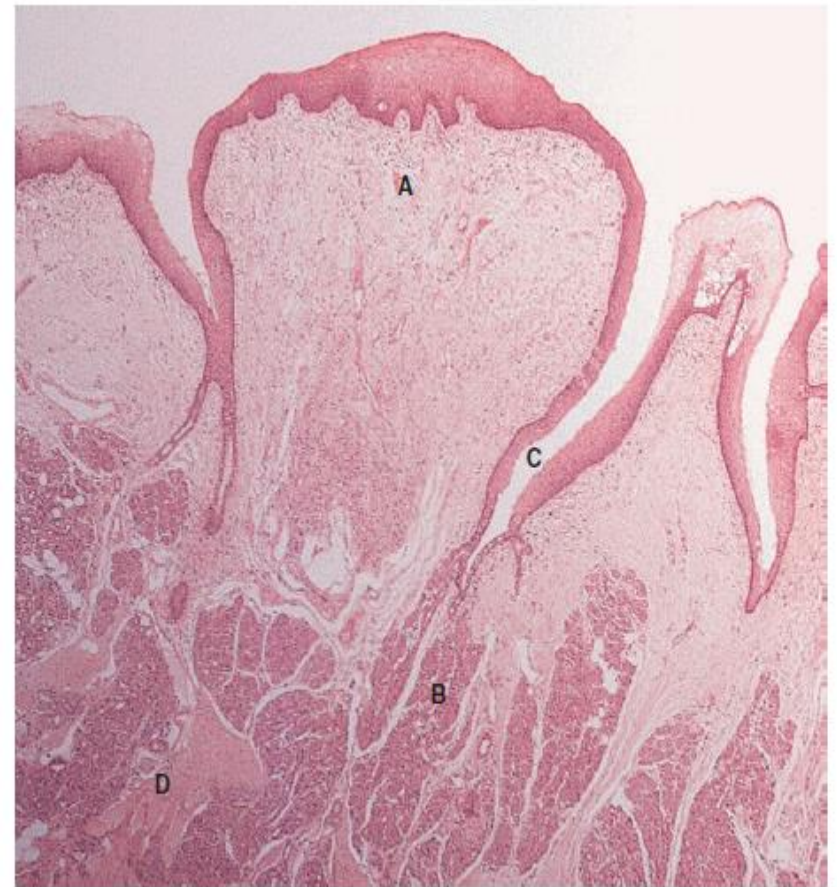


Fig. 14.72 Section of circumvallate papilla (A). Serous glands (B) of von Ebner empty via ducts into the base of the trench (C) surrounding the papilla, which is not raised above the surface of the tongue. D = muscle of tongue (H & E; $\times 35$). Courtesy Prof. P R Morgan.

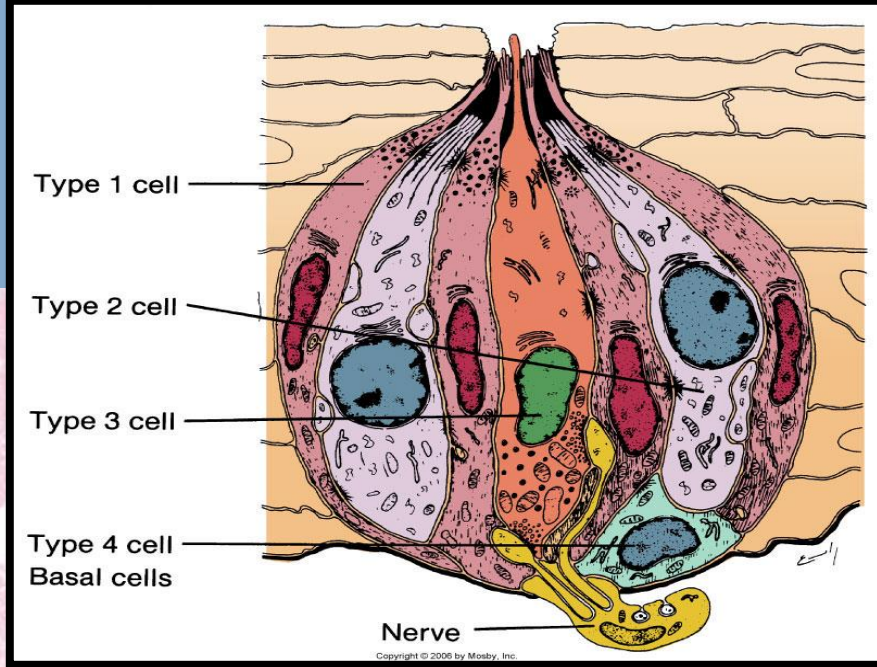
TASTE BUDS

The taste buds, the special chemoreceptive organs responsible for taste, particularly found:

- around the walls of the circumvallate papillae
- in small numbers on the upper surface of fungiform papillae
- lateral walls of foliate papillae
- mucosa of the soft palate and in the epiglottis.



Fig. 14.75 Electron micrograph of a taste bud. A = epithelial cells adjacent to taste bud; B = darker type I cells; C = lighter type II cells; D = lighter type III cell; E = underlying connective tissue; F = taste pore; arrows indicate where type I and type III cells form synapses with intragemmal nerves ($\times 11,000$). Courtesy Dr S M Royer and Dr J C Kinnamon.



Structure of a Taste Bud

Each taste bud has a **small pore** at the surface, where taste molecules enter. The bud is separated from deeper tissues by a **basal lamina (thin base layer)**.

Inside the taste bud are **two main types of cells:**

Supporting cells
Taste cells

Type of Cell	Function
Type IV cells	relatively undifferentiated cells which are distinguished by their basal position and the presence of intermediate filaments
Type I cells	have a dark appearance
Types II & III Cells	These cells are lighter.
Types I and III cells	may form synapses with intragemmal nerves.
Type IV cells	relatively undifferentiated cells which are distinguished by their basal position and the presence of intermediate filaments

Taste Buds in the Human Adult

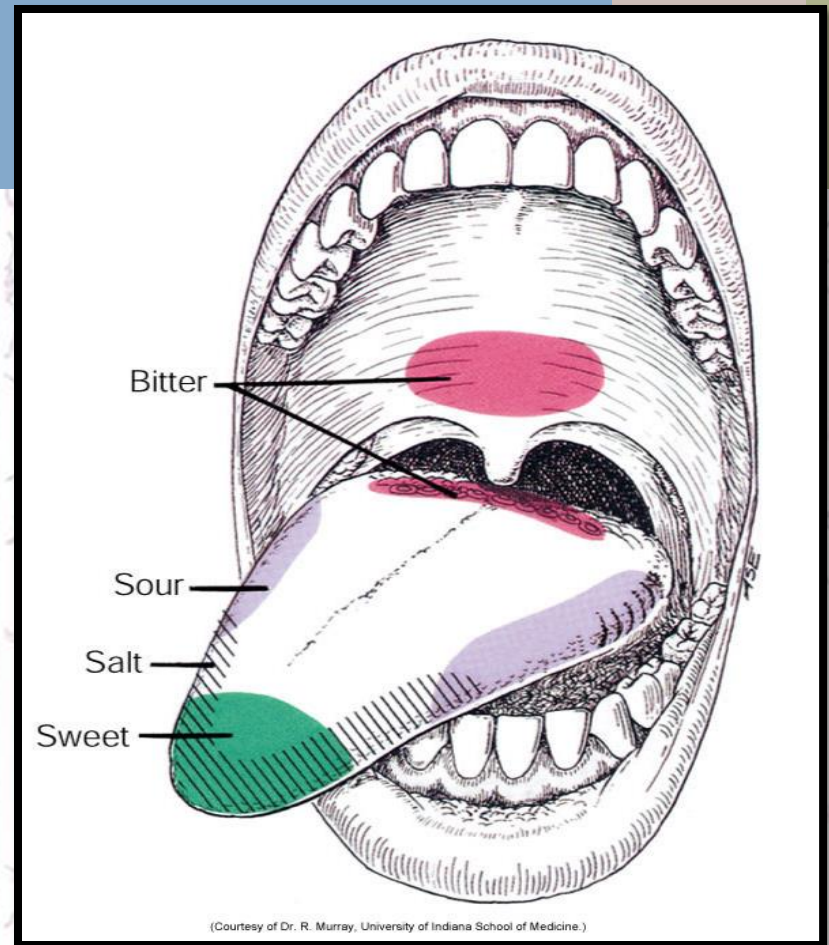
<u>Location</u>	<u>Number</u>
Tongue	10,000
Soft palate	2,500
Epiglottis	900
Larynx/pharynx	600
Oropharynx	250

4 taste sensations: Sweet, salty, sour and bitter

Sweet and salt: ant tongue

Sour: lateral tongue

Bitter: region of circumvallate papilla



Junction	Location / Interface	Key Features	Clinical Significance
Mucocutaneous Junction	Skin ↔ Oral mucosa at the lips	Vermilion zone: thin keratinized epithelium, long CT papillae with capillary loops; few/no glands → prone to dryness and cracking	Creates the characteristic red colour of the lips
Mucogingival Junction	Attached gingiva ↔ Alveolar mucosa	Abrupt transition from keratinized/parakeratinized, firm gingiva with dense collagen → non-keratinized, elastic, mobile alveolar mucosa	Clinically identified by the mucogingival groove and colour change
Dentogingival Junction	Gingiva ↔ Tooth surface	Junctional epithelium (JE) attaches to enamel/cementum via inner basal lamina + hemidesmosomes; highly permeable with continuous immune-cell migration	Principal epithelial seal and first line of defence against periodontal infection; sulcus > 3 mm is generally pathological

The background of the slide is composed of two distinct microscopic images of lung tissue. The left half shows a dense, pink-stained area, likely representing alveolar septa or a region of consolidation. The right half shows a more typical alveolar structure with large, clear spaces (alveoli) separated by thin, pink-stained walls. The text 'CLINICAL CONSIDERATIONS' is overlaid on the left side of the image.

CLINICAL CONSIDERATIONS

LINGUAL TONSILS

- localised symmetrical swelling,
- symptomless
- without the presence of any observable underlying pathology



Fig. 14.78 View of back of tongue showing a lingual thyroid (arrow). Courtesy Prof. J D Langdon.

LINGUAL THYROID

- Benign developmental condition.
- The dorsum of post. 1/3rd of tongue show → midline swelling representing active thyroid gland tissue (failed to migrate from its developmental foramen caecum site).
- Remnants of the migrating thyroid gland may also form **thyroglossal duct cysts** in neck.
- Diagnosis: radioactive iodine test & biopsy

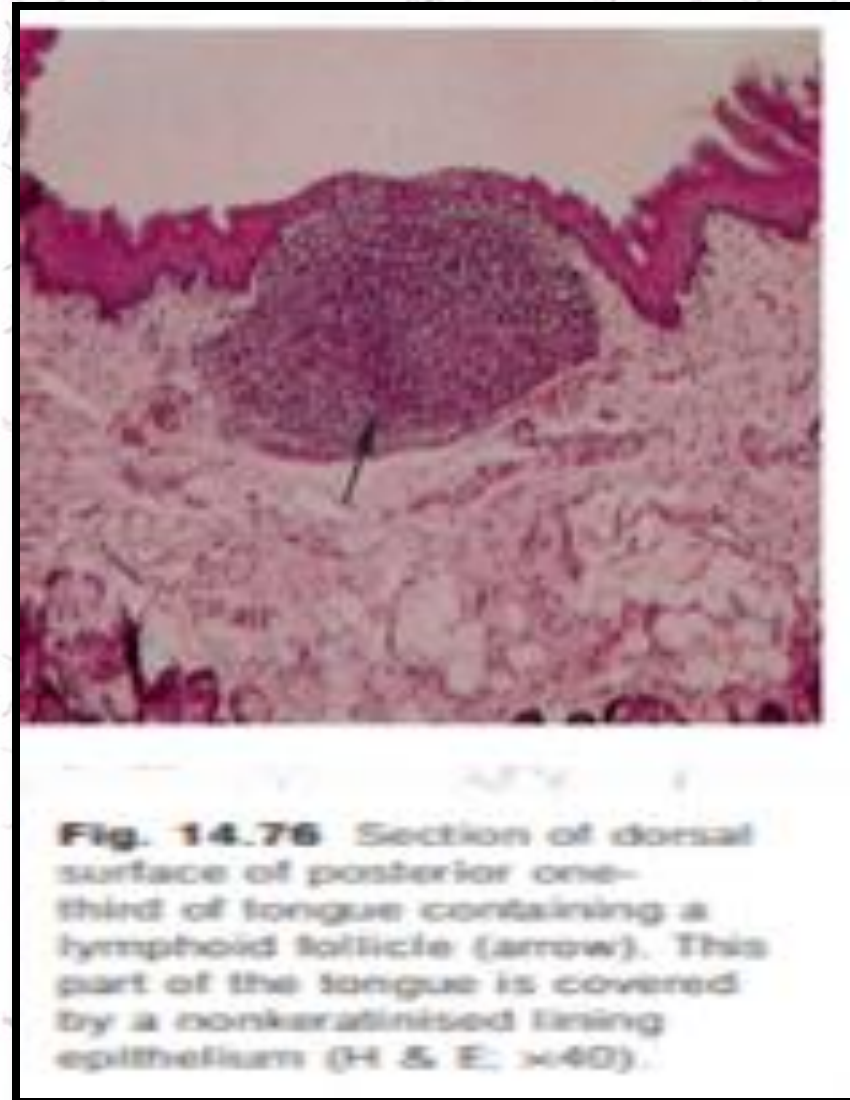


Fig. 14.76 Section of dorsal surface of posterior one-third of tongue containing a lymphoid follicle (arrow). This part of the tongue is covered by a nonkeratinised lining epithelium (H & E, $\times 40$).

ERYTHEMA MIGRANS

- Common benign inflammatory condition → cause (unknown) → characterised by recurrence & disappearance of red areas on tongue.
- Smooth, red regions on tongue related to filiform papillae atrophy surrounded by grey-white, irregular boundary → likened to outline of continents on map, → '**geographical tongue**'.



- **characteristic** → outlines of the 'map' change, even from day to day
- symptomless, burning sensation in some.
- 3% of population, adults more often affected than children.
- **Red coloration** → histologically coincides with → epithelial thickening & inflammatory cells presence
- In grey-white borders, ↑ concentration of neutrophils in epithelium

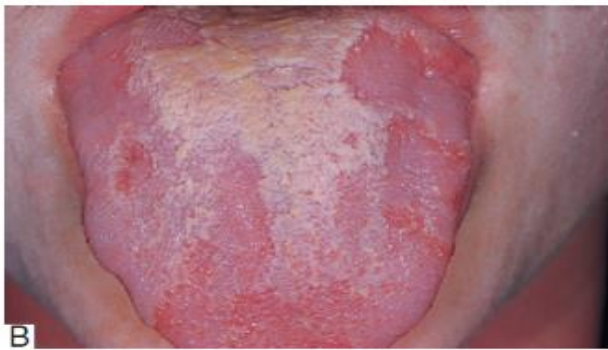


Fig. 14.80 (A) Typical appearance of erythema migrans (geographic tongue) showing irregular depapillated red patches centred on the lateral sides of the tongue, with each patch having a narrow red and white rim. (B) Tongue of the same patient shown in Fig. 14.80A but taken on a later occasion and showing a change in the pattern of red patches. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed, 2008.

MEDIAN RHOMBOID GLOSSITIS

- Median rhomboid glossitis → benign abnormality seen in midline of dorsum of tongue at junc. b/w ant. 2/3rd & post. third.
- Appears as nodular red or pink region devoid of papillae & showing infiltration of inflammatory cells.
- can sometimes mistakenly be diagnosed as carcinoma.



Fig. 14.82 The typical lozenge-shaped area of depapillation in the midline of the dorsum of the tongue representing median rhomboid glossitis. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed, 2008.

WHITE LESIONS

white patch → one of the most sites → floor of the mouth & ventral surface of tongue, here associated with smoking .

In one form a smooth surfaced, white patch shows a characteristic rippling pattern, with the normal nonkeratinised lining epithelium alternating with areas of keratinised epithelium

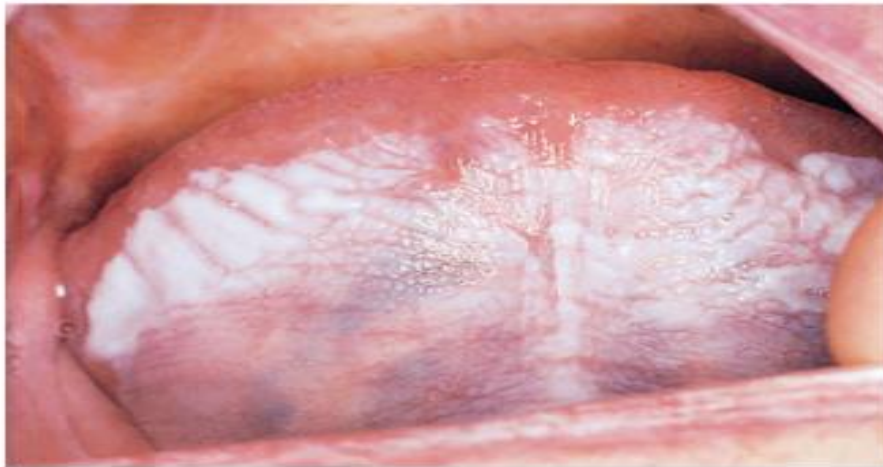


Fig. 14.83 White patch on the ventral tongue and floor of the mouth (sometimes referred to as sublingual keratosis) associated with smoking. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed, 2008.

WHITE LESIONS - LICHEN PLANUS

- A common chronic inflammatory disease that produces white, often striated lesions in mouth, usually in the buccal mucosa, is lichen planus
- the white appearance reflecting the presence of hyperkeratosis or parakeratosis.



Fig. 14.85 Lichen planus in the buccal mucosa, producing a lacy network of white striae. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed, 2008.

WHITE LESIONS - SQUAMOUS CELL CARCINOMA

- A change in form to speckled appearance associated with firmness on palpation (induration related to infiltration by tumour cells), ulceration and a rolled border may signify a malignant change, producing a squamous cell carcinoma.
- Confirmation → biopsy showing invasion of epithelium into deeper tissues



Fig. 14.86 Squamous cell carcinoma showing a speckled appearance, firmness on palpation (induration) and an ulcerated rolled margin. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed, 2008.

WHITE LESIONS - CANDIDA ALBICANS

- White patches associated with infections → *Candida albicans* (appear in lining mucosa)
- Diagnosis: stains to reveal the causal organism within epithelium



Fig. 14.88 White patch on cheek (arrow) resulting from infection with *Candida albicans*. Courtesy Prof. P R Morgan.

WHITE LESIONS - WHITE SPONGE NEVUS

- rare, benign, **autosomal dominant**; thick white soft plaques bilaterally on nonkeratinised mucosa (tongue, cheek).
- Histology: thickened, parakeratinised epithelium with widespread suprabasal vacuolation.
- Underlying cause = mutation affecting **CK13** and/or **CK4** structure.

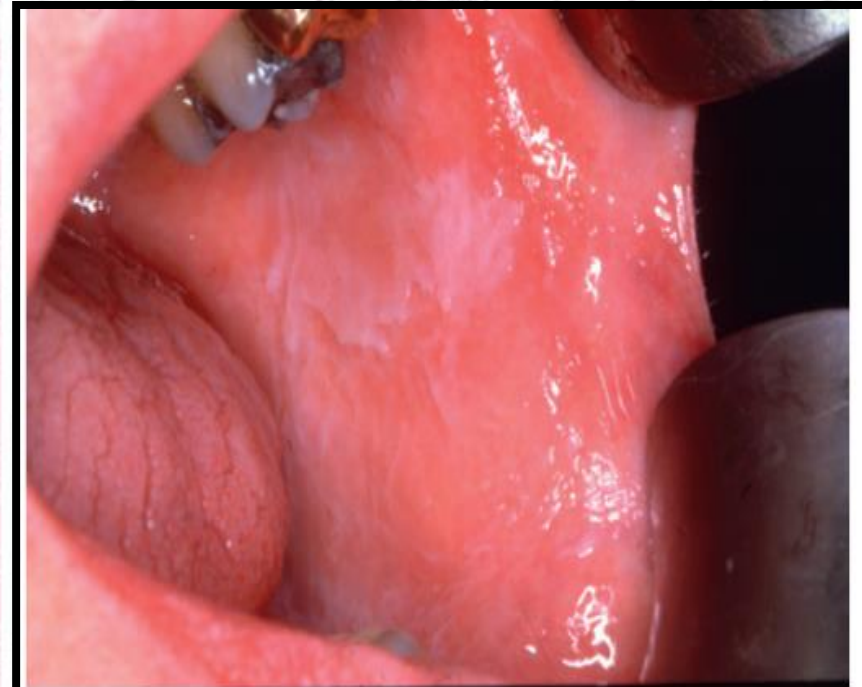


Fig. 14.90 White sponge naevus on the buccal mucosa. Courtesy Prof. P R Morgan.

BASEMENT MEMBRANE DISORDERS

- **Mucous membrane pemphigoid**
– autoantibodies against **BP180** & **BP230** (bullous pemphigoid antigens, hemidesmosome components) weaken epithelium-lamina propria bond → splitting at basement membrane → blisters/ulcers, inflamed gingiva.
- **Epidermolysis bullosa** (group of mucocutaneous conditions, all cause subepithelial blistering) – some forms show mutant **CK14**.



Fig. 14.92 Mucous membrane pemphigoid of the gingiva producing an inflamed appearance of the gingiva.
From Cawson's essentials of oral pathology and oral medicine, 8th ed, 2008.

CYTOKERATINS AND MALIGNANCY

It has been recognised that there is potential to exploit the epithelial specificity of cytokeratins in diagnostic histopathology, principally in the determination of the tissue of origin of poorly differentiated neoplasms.

THANKS FOR YOUR ATTENTION

Enter your sub headline here

